

Developing Electrohydraulic System Solutions using Adaptive Neuro - Sliding Mode Control and Vision Sensing

Thesis submitted by
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I, **Shri Shouvik Chaudhuri** registered on **14th July, 2015**, do hereby declare that this thesis entitled "**Developing Electrohydraulic System Solutions using Adaptive Neuro - Sliding Mode Control and Vision Sensing**" contains literature survey and original research work done by the undersigned candidate as part of Doctoral studies.

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.....

*Dedicated to struggles & failures,
which helped me to learn
& hopefully,
will continue to do so.*

.....

*“I would rather have questions
that can’t be answered,
than answers that
can’t be questioned”*

---Prof. Richard P. Feynman

* * * * *

The Road Not Taken

*Two roads diverged in a yellow wood,
And sorry I could not travel both
And be one traveler, long I stood
And looked down one as far as I could
To where it bent in the undergrowth;*

*Then took the other, as just as fair,
And having perhaps the better claim,
Because it was grassy and wanted wear;
Though as for that the passing there
Had worn them really about the same,*



*And both that morning equally lay
In leaves no step had trodden black.
Oh, I kept the first for another day!
Yet knowing how way leads on to way,
I doubted if I should ever come back.*

*I shall be telling this with a sigh
Somewhere ages and ages hence:
Two roads diverged in a wood, and I—
I took the one less traveled by,
And that has made all the difference.*

--- Robert Frost

(Painting Courtesy: Dr. Anusha Chaudhuri)

* * * * *

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As I start to pen down my journey during the past few years, spent in the pursuit of a doctoral thesis, I reminisce about the first time I came across the famous commencement address given by Mr Steve Jobs at Stanford University. In the first story of that address, he eloquently put his own experiences about the nature of curiosity in a few simple words and told us that,

“You cannot connect the dots looking forward, you can only connect them looking backward. So you have to trust that the dots will somehow connect in your future.”

At this point you must be wondering, why I am reiterating words from a famous speech in my thesis acknowledgement. The answer to that question is quite simple. In order to have a chance at connecting the dots in the distant future, it is obvious that you must be able to create or explore the dots in the first place in the present. Over the years, this philosophy has given me the necessary perspective and motivation to indulge in things I have found to be of interest. Having said that, I must admit that such experiences or indulgences would have never come to fruition, had I not come across certain individuals and environments along the way. This is my humble attempt to acknowledge their roles in this quest of mine.

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Abstract

Electrohydraulic actuation systems (**EHAS**) have propelled the development of many industrial sectors over the years, primarily owing to the gamut of preferable characteristics possessed by the **EHAS** over other actuation mediums such as large torque or force delivering capability, high load holding capacity without braking, fast response time and high power-to-weight ratio. Additionally, the use of Proportional valves (**PV**) in the **EHAS** lowers the machining costs, thereby, reducing the filtration costs. This further increases their resistance to dust-prone environments, thus, making them a lucrative industrial solution. Despite such features and wide scale industrial utilization, the real-time control of such industry-grade **EHAS** tend to be quite challenging. These issues arise due to the presence of several nonlinearities in the actuation and the loading subsystems, uncertainties in involved parameters and external disturbances which gives rise to unknown system dynamics, thus complicating the control aspects of **EHAS** and severely impacting the bounds of achievable performance. Therefore, the design and development of robust and intelligent control algorithms for such industry-grade **EHAS**, has gained prominence in recent times, with the objective of achieving equivalent performance to the servo-grade **EHAS**. The most widely used control candidate for such purposes has been the sliding mode control (**SMC**) approach. Also, the use of contactless vision sensing devices in the feedback loop or vision based feedback control in **EHAS** applications as an alternative to conventional sensor-based schemes has gained traction over the last decade. The present work is, thus, devoted to developing real-time control solutions for industry-grade **EHAS** by utilizing hybrid adaptive control strategies constructed around the sliding mode control approach and vision sensing based feedback schemes. In addition, the utility of the vision sensing approach is extended to a multi-actuator **EHAS** or manipulator, so as to develop a motion sensing application.

The first part of this work is concerned with the development of a real-time position control application of an **EHAS** by utilizing a synergistic combination of an integral sliding mode controller (**ISMC**), known for being insensitive to external disturbances and variation in system parameters, and a recurrent neural network (**RNN**) with dynamic mapping capabilities and different orthogonal Hermite polynomial basis functions (**OHPBFs**), defined over the interval $(-\infty, \infty)$, as the activation function for each hidden neuron. This network structure is termed as the recurrent Hermite neural network with integral sliding mode control (**RHNN-ISMC**). In the second part of this work, a real-time position control application of a linear **EHAS** with vision based feedback from a monocular CCD

camera is developed and controlled with the aid of first-order sliding mode (or **1-SMC**) scheme. Both the control strategies mentioned above, are augmented with a tan-hyperbolic bias controller for tackling the hard nonlinearities of the system like stiction in the actuator and deadband in the **PV**. Also, the continuous variation of the dynamic (or free) parameters of the system is tackled by using adaptive laws based on a Lyapunov functional to update such parameters, in both the control configurations. The objective of the third part of this work is to expand the utility of the vision sensing approach to multi-actuator **EHAS** or manipulator domain. Therefore, in this work, a real-time target motion sensing and trajectory replicating system for moving objects, using a virtual test target, is developed by using a 6-DoF Electrohydraulic Parallel Manipulator or Stewart Platform (**SP**) with an eye-in-hand camera configuration, based on the dynamic image based look-and-move visual servoing scheme.

The results obtained from real-time experiments on a linear **EHAS** for the **RHNN-ISMC** scheme, establishes its ability to provide satisfactory tracking responses for standard test signals of 0.02m and 0.04m peak-to-peak amplitude, particularly in the low-frequency regime (up to 1.0Hz). The evaluation of the real-time performance of the proposed controller against other previously established control schemes, based on normalised performance indices, suggests that the **RHNN-ISMC** scheme requires much less control effort to achieve a similar tracking performance. Similar real-time performances are obtained for the second part of the work, involving the **1-SMC** scheme, whereby the vision based scheme seems to be at par with the **LVDT** based feedback scheme. A comparison of the real-time performances for standard regulation and tracking demand signals up to 1.0Hz frequency, indicates that the adaptive **1-SMC** scheme with deadband controller (**ASMCDB**) outperforms the conventional **PI** controller. Lastly, the real-time performance of the motion sensing system is analysed by subjecting it to simple 1D and 2D target trajectories with different temporal velocities. The corresponding performance indices calculated for each of these trajectories indicate that this particular laboratory scale parallel manipulator based servoing system (**PMBSS**) is suitable for operating around 0.1Hz for 1D and 2D trajectories. The real-time application of the proposed controllers and the motion sensing system involving **EHAS**, hints at the possibility of developing low-cost industrial solutions with reasonably good performances.

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List of Abbreviations

A

ACSMC	1st order-Sliding Mode Control
ACSMC	Adaptive Complementary Sliding Mode Control
ADC	Analog-to-Digital Converter
AFSMC-AB	Adaptive Fuzzy Sliding Mode Control with Adaptive Bias
ANN	Artificial Neural Network
APF	Active Power Filter
ASMCDB	Adaptive Sliding Mode Control with DeadBand

B

BSM	Ball Shaped Marker
-----	--------------------

C

C	Monocular Camera
CCD	Charge-Coupled Device
CE	Control Effort
CMACNN	Cerebellar Model Articulation Controller Neural Network
CNC	Computer Numerical Control
cRIO	Compact Reconfigurable Input Output

D

DAAE	Discrete Average Absolute Error
DAC	Digital-to-Analog Converter
DACE	Discrete Average Control Effort
DADS	Dynamic Analysis and Design Software

DATE	D iscrete A verage T rajectory E rror
DHLRNN	D ouble H idden L ayer R ecurrent N eural N etwork
DLRNN	D ouble L oop R ecurrent N eural N etwork
DoF	D egree o f F reedom
E	
ECL	E nd- P oint C losed L oop S ystem
EHAS	E lectro H ydraulic A ctuation S ystem
EOL	E nd- P oint O pen L oop S ystem
F	
F	I ncline F ilter
FFNN	F eed- F orward N eural N etwork
FFT	F ast F ourier T ransform
FNN	F uzzy N eural N etwork
FPGA	F ield- P rogrammable G ate A rray
G	
GA-OFFB	G enetic A lgorithm O ptimized F uzzy F eedforward B ias
H	
HC	H older C up for S pring
HPP	H ydraulic P ower P ack
I	
IAE	I ntegral A bsolute E rror
IAQ	I ndoor A ir Q uality
IBVS	I mage B ased V isual S ervoing
ICE	I ntegrated C ontrol E lectronics
IM	I nduction M otor
iM	I ncput M odule
ISM	I ntegral S liding M ode C ontrol
ISO	I nternational S tandards O rganization

ITAE	I ntegral T ime-weighted A bsolute E rror
IVSC	I ntegral V ariable S tructure C ontrol
L	
LabVIEW	L aboratory V irtual I nstrument E ngineering W orkbench
LVDT	L inear V ariable D ifferential T ransformer
M	
MATLAB	M ATrix L ABoratory
MEMS	M icro- E lectro- M echanical S ystems
MIS	M inimally I nvasive S urgery
MLP	M ulti- L ayer P erceptron
MRAC	M odel R eference A daptive C ontrol
N	
NCE	N ormalized C ontrol E ffort
NDAE	N ormalized D iscrete A bsolute E rror
NDTAE	N ormalized D iscrete T ime-weighted A bsolute E rror
NRV	N on- R eturn V alve
NTSMC	N on-singular T erminal S liding M ode C ontrol
O	
OHP	O rthogonal H ermite P olynomial
OHPBF	O rthogonal H ermite P olynomial B asis F unction
oM	O utput M odule
P	
P	P ump
PBVS	P osition B ased V isual S ervoing
PD	P roportional- D erivative C ontroller
PI	P roportional- I ntegral C ontroller
PID	P roportional- I ntegral- D erivative C ontroller
PMBSS	P arallel M anipulator B ased S ervoing S ystem

PRV	P ressure R elief V alve
PV	P roportional V alve
R	
RBFNN	R adial B asis F unction N eural N etwork
REM	R elative E rror in M agnitude
RFNN	R ecurrent F uzzy N eural N etwork
RHNN	R ecurrent H ermite N eural N etwork
RNN	R ecurrent N eural N etwork
RTS	R ea T - T ime S ystem
S	
S	S pring
SISO	S ingle I nterface S ingle O utput
SM	S upply M anifold
SMC	S liding M ode C ontrol
SMIC	S liding M ode C ontrol with I ntegral C ompensator
SP	S tewart P latform
SRA	S ingle R od A ctuator
T	
T	T ank or R eservoir
T2FNN	T ype- 2 F uzzy N eural N etwork
TAMB	T hrust A ctive M agnetic B eating
TT	T est T arget
V	
VEE	V isual E ngineering E nvironment
VS	V isual S ervoing
VSC	V ariable S tructure C ontrol

List of Symbols

A list of all the symbols, their representative quantities and units has been provided below so as to familiarize the readers with the frequently used symbols within this thesis.

Symbols for Physical Systems

a, b	Control valve solenoids
a_1	Hydraulic Actuator cap-end port
a_2	Hydraulic Actuator rod-end port
Ch_1	Hydraulic Actuator cap-end chamber
Ch_2	Hydraulic Actuator rod-end chamber
A	Metered port of Control valve
B	Metered port of Control valve
P	Pump port of Control valve
T	Tank port of Control valve

Scalar Symbols (English Alphabets)

$\ddot{e}$	Second order Piston displacement error dynamic	m/s ²
$\ddot{x}$	Piston acceleration	m/s ²
$\dot{\hat{E}}$	Rate of change of estimate of E	m/s ³
$\dot{\hat{u}}_{sat}$	Rate of change of estimate of u_{sat}	m/s ³
$\dot{e}$	First order Piston displacement error dynamic	m/s
$\dot{s}$	Time derivative of sliding surface	m/s ²
$\dot{W}$	Rate of change of the positive definite Lyapunov function, W	
$\hat{E}$	Estimate of E	
$\hat{u}$	Estimate of input linearized control, u	m/s ²

$\hat{u}_b$	Estimate of bias controller, u_b	m/s ²
$\hat{u}_c$	Estimate of compensation controller, u_c	m/s ²
$\hat{u}_{db}$	Estimate of tan-hyperbolic deadband controller, u_{db}	m/s ²
$\hat{u}_{RHNN}$	Estimate of u_{RHNN}	m/s ²
$\hat{u}_{sat}$	Estimate of u_{sat}	m/s ²
$\hat{u}_{sm}$	Estimate of first-order sliding mode controller, u_{sm}	m/s ²
$\tilde{E}$	Difference between E and $\hat{E}$	
$\tilde{u}_{RHNN}$	Difference between u_{RHNN}^* and $\hat{u}_{RHNN}$	m/s ²
$\tilde{u}_{sat}$	Difference between u_{sat}^* and $\hat{u}_{sat}$	m/s ²
A_{a1}	Actuator cap-end chamber area	m ²
A_{a2}	Actuator rod-end chamber area	m ²
c	Coordinate of centroid of test target along x-axis in image plane	
c_c	Coordinate of image center along x-axis	
C_{vA}	Valve coefficient of metered port A	m ³ /(s.V√Pa)
C_{vB}	Valve coefficient of metered port B	m ³ /(s.V√Pa)
d	System disturbance	m/s ²
d_b	Cylinder bore diameter	m
d_r	Piston rod diameter	m
E	Small positive constant for the switching controller, $\hat{u}_c$	
e	Actuator piston displacement error	m
f	Frequency	Hz
F_0	Static friction or stiction	N
F_b	Maximum boundary lubrication friction	N
F_c	Coulomb friction	N
F_f	Actuator Friction	N
f_x	Focal length of camera in pixels along x-axis	
f_y	Focal length of camera in pixels along y-axis	
h	Height between base and top platform of parallel manipulator	m
H_k	k^{th} order orthogonal Hermite polynomial for the k^{th} hidden neuron	

h_k	Orthogonal Hermite polynomial basis function (k^{th} neuron)	
k	Spring Constant	N/m
k_I	Integral Gain of PI controller	V/(ms)
k_P	Proportional Gain of PI controller	V/m
m	Mass of actuator piston	Kg
N	Iteration number	
n	Order of the system	
N_s	Total number of samples	
num	Total number of actuators in the parallel manipulator	
O_c	Coordinate of optical center of camera in pixels along x-axis	
O_r	Coordinate of optical center of camera in pixels along y-axis	
P_1	Actuator cap-end chamber pressure	Pa
P_2	Actuator rod-end chamber pressure	Pa
P_P	Pump pressure	Pa
P_T	Tank pressure	Pa
q	Number of hidden neurons	
Q_1	Actuator supply flow rate	m^3/s
Q_2	Actuator return flow rate	m^3/s
r	Coordinate of centroid of test target along y-axis in image plane	
r_c	Coordinate of image center along y-axis	
r_j	j^{th} input signal to the RHNN	
s	Sliding surface	m/s
s_{ei}^*	Error in actuator strokes of the parallel manipulator ($i = 1, \dots, 6$)	m
s_i^*	Actuator demand strokes of the parallel manipulator ($i = 1, \dots, 6$)	m
s_c	Skew coefficient of camera	
s_i	Actuator response strokes of the parallel manipulator ($i = 1, \dots, 6$)	m
s_{max}	Maximum available Actuator stroke	m
T	Time period of target frequency	sec
t	Time	sec

t^*	Demand trajectory of the parallel manipulator	m
t_0	Initial time instant	sec
t_e	Trajectory error of the parallel manipulator	m
u	Control Input to the system	m/s ²
u^*	Ideal value of input linearized control, u	m/s ²
u_{RHNN}^*	Ideal value of u_{RHNN}	m/s ²
u_{bl}	Estimate of u_{bl} ($l = 1, 2$)	m/s ²
u_{bl}	Saturation values of the bias controller ($l = 1, 2$)	m/s ²
u_b	Control input from the bias controller	m/s ²
u_c	Control input from the compensation or switching controller	m/s ²
u_{db}	Control input from the tan-hyperbolic deadband controller	m/s ²
u_{RHNN}	Control input from the RHNN	m/s ²
u_{sat}	Saturation level of the controller, u_{db}	m/s ²
u_{sm}	Control input from the first-order sliding mode controller (1-SMC)	m/s ²
V	Control voltage to the valve	Volts
$v, \dot{x}$	Piston velocity	m/s
v_0	Slip velocity	m/s
V_b	Bias voltage	Volts
v_b	Lubrication friction velocity	m/s
V_{in}	Input-Linearized voltage	Volts
V_i	Control voltages to actuators of parallel manipulator ($i = 1, \dots, 6$)	m
V_{max}	Maximum available control voltage	Volts
v_s	Stribeck velocity	m/s
V_{th}	Threshold voltage	Volts
W	Positive definite Lyapunov function	
w	RHNN scalar output	
x	Displacement of actuator piston	m
x_0	Initial position of actuator piston	m
X_d	Demand signal magnitude from FFT analysis	m

X_d	Response signal magnitude from FFT analysis	m
Z	Camera Depth from target	m
z^{-1}	Time delay	sec
z_k	Input to the k^{th} hidden neuron of RHNN	
$\dot{u}_{bl}$	Rate of change of estimate of u_{bl}	m/s ³
u_{sat}^*	Ideal value of u_{sat}	m/s ²

Scalar Symbols (Greek Alphabets)

α_b	Large constant of bias or deadband controller, $\hat{u}_b$	m/s ²
α_v	Viscous friction coefficient	kg/sec
χ	Small positive number	
δ	Pre-compression of Spring	m
$\dot{\eta}_{sm}$	Rate of change of estimate of η_{sm}	m/s ³
η_θ	Positive learning rate for $\dot{\hat{\Theta}}$	s ⁻²
η_b	Positive learning rate for $\dot{\hat{u}}_{bl}$	s ⁻²
η_E	Positive learning rate for $\dot{\hat{E}}$	s ⁻²
η_{sm}	Toggle limit of the first order sliding mode controller	m/s ²
γ_k	Output of the k^{th} hidden neuron of RHNN	
$\hat{\eta}_{sm}$	Estimate of η_{sm}	m/s ²
κ	Frequency index of signal frequency, f	
λ_i	Hurwitz coefficients ($i = 0, \dots, n - 1$)	
λ_x	Constant gain of feature space control law for x-axis	s ⁻¹
λ_y	Constant gain of feature space control law for y-axis	s ⁻¹
μ_j	Weight between j^{th} input neuron and a hidden neuron of RHNN	
ψ	Fixed recurrent gain	
σ_{sat}	Positive learning rate for $\dot{\hat{\eta}}_{sat}$	s ⁻²
σ_{sm}	Positive learning rate for $\dot{\hat{\eta}}_{sm}$	s ⁻²
τ	Sample time	sec
θ_k	Weight between k^{th} hidden neuron and output neuron of RHNN	
$\tilde{\eta}_{sm}$	Difference between η_{sm}^* and $\hat{\eta}_{sm}$	m/s ²

ε	Minimum reconstruction error	
η_{sm}^*	Ideal value of η_{sm}	m/s ²
Δ	Unknown approximation error	
Δ_{db}	First-order terms in Taylor series expansions of $\hat{u}_{db}$	m/s ²
Δ_h	Higher-order terms in Taylor series expansions of u^*	m/s ²
Δ_{sm}	First-order terms in Taylor series expansions of $\hat{u}_{sm}$	m/s ²

Vector Symbols (English & Greek Alphabets)

$\dot{\theta}$	Angular velocity vector of camera frame
Γ	Hidden layer output vector of the RHNN
$\dot{\mathbf{d}}$	Translational velocity vector of camera frame
$\dot{\mathbf{e}}$	Image pixel error rate
$\hat{\mathbf{e}}_0$	Unit vectors for translation motion of parallel manipulator
$\mathbf{d}$	Translational vector of parallel manipulator
$\mathbf{e}$	Error vector
$\mathbf{f}$	Current image feature
$\mathbf{f}^*$	Desired image feature
$\mathbf{k}$	Visual Servoing parameter vector
$\mathbf{l}_0$	Initial Leg length vector of parallel manipulator
$\mathbf{l}_i$	Current Leg length vector of parallel manipulator ($i = 1, \dots, 6$)
$\mathbf{M}$	Weight vector between input and hidden layer of the RHNN
$\mathbf{p}_{T_i}$	Position vector of T_i in the moving coordinate system ($i = 1, \dots, 6$)
$\mathbf{q}$	End-effector pose vector of parallel manipulator
$\mathbf{R}$	Input vector to the input layer of the RHNN
$\mathbf{s}$	Image coordinates of target centroid
$\mathbf{s}_c$	Coordinates of image center
$\mathbf{v}, \dot{\mathbf{q}}$	Camera (or end-effector) velocity vector of parallel manipulator
$\mathbf{x}$	State vector
$\mathbf{x}^*$	Demand vector
$\mathbf{x}_{B_i}$	Position vector of B_i in the global coordinate system ($i = 1, \dots, 6$)

$\dot{\mathbf{f}}$	Current image feature rate vector
Θ	Weight vector between hidden and output layer of the RHNN
θ	Rotational vector of parallel manipulator
Θ^*	Ideal values of Θ vector of the RHNN
$\dot{\hat{\Theta}}$	Rate of change of estimate of Θ
$\hat{\Theta}$	Difference between Θ^* & $\hat{\Theta}$
$\hat{\Theta}$	Estimates of Θ vector of the RHNN

Matrix Symbols (English Alphabets)

$\mathbf{J}$	Interaction matrix or Feature Jacobian matrix
$\mathbf{J}^+$	Moore-Penrose pseudoinverse for $\mathbf{J}$
$\mathbf{K}$	Intrinsic camera matrix
$\mathbf{R}_\alpha$	Roll transformation due to rotation by α about p-axis
$\mathbf{R}_\beta$	Pitch transformation due to rotation by β about q-axis
$\mathbf{R}_\gamma$	Yaw transformation due to rotation by γ about r-axis
$\mathbf{R}_m$	Rotational matrix

CHAPTER 1

Introduction

The science of Hydraulics or liquid flow and the field of automatic control share a collaborative past and have developed alongside each other over time. Some of the earliest documented examples of their union can be found in the works of two Alexandrian inventors, namely, Ktesbios from the 3rd century BC credited with the invention of a water clock [1] and Hero from the 1st century AD, who used water-regulated devices to accomplish various control tasks involving the opening of temple doors, rotational motion of an *Aeolipile* or a steam-engine and automatic production of musical notes [2]. A branch of this ancient discipline evolved into the field of *Hydraulic Control* during the past few centuries, concerning the use of modern machinery. This in turn led to parallel developments in various industrial sectors such as machine tools, ship control, aircraft control, earth moving equipment, transportation and many others [3]. Recent advancements in solid state electronics, the advent of electrically-interfaced hydraulic systems and the rise of modern micro-computational devices has further accelerated the progress of electrohydraulic control by enhancing their practicality [1]. The present work described in this thesis contributes to this ever-growing technological field by developing real-time motion control solutions for electrohydraulic systems, with the aid of newly proposed nonlinear control strategies and vision guided feedback algorithms, which in turn are used to lay the ground work for a novel motion sensing electrohydraulic platform.

In this introductory chapter of the thesis, an overview of electrohydraulic systems and their control aspects have been presented at the beginning. This is followed by a detailed survey of the research available in the literature about the topic, based on which the outstanding issues have been identified. This further leads to the motivation behind the present work, followed by the objective and contributions of this research. Finally, the organization of the rest of the thesis has been described.

1.1 Overview of Electrohydraulic Systems

Research in hydraulic technology spanning over several decades has led to the development of hydraulic systems intertwined with electrical interfaces such as electrically operated control valves and sensors for precise actuation of various systems. As a whole, these electrically-interfaced precision hydraulic actuation systems are often referred to as electrohydraulic systems or more specifically *Electrohydraulic Actuation Systems* (**EHAS**). These **EHAS** play an indispensable role in most modern industrial sectors primarily owing to the desirable features of hydraulic power which includes a better dynamic response, high power density and fault tolerance as compared to electrical or equivalent electro-mechanical systems. This section presents a brief description of these **EHAS**, their applications in various sectors and the control issues associated with them.

1.1.1 Electrohydraulic Actuation Systems

In general, a typical **EHAS** can be thought of as a combination of two distinct sections *viz.* the Hydraulic section and the Electrical/Electronics section, interfaced with the aid of a control element. Each of these sections comprises some major elements/devices and the interplay among these elements leads to the generation of precise real-time motions. These sections and their major elements are described below in detail.

1. **Hydraulic Section:** This domain of the **EHAS** primarily consists of three major elements and an assembly, as described below.

- (a) *Hydraulic Fluid:* The hydraulic fluid refers to any liquid which has the features to act as the medium of power transmission in a hydraulic system. In general, water and certain grades of mineral oil based liquids (HL-, HF-, HE-, HG-, HV- etc.) standardized by the **ISO** are used as the hydraulic fluid. These hydraulic fluids ideally tend to be non-compressible, fire-resistant, non-corrosive, thermally stable and have a long life span [3].
- (b) *Power Supply Assembly:* A hydraulic power supply assembly or simply *Hydraulic Power Pack* (**HPP**) is used to drive the hydraulic fluid through the entire hydraulic circuit. An **HPP** may be used either as a source of constant pressure or flow or both. A typical constant flow **HPP**, which are most prevalent, comprises an oil reservoir or tank (**T**) which holds sufficient supply of the working fluid, a hydraulic pump for driving the fluid from the reservoir to the circuit under high pressure, a prime mover

(generally an electric motor for static supplies) to drive the pump, particle oil filters for avoiding oil contamination, a pressure relief valve for safety purposes, non-return or check valves for unidirectional flow and occasionally an oil cooler to dissipate heat generated during operation of the circuit. The **HPP** serves as an open-circuit pumping system where the control valve and the actuation element serve as the load for the hydraulic pump [4].

- (c) *Control Valve*: A hydraulic control valve is a device which is used to control the fluid power source by utilizing mechanical motion which may be triggered by solenoids. In a simple industrial hydraulic circuit, the direction and speed of movement of the actuation element are controlled by the directional control valve and the flow control valve, respectively. The spool in a conventional solenoid operated direction control valve has only specified states and can be thought of as a switching valve just like an electrical on/off switch. Those applications which require better control often tend to use solenoid-operated proportional directional valves or simply *Proportional valves (PV)*, whereby the valve spool can achieve any intermediate state between fully open or closed, analogous to an electrical dimmer switch. This is achieved by precise positioning of the valve spool within these solenoid-operated valves, which act as a transducer between the hydraulic and electrical domains of the **EHAS** [1]. In a **PV**, the direction of the spool movement away from the central position determines the direction of the movement of the actuation element and the amount of spool movement controls its speed. So in effect, the **PV** acts as both a directional control valve and a flow control valve. Additionally, **PV** are more robust compared to servo-grade valves owing to the presence of higher radial clearances between the spool-bush pair [5], leading to minimum filtration costs. This implies that they can be used in dust-prone external environments involving forestry and agricultural applications. Moreover, due to large clearance between the mating parts, the machining /assembly costs are also lower. This makes them a lucrative industrial solution.
- (d) *Hydraulic Actuator*: The actuator functions as the final motional element of the **EHAS** which uses pressure supplied by the hydraulic fluid and regulated by the control valve to facilitate mechanical operation of a mechanism or system, thereby generating desired motion or force. The output of these actuators are typically linear or rotary in nature, depending upon the application. The most commonly used hydraulic actuator is the linear

actuation cylinder or jack, which is constructed as a hollow cylindrical tube along which a piston, fitted with a sliding solid piston-rod. They may be single-acting or double-acting depending upon the unidirectional or bidirectional nature of application of fluid pressure on the cylinder piston, respectively [6]. Hydraulic actuators may also be servo-grade or industry-grade based on their friction characteristics. The construction of a typical hydraulic cylinder may be either single-ended where a piston rod is affixed to just one end of the piston or double-ended where piston-rods are affixed to both ends of the piston. This, in turn, leads to the fact that the pressure-bearing area of the two inner cylinder chambers is asymmetrical for single-ended actuators and symmetrical for double-ended actuators. Considering similar work outputs, single-rod actuators are less expensive and occupy smaller workspaces, which makes them suitable for industrial and aerospace applications [6].

2. **Electrical/Electronic Section:** This domain of the **EHAS** primarily consists of four major elements, as given below.

- (a) *Transducers:* The output of the hydraulic actuator *viz.* the instantaneous position of the piston rod, the force applied by the piston on the load or the pressure inside the cylinder chambers, can be detected with the aid of transducers which are sensing elements. This output data is converted into an output signal by these elements, which may be used as a feedback for a closed-loop control system or simply to monitor the output in an open-loop control system. In most **EHAS** applications, transducers are used as the sensing elements to measure quantities such as piston position, piston velocity, force exerted by the piston rod, delivery pressure of the pump and flow rate of the hydraulic fluid, thereby converting them into equivalent low power analog electrical signals for system monitoring or control. The linear piston displacement is routinely measured with *Potentiometers*, which are resistive DC output devices or *Linear Variable Differential Transformer (LVDT)*, which are inductive AC output devices. The piston velocity is often measured with the help of AC or DC output *Tachometers*. Although sometimes, *Accelerometers* are also used to record system acceleration for control purposes and extraction of piston velocities. The force imparted by the piston is measured with the aid of *Load Cells* which are strain gauges arranged in bridge configurations. Pressure and flow in the hydraulic circuits are measured with the aid of single or differential *Pressure Transducers* and *Flow Transducers*, respectively, which are either piezoresistive or piezoelectric in nature.

- (b) *Data Acquisition & Control Devices*: The output signal produced by the transducers are captured and stored for analysis, with the aid of real-time data acquisition and control devices, which may be driven by microprocessors or Field-Programmable Gate Arrays (**FPGA**). These devices are often equipped with domain converters i.e. Analog-to-Digital Converters (**ADC**) used to digitize the incoming analog feedback signals from the transducers and Digital-to-Analog Converters (**DAC**) to transform digital control commands to analog electrical control signals to be sent to the control valve, alongside other signal conditioning circuits involving elements such as filters and amplifiers. These acquisition systems can work either in standalone mode whereby the control commands and the available data are stored within their internal memory which may be later retrieved for analysis or they may work in association with a computer, whereby the control data is directly stored within the computer's memory. Although, single-board computers like Arduino Mega or Raspberry Pi may be used for such purposes, many industrial sectors tend to use industry-grade data acquisition and control devices such as Compact Reconfigurable Input Output (**cRIO**) system by National Instruments and Data Acquisition system by Keysight Technologies, since they are quite robust and come with a user-friendly developmental software environment.
- (c) *Control Computer*: The control computer is primarily associated with the data acquisition systems via an ethernet cable or a wireless network and acts as a single point-of-contact for the user from which they can monitor and control the **EHAS**. Since, the digital version of feedback signals from the transducers are available to the user via the acquisition systems, it is possible to design a closed-loop control system with the aid of the control computer, wherein the control algorithm is developed within a software environment, from which the control command corresponding to the feedback signal can be generated. Most commercially available system-design platform and development environments such as Laboratory Virtual Instrument Engineering Workbench (**LabVIEW**) by National Instruments and Visual Engineering Environment (**VEE**) by Keysight Technologies are quite user-friendly and provide a visual programming style. The control command corresponding to a particular user-defined demand can be formed within such development environments and sent to the control valve via the data acquisition and control devices.
- (d) *Valve Amplifier*: The command signal from the control computer which emerges from the data acquisition and control device (coupled with **DAC**)

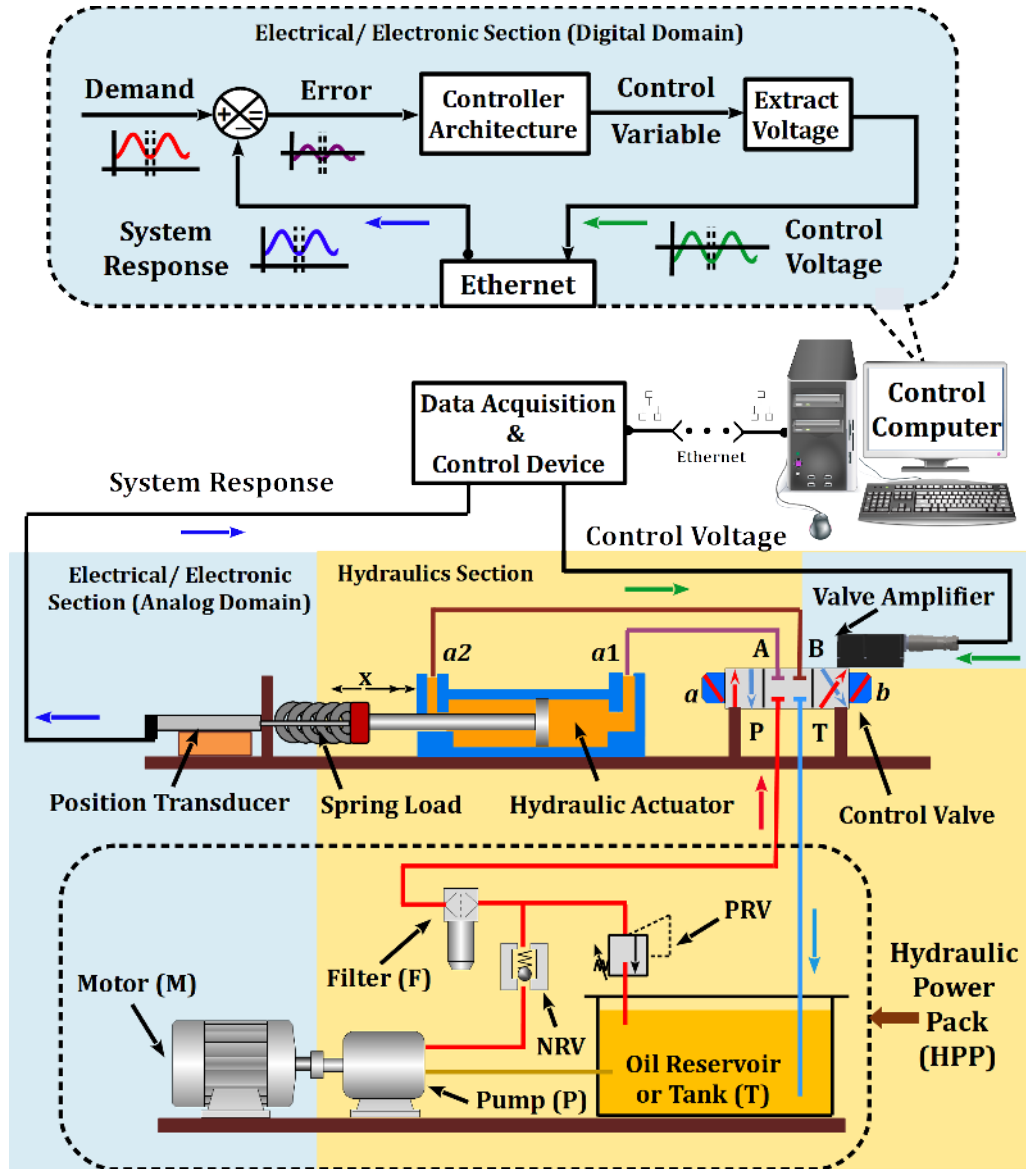


Figure 1.1: Schematic of a typical closed-loop Electro-Hydraulic Actuation System

in an analog form, is passed on to an electronic amplifier, which in turn controls current flowing through the solenoid of the valve and in turn the movement of the valve-spool. Though the valves may either be voltage driven or current driven, a valve solenoid typically draws a current of 2A to 3A, which is provided by the DC power supply (12V or 24V) to the amplifiers. The output of the amplifier i.e. the electrical current to the solenoid is controlled by the value of the incoming command signal. In a proportional valve, an increase in the command input signal to the amplifier results in a corresponding proportional increase in output current to the valve solenoid. Although, the required value of incoming command signal (in the form of electrical current) from the data acquisition and control device is just a few milliamperes. Therefore, the control device

can even be substituted with a potentiometer for the purposes of simple experiments.

The interplay between these two domains and their elements leads to a highly sophisticated industrial motion control system driven by ever-changing user-defined demands. A schematic of a typical closed-loop **EHAS**, highlighting the aforementioned elements and their interactions has been shown in Fig. 1.1.

1.1.2 Applications of EHAS

EHAS possess a gamut of preferable industrial characteristics such as large torque or force delivering capability, high load holding capacity without braking, fast response time, speed reversal and stalling capabilities, presence of higher stiffness compared to their electrical counterparts and high power-to-weight ratio [7]–[11]. Most modern industries have come up with their specific applications utilizing the high degree of accuracy and performance offered by the **EHAS**, some of which are enlisted below.

1. **Aerospace:** In the aerospace sector, **EHAS** are often used for actuation of primary flight control surfaces in an aircraft [12], [13]. Apart from aircrafts, they are also used in space shuttles to control different propellant valves, which in turn control thrust and mixture ratio of its main engine [14] and for actuation of the landing gear, brakes, steering and jet engine deployment [15].
2. **Automobile:** The automobile industry uses the capabilities of the electrohydraulic systems for purposes of auto-steering [16], braking [17], auto-transmission [18], double-clutch transmission [19] and active suspension [20], [21] in land vehicles.
3. **Construction & Mining:** Most construction and mining corporations tend to use **EHAS** for controlling the thrust system of shield tunneling machines [22], for designing heavy-duty excavators [23]–[25], for road grading and paving machines [1], for maintenance, rescue & fire-fighting purposes in mobile elevating work platforms or skylifts [26] and for controlling the bucket motion of wheel loaders [27]. They are also used in structural control for providing protection against environmental loads such as strong earthquakes and high winds [28].
4. **Manufacturing & Machine tools:** In the manufacturing sector, **EHAS** are used in press brakes of Computer Numerical Control (**CNC**) machines [29] and closed loop position control of their feed drives [30]. They are also

utilized as a feed drive mechanism in various machine tools used for turning, milling and drilling purposes [30], [31] and in injection molding machines as a driving system [32].

5. **Agriculture & Earth moving machinery:** In agriculture, they are often used in the steering systems of agricultural vehicles [33], hydraulically actuated tractors equipped with front-end loaders [34] and in the drive systems of combine harvesters [35]. They are also used in harvester manipulator machines used for tree-harvesting operations in forestry [36].
6. **Robotics:** In the field of Robotics, they were primarily used in industrial robots for delivering powerful and robust actuation, owing to their robustness and ability of generate significant forces/torques for their size without overloading the system. In fact, the very first industrial robot *Unimate* was hydraulically actuated [37]. Recent developments in legged robotics have utilized **EHAS** in bipeds or humanoids [37], quadrupeds [38], [39] and hexapods [40] for traversing rough terrains. Robotic manipulators in serial and parallel configurations also utilize them for purposes of motion simulation of aircrafts, ships, rough terrain, spacecraft docking mechanisms and environmental manipulation [1], [41], [42].

Apart from the major applications mentioned above, **EHAS** are frequently employed in various small scale industrial applications with or without automated control schemes to carry out specific tasks.

1.2 EHAS Control

Despite having a strong industrial presence, the real-time control of **EHAS** tend to be quite challenging. The presence of system nonlinearities in the actuation and the loading subsystems, uncertainties in involved parameters and external disturbances [3], [7], [43] gives rise to unknown system dynamics, which complicates the control issue of an industry-grade **EHAS** and severely impacts the bounds of achievable performance. In order to design intelligent and robust control schemes for such systems, a thorough knowledge of the sources of these complexities in the **EHAS** must be available. Hence, a consolidated list of the most commonly encountered complexities afflicting the performance of the **EHAS**, are given below.

1. **Continuous or Soft Nonlinearities:** Continually occurring nonlinearities in an **EHAS** manifest themselves in the form of dynamic or kinetic friction between the actuator cylinder and piston [8], [44], the compressibility of the hydraulic oil, square root pressure-flow relations in the metered ports

of the proportional valves, transmission nonlinearities resulting from system geometry, asymmetric actuation due to unequal piston cross-sectional area, varying trapped fluid volume between the actuator & control valve and flow leakage through the moderate radial clearance between the spool and the bush of the proportional valve [3], [43].

2. **Discontinuous or Hard Nonlinearities:** Nonlinearities of discontinuous nature are present in an **EHAS** in the form of discontinuities in the static friction of the actuator, deadband in the valve spool of the proportional valves resulting from axially wider lands compared to the bush metering ports (over-lapped spool), which in turn fails to trigger any activity of the valve when a command signal is applied within a particular range [3], [9], directional change of valve opening and saturation of control input applied to the valve [45].
3. **System or Parametric Uncertainties:** Deviation of the system parameters from their nominal operating conditions due to the changing environmental factors and component wear, leads to uncertainties in the plant or system behaviour over time [46]. These uncertainties generally tend to arise in the behaviour of an **EHAS** due to change in valve flow and pressure gains over a range of operating pressures and flow rates; change in the bulk modulus of the hydraulic oil with temperature; variation in the cylinder friction characteristics with external loading and long operational hours; changing spool valve position based on large hydraulic flow reaction forces and variation in fluid capacitance on either side of the actuator piston [43], [47], [48].
4. **Measurement Uncertainties:** Uncertainties in the values of the measurands of the **EHAS** such as the piston rod position, piston load, flow rate and operating pressure arise primarily due to inherent systematic and random errors in the measurement processes of the transducers and the possibility of inaccurate measurements due to the sensor noise arising from physical abrasion or obstruction of the sensing equipment [49]. Additionally, variation in the calibration of the transducers and data acquisition systems over time may further lead to uncertainties in measurement. It is also to be noted the conversion uncertainties are induced in measurement due to the variation in the scaling process parameters of the incoming measurand data which is often necessary to perform the control tasks, owing to the presence of nonlinearities in system behaviour.

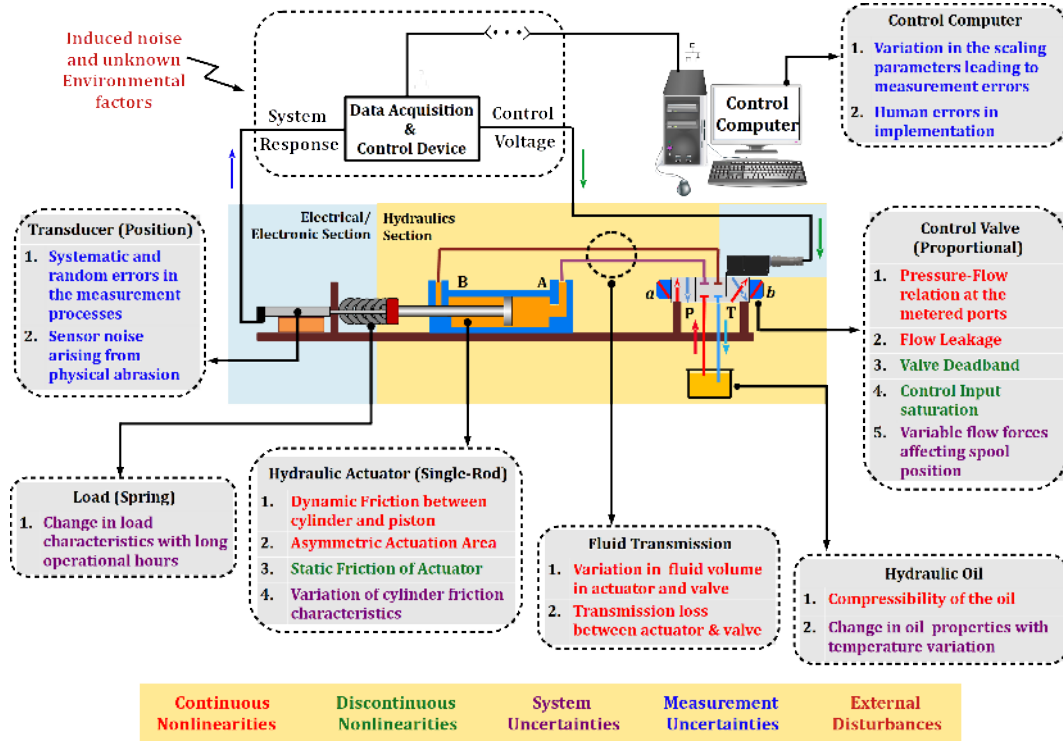


Figure 1.2: Schematic representation of the associated nonlinearities, uncertainties and disturbances in motion control of Electro-Hydraulic Actuation Systems

5. **Model Uncertainties & Approximations:** The presence of uncertain nonlinearities and parametric uncertainties in the **EHAS** leads to plant dynamics which cannot be modeled appropriately since the equations describing these behaviours are not completely known [45], [46], which in turn leads to approximations in the plant model. It is also possible that the system designer may assume linear behaviour of certain plant phenomena in order to simplify the system model, which leads to approximations in the control schemes.

6. **External Disturbances:** The presence of electrical noise in the sensor data caused by interference with nearby electrical sources, mechanical disruptions and other undetermined environmental factors may serve as external sources of disturbances. This leads to the possibility of deviation of the measured signals from their original value during the transference of signals from the transducers.

A visual account of all these sources of nonlinearities, uncertainties and disturbances which afflicts the performance of an **EHAS** and therefore makes it challenging to control, are so illustrated in Fig.1.2 such that their effects on the primary components of the EHAS mentioned in *Section 1.1.1* are clearly understandable.

Apart from the discernible factors mentioned above, the performance of an **EHAS** also depends on the form of control, in which it is utilised. For instance, the motion control performance of articulated systems with multiple degrees of freedom (**DoF**) viz. a serial or parallel manipulator, may be hindered not only by the aforementioned control complexities of single **EHAS** but also due to the nonlinearities arising from the associated multi-body dynamics arising from the interlocking of multiple **EHAS** or coupling disturbances in the joint, which may or may not be apparent to the system designer. To efficiently control such **EHAS** applications, it is necessary to design robust control schemes like Sliding Mode Control (**SMC**), which take into account the dynamics, approximations, uncertainties and disturbances of the system. The next section describes a simplified mathematical model of the **EHAS** motion involving some of the nonlinearities, which is the cornerstone for designing and deploying its control schemes.

1.3 Mathematical Formulation of EHAS Motion

In order to formulate a simplified mathematical model of **EHAS** motion, a typical linear **EHAS** comprising a double-acting single rod actuator (**SRA**) with a compression spring (**S**), driven with the aid of a solenoid-operated proportional valve (**PV**) is considered similar to the one shown in Fig. 1.1. A schematic of the major system components mentioned above, the system parameters affecting the **EHAS** motion and the internal structure of the valve along with its symbolic representation have been illustrated in Fig.1.3 below. The motion of the piston rod of the **SRA** is governed by the flow of hydraulic oil into either of its chambers, namely, cap-end (Ch_1) and rod end (Ch_2). The flow volume and flow direction to the chambers, are in turn controlled by the solenoid-operated **PV**, as mentioned earlier in *Section 1.1.1*.

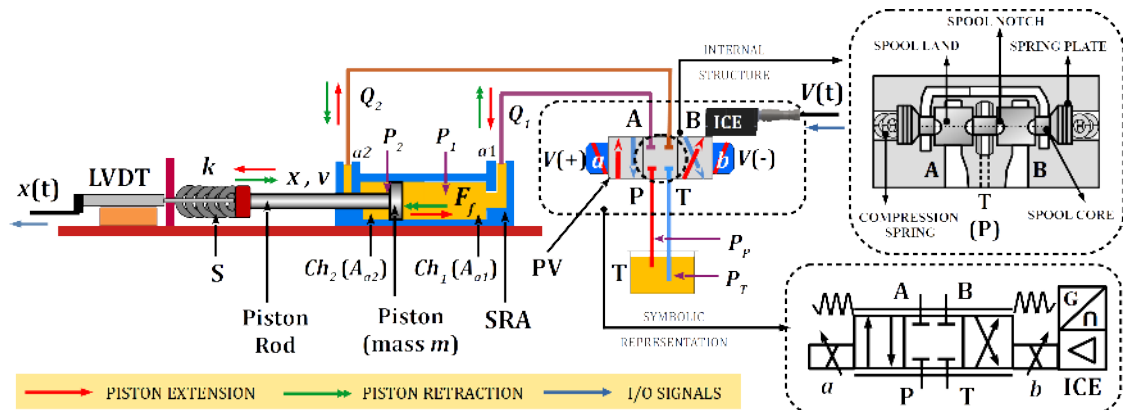


Figure 1.3: Schematic diagram showing system parameters of an **EHAS** with a spring-loaded double-acting **SRA** along with an internal view of the **PV**

In Fig.1.3, such a **PV** in a 4/3 configuration has been shown, comprising 3 flow positions or paths and 4 flow ports, namely, P, T, A and B which are connected to the hydraulic pump (**P**), the reservoir or tank (**T**), the cap-end actuator chamber port $a1$ and the rod-end actuator chamber port $a2$, respectively, in the hydraulic circuit. The **PV** functions by precisely controlling the motion of its internal spool-land arrangement shown in the internal structure inset of Fig.1.3 and the excitation electrical signal, such that connections are established between the ports P to A and B to T or alternatively between P to B and A to T, depending upon the required flow direction. The motion of the spool-land arrangement is, thus, established with the aid of the solenoids (a or b), either of which, when energized under the action of the incoming control voltage $V(t)$ exerts an electro-magnetic force on the armature, integrated with the spool, causing the assembly to move. Two compression type solenoid return springs are also provided on either sides of the spool end, as shown in the internal structure inset of Fig.1.3, for the purpose of holding the control spool in a mechanical central position within the valve when the solenoids have been de-energized. It has been considered that the solenoid a , when energized by a positive control voltage, leads to a rightward movement of the control spool in proportion to the magnitude of the voltage, thus establishing flows from the ports P to A and B to T, which in turn causes a rightward movement of the actuator piston. This movement is, hereby, defined as the *extension motion* of the piston. Similarly, an application of negative control voltage signal to the solenoid b , generates a leftward control spool movement, connecting the ports P to B and A to T and ultimately leading to a leftward movement of the actuator piston, hereby, termed as the *retraction motion* of the piston. The flow directions corresponding to the extension and retraction motions of the piston has been shown in Fig.1.3 with the aid of red colored single arrowheads and green colored double arrowheads, respectively.

This linear displacement, say x , with corresponding velocity, say v , of the cylinder piston of mass m of the **SRA**, against the system load represented by the compression spring (**S**) of stiffness k , can be expressed by the piston dynamics equation,

$$m\ddot{x} = P_1A_{a1} - P_2A_{a2} - kx - F_f \quad (1.1)$$

Here, P_1, P_2 represent the pressures acting within the cap-end cylinder chamber (Ch_1) of area A_{a1} and rod-end cylinder chamber (Ch_2) of area A_{a2} , respectively. The term F_f stands for the friction within the actuator. These variables can be further expressed in terms of typical hydraulic system parameters.

- (a) *Cylinder Chamber Areas*: The cap-end and rod-end chamber areas, A_{a1} and A_{a2} , may be expressed in terms of the cylinder bore diameter, say d_b , and

piston rod diameter, say d_r , in the form,

$$A_{a1} = \pi d_b^2/4 \quad (1.2a)$$

$$A_{a2} = \pi(d_b^2 - d_r^2)/4 \quad (1.2b)$$

- (b) *Friction Characteristics*: In a typical **SRA**, the major sources of friction comprise, but are not limited to, the movement of the piston and the rod, contact between the rod and seals, contact between piston O-rings, seal and the cylinder & the viscous effects of the hydraulic fluid. The term F_f is used to denote these sources of friction, which can be modeled based on the equations put forth by Sarkar *et al.*, corresponding to the piston velocity v in the form,

$$\begin{aligned} F_f &= F_0, |v| \leq v_0 \\ &= F_c + (F_b - F_c) \exp\{-(v - v_b)^2/(v_s - v_b)^2\} + \alpha_v(v - v_0), |v| > v_0 \end{aligned} \quad (1.3a)$$

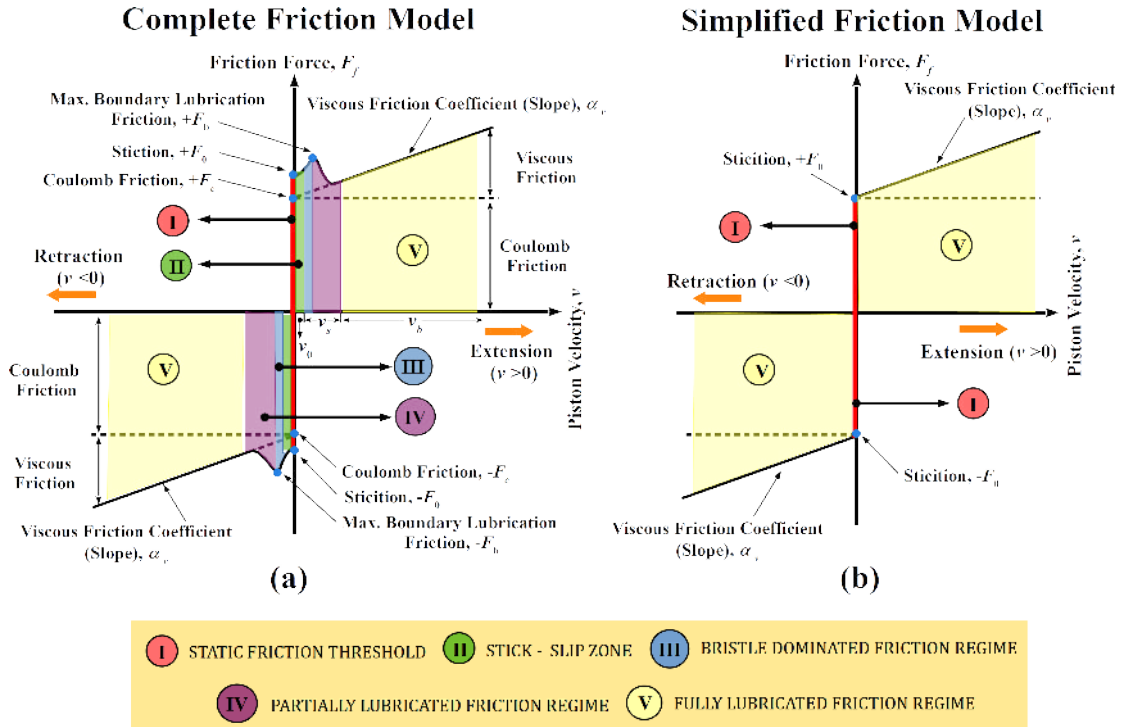


Figure 1.4: Friction Characteristics of single-rod hydraulic actuators

Here, F_0 represents the static friction or stiction, F_b is the maximum boundary lubrication friction, F_c is the Coulomb friction, v_0 stands for the slip velocity, v_s denotes the Stribeck velocity, v_b stands for the lubrication friction velocity and α_v represents the viscous friction coefficient. By assuming

a linear friction model, ignoring the Stribeck friction, (1.3a) can be further simplified to the form,

$$F_f = \text{sgn}(v)\{F_0 + \alpha_v v\} \quad (1.3b)$$

The complete and simplified friction models described by (1.3a) and (1.3b) are shown in Fig.1.4(a) and Fig.1.4(b), respectively.

- (c) *Chamber Pressures:* Assuming the hydraulic fluid as incompressible and negligible leakage in the **PV**, and the spool valve to be critically lapped, the chamber pressures P_1 and P_2 in (1.1) can be extracted by equating the Continuity equation for the supply and return flow, Q_1 and Q_2 , respectively and the pressure-discharge relations for a simple **PV** model [8], [9], whereby the ports A and B of the **PV**, meter both the flows between the actuator ports, $a1$ and $a2$, leading to the actuator chambers, Ch_1 and Ch_2 . This can be expressed for both the piston extension ($v \geq 0$) and retraction motions ($v < 0$), in the form,

$$\begin{aligned} Q_1 &= A_{a1}v = C_{vA}V\sqrt{P_P - P_1}, v \geq 0 \\ &= C_{vA}V\sqrt{P_1 - P_T}, v < 0 \end{aligned} \quad (1.4a)$$

$$\begin{aligned} Q_2 &= A_{a2}v = C_{vB}V\sqrt{P_2 - P_T}, v \geq 0 \\ &= C_{vB}V\sqrt{P_P - P_2}, v < 0 \end{aligned} \quad (1.4b)$$

where, v is the piston velocity, P_P is the pump pressure, P_T is the tank pressure, V is the control voltage to the **PV** and C_{vA}, C_{vB} are the **PV** valve coefficients for Ports A and B, respectively. The valve coefficients C_{vA} and C_{vB} are considered unequal owing to the asymmetric nature of the double acting **SRA** used in this model. It is a usual practice to choose the valve-actuator pair in such a way that the ratio of the valve coefficients (C_{vA} and C_{vB}) are approximately equal to the circumferential lengths of the metered ports, thus ensuring identical pressure drops at the these ports. Therefore, the chamber pressures can now be derived from (1.4a) and (1.4b) as,

$$\begin{aligned} P_1 &= P_P - (A_{a1}v/C_{vA}V)^2, v \geq 0 \\ &= P_T + (A_{a1}v/C_{vA}V)^2, v < 0 \end{aligned} \quad (1.5a)$$

$$\begin{aligned} P_2 &= P_T + (A_{a2}v/C_{vB}V)^2, v \geq 0 \\ &= P_P - (A_{a2}v/C_{vB}V)^2, v < 0 \end{aligned} \quad (1.5b)$$

The dynamic equation of motion in (1.1) for **EHAS** can be recast by substituting the equations for chamber pressures in (1.5a) and (1.5b) along with either of the friction models expressed in (1.3a) and (1.3b) for both piston extension and retraction. This expanded form of the dynamic equation can be represented in a feedback linearized form as,

$$\ddot{x} = -\dot{x} + u + d \quad (1.6a)$$

where, $\dot{x} = v$ is the piston velocity, u is the control or manipulated variable and d represents the disturbances in the system. The control variable u for this **EHAS** encompasses the term involving the control voltage V in the expanded form of the dynamic equation in (1.1) and can be represented, in general, as

$$u = (\dot{x}^2/mV^2)(A_{a1}^3/C_{vA}^2 + A_{a2}^3/C_{vB}^2) = K_{vc}(\dot{x}^2/mV^2) \quad (1.6b)$$

where, $K_{vc} = A_{a1}^3/C_{vA}^2 + A_{a2}^3/C_{vB}^2$ is a constant quantity for a particular **EHAS**. The control voltage to be administered to the **PV** can be extracted from this equation in terms of the control variable u , which can be expressed as,

$$V = \text{sgn}(v)\dot{x}\sqrt{K_{vc}/mu} \quad (1.6c)$$

Hence, it becomes quite obvious that the major focus in designing effective control schemes for **EHAS**, is on developing suitable formulations for the control variable u . The upcoming section entails the background of the methodologies used for designing the control variable u for **EHAS** applications, utilized in the course of this thesis work.

1.4 Literature Survey on Electrohydraulic Control

The modern concepts and hardware behind motion control of electrohydraulic systems were developed in the 1940s following the filing of several patents by corporations involved in the development of aerospace and defence grade technologies [1]. However, the advent of electronically controlled computing systems in recent times has provided the necessary boost for the rapid research and development of electrohydraulic systems in motion control by utilizing relatively low-cost powerful microcomputers, based on which complex control algorithms may be implemented [50]. Based on this technological advancement, research on designing control solutions for **EHAS** has emerged as a prominent field of study during the past few decades to overcome the nonlinearities and uncertainties mentioned in *Section 1.2* which tend to degrade the position, force or pressure tracking performances of

these systems. Over the years, numerous designs for the control variable u as mentioned in *Section 1.3*, have been devised based on linear, adaptive, robust and intelligent (or model-free) control principles [51], [52].

The most widely popular amongst these designs has been the sliding mode control (**SMC**) approach either in standalone mode or augmented with the aid of other adaptive & intelligent control approaches. In addition to these control approaches, the practice of utilizing contactless Visual Sensing devices in the feedback loop or Vision based feedback control has emerged as an alternative to conventional sensor-based schemes in industrial sectors, primarily owing to the possibility of a reduction in uncertainties and induced noise in the transferred signals associated with contact-type transducers as shown in Fig.1.2. A detailed literature survey regarding the development and impact of these methodologies on the field of electrohydraulic control, which fall under the purview of this thesis, has been furnished in the following subsections.

1.4.1 Sliding Mode Control Approach

Sliding Mode Control (**SMC**) emerged in the early 1950s as a new class of nonlinear control, discontinuous in nature, following the pioneering works of Emel'yanov and his colleagues in the field of autopilot design at the Institute of Control Sciences (Moscow). The approach was based on the *principle of structure variability*, whereby it is possible to switch back and forth between unstable and stable control structures based on a *switching function*, to achieve stability in the large for a nonlinear dynamical system faced with parameter variations and external disturbances. During studies on such systems, it was observed by researchers that the end-points of the phase plane trajectories alongside the *switching or sliding surface* in the complete phase portrait, constitutes a special trajectory which in turn guides the system towards an equilibrium or demand point, most generally represented by the origin of the phase plane. The system motion along this special trajectory was termed as *sliding mode* and therefore, this approach came to be known as *Variable Structure Control with sliding mode*, which was later simplified to **SMC** [53]–[55].

The phase plane plot of this conventional **SMC** scheme has been shown in Fig.1.5(a). The most discernible feature of this conventional **SMC** scheme is its robustness or *invariance* to deviations in system parameters and external disturbances as soon as the system trajectory reaches the sliding surface. In order to achieve global robustness, a modified *Integral Sliding Mode Control (ISMC)* scheme was developed, which ensured robustness throughout the system response by eliminating the reaching phase and applying sliding mode across the entire phase plane as shown in Fig. 1.5(b). This feature made the **SMC** scheme widely

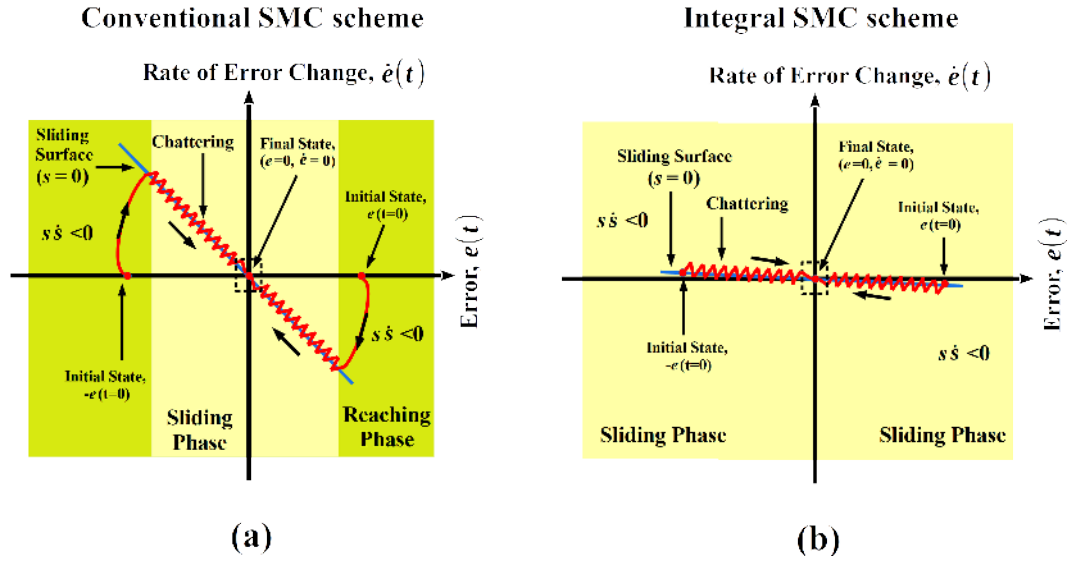


Figure 1.5: Phase portrait of Sliding Mode Control (a) Conventional Form (b) Integral Form.

popular from the practical perspective since the existing linear or adaptive control structures mostly could not fulfil the robustness objective required for real-time dynamical systems [54], [56]. Despite the practical benefits of **SMC**, it is not completely free from drawbacks. The major practical criticism of **SMC** has been the phenomenon of *Chattering*, as shown in Fig.1.5(a) and 1.5(b), whereby undesired system oscillations of finite frequency begin to occur due to the switching of the control input in sliding and steady-state modes of the scheme, which may cause potential physical damage to the system. The occurrence of this phenomenon has been attributed to the inability of practical systems to switch between control structures infinitely fast, which is an underlying assumption of **SMC** based schemes, primarily due to the presence of unmodelled dynamics in the control loop, finite computation time delays in microcontrollers with fixed sampling rates and the physical limitations of the actuation systems [54], [56]. In the last few decades, numerous approaches have been developed to suppress or reduce the chattering issue of **SMC** [57], noteworthy amongst which is the high-gain control with saturation method [58], the 2nd order **SMC** principles [59]–[61], observer based chattering suppression [57] and the sliding-sector method [62]. Henceforth, **SMC** has been extensively used to solve control issues related to numerous nonlinear dynamical systems faced with parametric variations and disturbances.

1.4.1.1 SMC of Electrohydraulic Actuation Systems

Electrohydraulic systems present researchers with a welcome opportunity to assess the strengths of **SMC** based schemes, owing to the presence of a plethora of nonlinearities, uncertainties and disturbances in **EHAS** subsystems as shown

in Fig.1.2. This section of electrohydraulic control, can be categorized into two branches depending on the type of control valve employed for the research, as discussed below.

- (a) *Servo-Valve Driven EHAS*: An early application of **SMC** for velocity control of **EHAS** was proposed by Chern & Wu, whereby a variable structure control scheme was combined with an integral compensator, termed as integral variable-structure control (**IVSC**), designed for robustness and zero steady-state error under step inputs. The simulation results of this investigation showed that accurate regulation and tracking control of such servosystems was possible with this approach, even with a wide range of parametric variations and external disturbances [63]. Later, this **IVSC** control scheme was extended and numerically simulated by Fung *et al.* to include a proportional compensator to achieve a faster output response for an electrohydraulic position servo control system compared to the original scheme [64].

In a different approach, Hwang *et al.* employed a variable structure control scheme with time-varying switching gains, the sliding surface and uncertainties related by a second-order equation, and a fixed boundary layer around the sliding surface, for addressing the position control issue of an electrohydraulic servo mechanism subjected to parametric uncertainties and disturbances [65]. Later, Hwang suggested that a **SMC** scheme with time-varying boundary layer and switching gain is required to ensure asymptotic tracking with reduced control effort for an electrohydraulic servo system with a weighed combination of position and differential pressure control [66]. The simulation results of both these studies showed promising results for such **SMC** schemes compared to the conventional ones. A similar proposition was made by Chen *et al.* concerning the use of an **SMC** scheme with varying boundary layers to improve the tracking performance of an electrohydraulic position servo system [67]. The experimental results of this study validated the previous simulations by Hwang, whereby the varying boundary-layer thickness scheme improved the system response by reducing the chattering, compared to the fixed boundary layer scheme. It was also observed that the tracking error for sinusoidal demands were much lower for this scheme as compared to a conventional **PID** controller.

In another version, Fung *et al.* devised a two-phase variable structure control (**VSC**) scheme to control the position of an electrohydraulic servo system described as a 3rd order nonlinear model with time-varying coefficients, involving separate reaching phase and sliding mode control laws [68]. The simulation results of the study showed that the proposed scheme provided accurate position control, sped up the output response and eliminated chat-

tering. Liu *et al.* suggested another **SMC** scheme suitable for controlling hydraulic servo systems with flexible loads and demonstrated the robustness of the scheme with simulation and experimental results [69]. Considering a third order servo valve operated electrohydraulic system model, Ghazali *et al.* reported an **SMC** scheme for its position tracking control and established its robustness and tracking superiority compared to a conventional PID scheme based on numerical simulations [70]. Recently, Kolsi Gdoura *et al.* proposed an adaptive sliding surface and gain based **SMC** scheme to control the position of a servo valve driven double rod **EHAS** and demonstrated its asymptotically stable low chattering behaviour with numerical simulations [71].

- (b) *Proportional-Valve Driven EHAS*: Most of the **SMC** approaches described above, dealt with servo valve driven electrohydraulic systems which are free from large deadbands. However as mentioned in *Section 1.1*, most industries tend to prefer proportional valves (**PV**) due to their robust nature and minimum filtration needs [5], despite the presence of large deadbands in them. Although, literature pertaining to this specific configuration of the **EHAS** is sparse, some researchers have proposed **SMC** based control methodologies to bridge this industry-academia gap. An early attempt at addressing the control issue of proportional solenoid valves was made by J.B. Gamble of the Vickers System division using a robust **SMC** scheme [47]. The author later followed this up with a comparative study regarding the control performance of a direct acting proportional solenoid valve using an **SMC**, a **PID** and a state feedback control technique. The experimental results of this study for step demands demonstrated that the **SMC** scheme provides a much faster and robust closed-loop response in comparison to the other schemes with minimal knowledge of the valve dynamics [72]. Bonchis *et al.* designed a novel **SMC** scheme to address the position control problem of an **EHAS** afflicted with nonlinear friction characteristics. The performance of the controller was demonstrated on a 4-DoF mining manipulator with asymmetric cylinders, **PV** and sensors mounted on the cylinder for measuring chamber pressures and piston displacement. The experimental results showed that the proposed scheme produced better tracking responses even in the presence of unmodeled friction [73]. The authors also performed an experimental comparative evaluation of ten position control strategies for a **PV** driven **EHAS** with single-rod asymmetric cylinder, available at the time. A 4-DoF mining manipulator was used as the test rig for these experiments, which showed that **SMC** scheme was one of the best strategies alongside acceleration feedback with experimentally identified friction model

(**FRID**) and model reference adaptive control (**MRAC**) [74].

Later, Bessa *et al.* developed an **SMC** scheme with an adaptive-fuzzy compensator for addressing the unknown dead-band nonlinearity present in a four-way **PV** of an **EHAS**. The numerical simulations presented in the study confirmed that the adaptive compensator could recognize and compensate for the undesirable effects of this dead-zone nonlinearity [75]. Recently, Das-mahapatra *et al.* proposed an adaptive fuzzy-bias **SMC** scheme for motion control of a **PV** driven **EHAS**, whereby a bias compensator is designed to take care of the discontinuous nonlinearities including the dead band of the valve. The experimental results for step and sinusoidal demands of various frequencies demonstrate the robustness and adaptability of the controller [9]. Low chattering input linearized **SMC** schemes of first and second orders were designed by Das Mahapatra *et al.* by coupling a simple integral compensator (**SMIC**), similar to the **IVSC** scheme proposed by Chern *et al.* for the servo valve **EHAS** configuration [63], so as to attenuate the input signal chatter and eliminate steady state errors in the linear motion tracking performance of an **EHAS** driven by a large deadband **PV**. The experimental results revealed that both the first and second order **SMIC** schemes yielded lower tracking errors and control effort for a wide bandwidth of sinusoidal signals [76]. Later, the authors extended the basic **SMIC** scheme by modifying the integral controller as a hyperbolic tangent function of an integral sliding variable with anti-windup feature and designing adaptive laws for the controller gains. The experimental investigations of this scheme was conducted on a real-time **PV** driven **EHAS** setup for providing angular motion in a typical wing maneuvering system of an aircraft. The results validated the premise that the proposed controller is capable of real-time chattering-free responses for a variety of loading conditions and motion signals [77].

It is quite obvious from the above survey of literature that much of the progress in the field of developing sophisticated **SMC** strategies to achieve servo-class performance with **PV** driven **EHAS** has occurred only in the last decade. Hence, there is much opportunity to further advance this field from the research perspective.

1.4.1.2 Neuro-SMC of Electrohydraulic Actuation Systems

In the last few decades, Artificial Neural networks (**ANN**) have also been employed in analysis and control of fluid power systems such as the **EHAS** [51], owing to their model-free architecture, nonlinear mapping and approximation abilities based on self-learning [78]. Some of the early applications of neural networks to the field of fluid power involved the use of a simple two-neuron, two layer network

to control a servo-valve driven electrohydraulic drive [79], the use of multilayer perceptron (**MLP**), a form of Feedforward neural networks (**FFNN**), for leakage detection in an electrohydraulic drive [80] and the use of **MLP** to diagnose experimental faults in a servo valve driven **EHAS**, trained only on simulation data [81]. **ANN** was first integrated with **SMC** schemes in order to alleviate some practical issues concerning sliding mode, leading to an intelligent-robust control methodology. In these integrated designs, **ANN** has been used either for online adaptation of the **SMC** parameters to avoid the chattering or in parallel to it for computing the equivalent control part [82]. Although, numerous **ANN** architectures have been developed over the years [83], only a few have become popular for such integrated controller developments [84], which are discussed below and thereafter, shown in Fig.1.6.

- (a) *Radial-basis-function Neural Network structure*: The earliest application involving a synergistic combination of the widely used **RBFNN** with **SMC** for controlling a servo valve driven **EHAS** was proposed by Hwang, whereby the **RBFNN** is used to model the uncertainties arising from parametric variations, friction and load. The numerical simulations suggest that the scheme provides excellent tracking, while avoiding the phenomenon of chattering control input [85]. Cao *et al.* reported the use of a Gaussian **RBFNN** with **SMC** to control a missile electrohydraulic servo mechanism, whereby the weights of the Gaussian **RBFNN** were adapted online to estimate the equivalent control of **SMC** [86]. Huang *et al.* improved the control structure suggested by Hwang, by developing a **RBFNN** structure with adaptive weights in a synergistic combination with **SMC** and applied it to a quarter-car hydraulic active suspension system involving a servo valve [87]. Similarly, Wu *et al.* suggested a **RBFNN** based **SMC** controller with the **RBFNN** being used to estimate the uncertain parameters of an electrohydraulic position servo system and established its better tracking performance with reduced chattering compared to a traditional **SMC** controller [88]. Recently, Sun *et al.* proposed an **RBFNN-SMC** scheme for providing optimal control force to an active suspension system of a quarter vehicle model along with a kalman filter for improved performance [89]. Also, Du *et al.* suggested the use of an adaptive fuzzy integral **SMC** scheme combined with a **RBFNN** to estimate the mathematical model of an electro-hydraulic power steering systems for heavy vehicles by updating the weights. The simulations comparing the proposed scheme with **PID**, **ISMC** and adaptive **RBFNN-ISMC** demonstrate the effectiveness of the strategy [90].

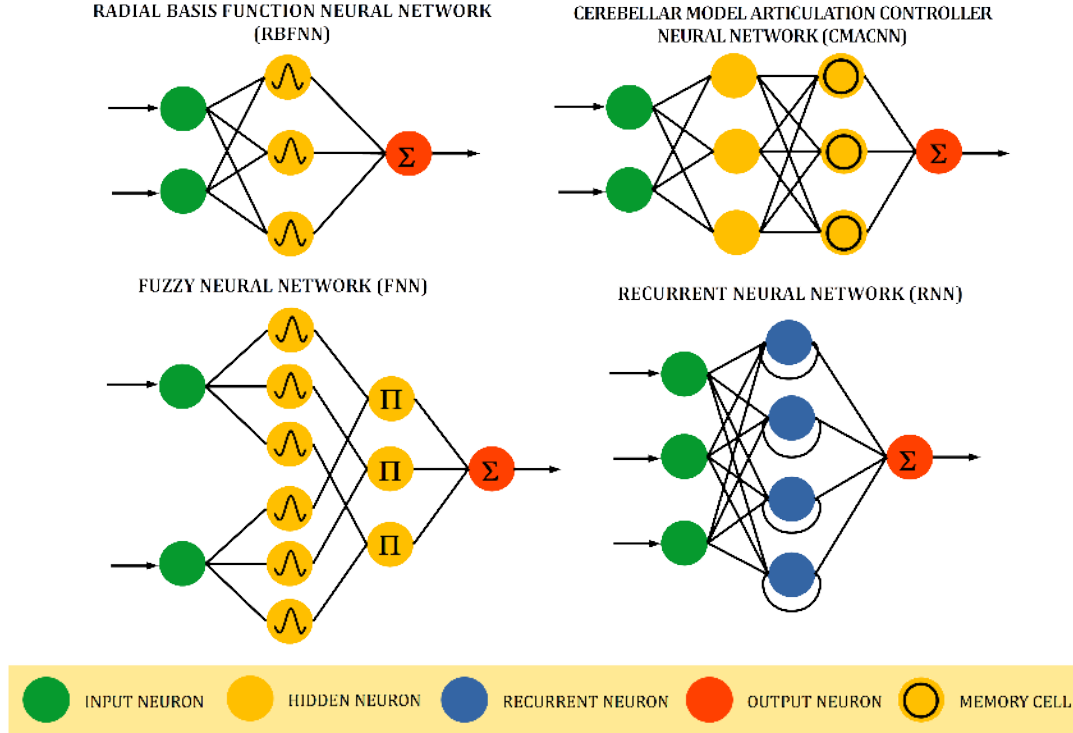


Figure 1.6: Some ANN structures used in integrated **SMC** schemes for **EHAS** control

- (b) *Fuzzy Neural Network structure*: A design method for position tracking control of an electrohydraulic servo system used for missiles, utilizing an **SMC** scheme based on fuzzy neural network (**FNN**) with back propagation algorithm was proposed by Yu *et al.* to ensure the existing **SMC** condition. The simulation results of the study verify that excellent tracking can be obtained without chattering for the proposed approach compared to the conventional **SMC** scheme [91]. He *et al.* modified the above scheme for a missile servosystem by introducing recurrence in the **FNN**, thus forming a **RFNN** based **SMC** scheme for position tracking control of the mechanism [92]. A novel variation to the above scheme was suggested for the first time by Kayacan *et al.* for velocity control of an electro hydraulic system with flow nonlinearities and internal friction, whereby a **SMC** based learning scheme is used to train a type-2 **FNN** (**T2FNN**) using triangular membership functions of type-2. The simulation results show that when the **T2FNN** scheme is used in parallel with a Proportional-Derivative (**PD**) controller, the system response improves in terms of settling time and steady state error [93].
- (c) *Other Neural Network structures*: Besides these commonly used networks, other **ANN** structures have also been suggested by researchers. Zhiming *et al.* reported the use of a Cerebellar model articulation controller neural network (**CMACNN**), an extension of the **MLP** model, along with fuzzy

sets as a learning technique to reduce the effects of uncertainties on a **SMC** scheme, in order to control the position response of a servo driven **EHAS** [94]. A solution to the force tracking problem of a servo valve driven **EHAS** was suggested by Lizalde *et al.* by combining a sigmoidal function activated Recurrent Neural Network (**RNN**) based system identifier with a 2-**SMC** scheme. The proposed scheme gave chatter-free and robust closed loop force response as indicated by the simulations [95], [96]. Recently, Truong *et al.* proposed a control scheme where a neural network with back-propagation is used to directly adjust the gains of the **SMC** with the aim of addressing the problem of payload variation for a 3-DoF electrohydraulic serial manipulator [97].

It is curious to note that most of the popularly used **ANN** structures in integrated Neuro-Sliding mode control schemes fall under the broad category of **FFNN**'s *viz.* the **RBFNN**, **FNN** and **CMAC**. The exception to this trend is the **RNN**, which is known to possess seamless dynamic mapping capabilities [98], [99] and is being increasingly used in other fields of study.

1.4.2 Vision based Feedback Control Approach

Vision has become an indispensable technological tool over the past few decades since it mimics the human sense of vision and allows for non-contact measurement of the environment. Although the paradigm of object tracking in computer vision has existed for several decades and has been widely researched, the notion of using vision as a feedback signal by utilizing this visual information from computer vision algorithms to provide closed-loop position control for a manipulator end-effector was first suggested by Hill and Park for the first industrial robot named *Unimate* [100]. They termed this vision based feedback control strategy as *Visual Servoing* (**VS**) to distinguish their approach from earlier *Blocks World* experiments where the visual control of the manipulator occurred in open-loop and the system alternated between picture taking and moving. Despite lending increased accuracy to the overall system, research in **VS** was hampered by the heavy computation load required for image processing at the time, thus limiting its growth. However, with the rise of modern computational technology, the idea of applying vision based feedback control or **VS** to practical industrial tasks has attracted considerable attention over the last two decades [101]–[107]. Most modern industries, routinely use manipulators with integrated vision systems to correct their position (or pose) or that of their end-effector to increase the accuracy of their control tasks [108]–[111].

The Visual Servoing task involves the use of one or more image sensors or camera(s) and a computer vision system to control the position of a device such as a robotic arm relative to a part it has to manipulate, which requires detecting, tracking, servoing and grasping (if necessary). Formally, the servoing task can be defined as the use of visual information to control the pose of a robot's end-effector relative to a target object or a set of target features [104]. Therefore, visual servoing is an interdisciplinary field spanning high-speed image processing (computer vision), robotics, kinematics, dynamics, control theory and real-time computing. The strength of the visual servoing approach results from the combination of an intuitive, human-oriented task description with the robustness and accuracy of the control action [104], [112]. Based on this diverse background, there has been a steady rise in visual servoing applications in many industrial sectors [113], for instance, in manufacturing for grasping objects on conveyor belts and part mating [104], tele-operation for decommissioning in nuclear environments [108], medical robotics in computer assisted surgeries [114], [115], for grasping and placing operations in food industries [116], in mining automation [109], in automated guidance industrial vehicles [117], in intelligent forklift trucks such as *Robolift* [118], in quadrotor control for autonomous target search [119], [120], surveillance and security systems [121] and even military applications involving automated air-target tracking [122]. In view of the current COVID-19 pandemic scenario, it can be anticipated that visual servoing is going to play a crucial role in automating many industrial and commercial operations, thus minimizing human contact in the process.

1.4.2.1 Classification of Visual Servoing Architectures

Literature suggests that visual servoing architectures used in various applications may be classified based on four major features of the visual feedback process and its utilization in the control of the manipulator [112], [123], [124], which are given below.

1. **Camera Configuration:** This classification is based on the placement of the visual sensors or camera(s) with respect to the end-effector of a manipulator and can be categorized into two schemes as shown in Fig.1.7 [104].
 - (a) **End-Effector Mounted or Eye-in-Hand:** In this configuration, the camera is mounted on the manipulator end-effector and the pose of the camera(s) with respect to the end-effector is often defined by a constant relationship. An inherent feature of this configuration is that the target image observed by the camera(s) is dependent on the motion of the manipulator.

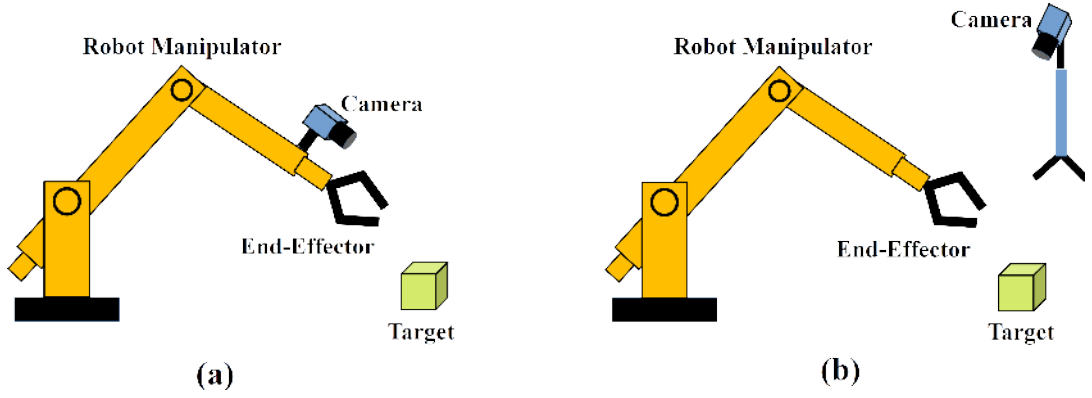


Figure 1.7: Camera Configuration for Visual Servoing (a) Eye-in-Hand (b) Eye-on-Hand.

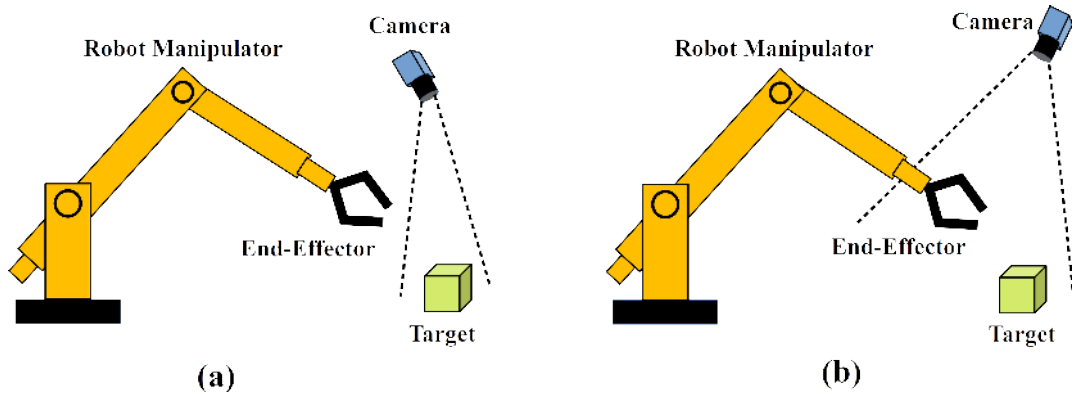


Figure 1.8: End-Effector positioning during Visual Servoing by (a) EOL System (b) ECL System.

- (b) **Fixed in Workspace or Eye-on-Hand:** In this configuration, the camera(s) is fixed in the workspace. In this case, the camera(s) are related to the base coordinate system of the robot, the robotic end-effector and to the target object by some geometric relations. Contrary to the previous case, the target image observed by the camera(s) is independent of the manipulator motion.

2. **End-Effector Positioning:** This classification scheme is based on whether the end-effector of the manipulator is positioned with respect to the target object based on a kinematic relationship or by end-effector observation. Based on this strategy, two system structures may be delineated [104] as shown in Fig.1.8.

- (a) **End-point Open Loop (EOL) System:** In this configuration, the camera is tasked with observing only the target object. This may lead to positioning errors of the manipulator end-effector owing to the presence of errors in the kinematic relationship between the camera(s) and

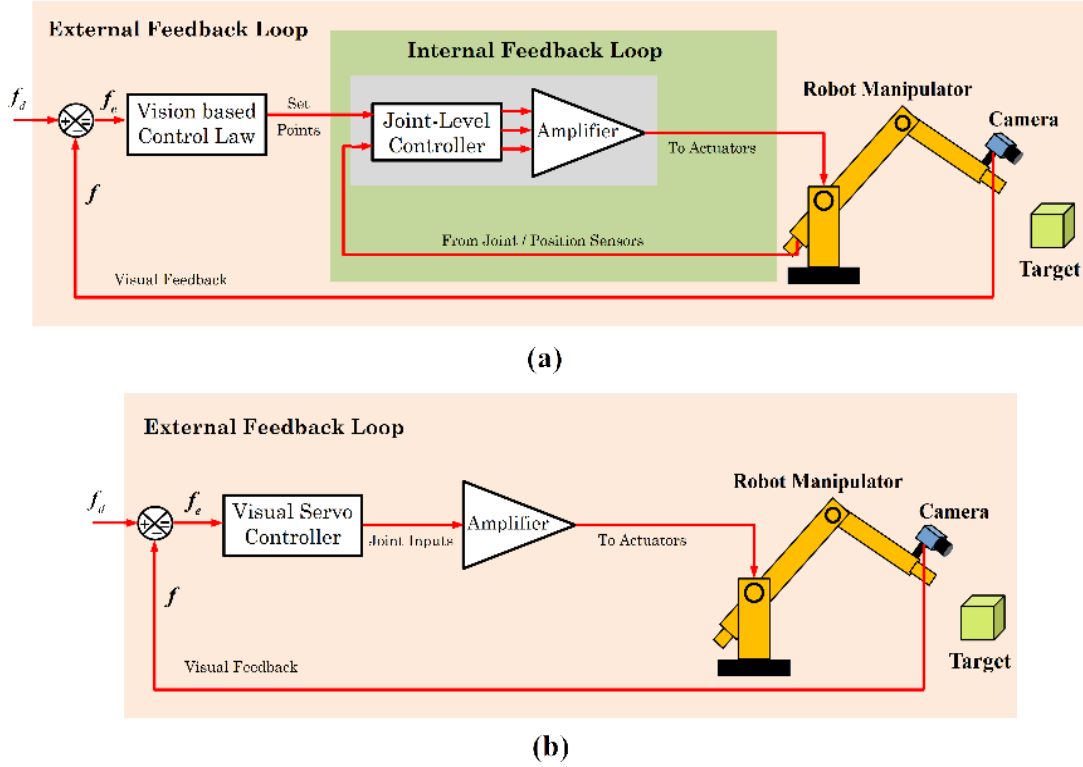


Figure 1.9: Control structure based Visual Servoing architecture with eye-in-hand manipulator (a) Dynamic look-and-move system (b) Direct visual servo system.

the end-effector, which is used to determine its position relative to the target object since the end-effector does not fall in the field of view of the camera(s).

- (b) **End-point Closed Loop (ECL) System:** In this approach, the camera is tasked with observing both the target object and the manipulator end-effector. This increases positioning accuracy of the mechanism by eliminating kinematic relationships, since control points on both the target object and manipulator end-effector can be observed by the camera(s).

3. **Control Structure:** This classification is based on whether the visual feedback system provides set-points as input to the joint level controller of the manipulator or the visual controller directly computes the joint-level inputs. Based on this approach, two categories may be defined, which are illustrated in Fig.1.9 [104].

- (a) **Dynamic Look-and-Move:** Here the control architecture is hierarchical and uses the vision system to provide set-point inputs to the joint-level controller, thus making use of joint feedback to internally stabilize the manipulator.

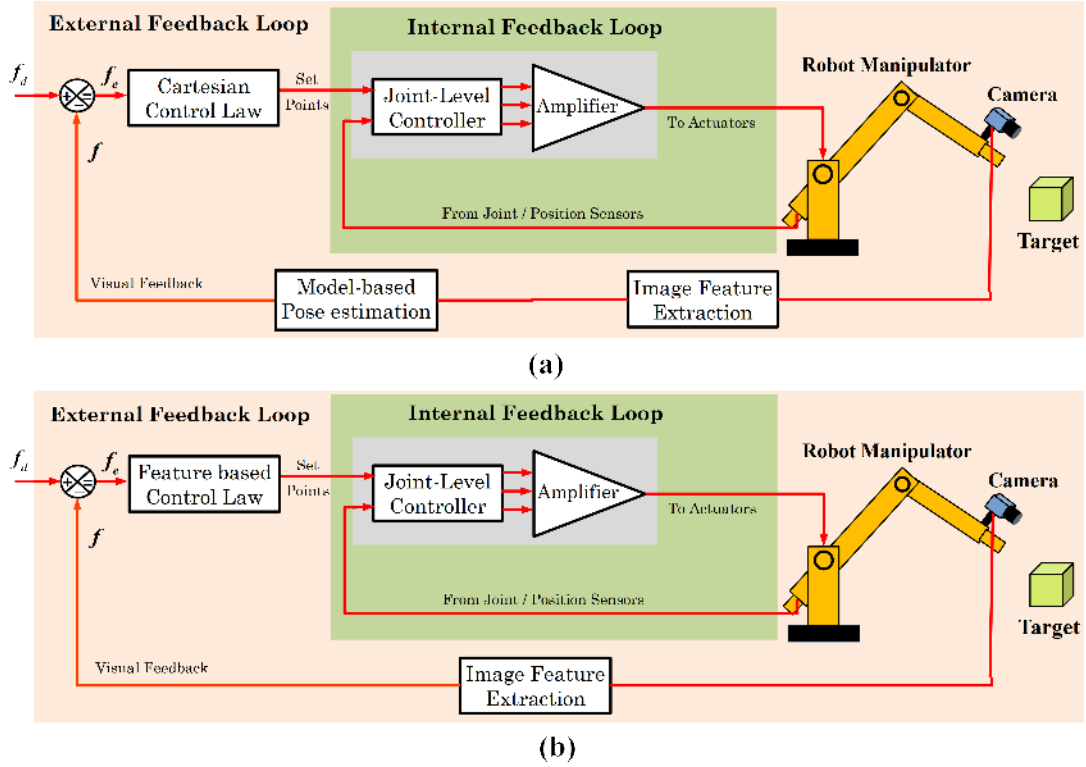


Figure 1.10: Error Signal based architecture with eye-in-hand manipulator (a) Position-based Visual Servoing (PBVS) (b) Image-based Visual Servoing (IBVS).

- (b) **Direct Visual Servo:** Contrary to the previous scheme, the control hierarchy is entirely replaced with a visual servo controller that directly computes joint inputs, thus using vision alone to stabilize the mechanism.

4. **Error Signal:** This classification scheme is based on whether the error signal is defined in the 3D task space coordinates or directly in terms of image features. Based on this approach, two basic categories may be outlined and illustrated as shown in Fig.1.10 [104].

- (a) **Position Based Control:** In this approach, features are extracted from the image and used in conjunction with a geometric model of the target and the known camera model to estimate the pose of the target in 3D Cartesian space with respect to the camera. This approach computes feedback by reducing errors in estimated pose space and is referred as *Position Based Visual Servoing* (PBVS).

- (b) **Image Based Control:** In this approach, image features from 2D image space, typically in pixels, are used to compute the control values directly. This approach may reduce computational delay, eliminate the necessity for image interpretation and eliminate errors due to sensor

modeling and camera calibration. However, it does present a significant challenge to controller design since the plant is nonlinear and highly coupled. It is often referred as the *Image Based Visual Servoing* (**IBVS**) and was put forth by Sanderson *et al.* in their seminal paper [124].

- (c) **Advanced Control:** In these advanced approaches, generally a combination of image-based and pose-based approaches (2D and 3D information) is used to obtain pose information of the target object by constructing a partial or complete model of the object with aid of epipolar geometry using multiple views or cameras as mentioned in the visual servoing chapter of the Handbook of Robotics [125]. Some of the advanced servoing approaches include 2.5D servoing [126] and partitional visual servoing [127].

1.4.2.2 Visual Servoing of Robotic Manipulators

Robotic manipulators have been routinely used for carrying out various pre-defined tasks in industrial environments. However, the use of visual servoing principles had a great impact on the autonomous operation of these manipulators as discussed below for two types of commonly used manipulator structures.

- (a) *Serial Manipulators:* These manipulators can be regarded primarily as open-loop kinematic chains [128] formed by interconnected joints and links driven by actuators. A survey of the available research on visual servoing of serially linked manipulators shows that they have been used for wide range of applications. The most common application amongst these have been grasping and placing operations in assembly lines. For instance, one of the earliest uses of vision guided robots in assembly tasks was proposed by Shirai *et al.* using industry grade serial manipulators [129]. Later, Hutchinson *et al.* proposed the use of vision enabled serial manipulators for grasping and part-mating of objects in the manufacturing industry [130]. Similarly, Wu *et al.* proposed the use of such manipulators for grasping and placing of food items from conveyor belts in food industries using **PBVS** [116]. Bandala *et al.* developed a hydraulically actuated dual arm robot for pipe cutting and grasping using vision assisted tele-operation in nuclear environments [108]. In the field of medical robotics, Pérez *et al.* developed a vision-based serially-configured autonomous robot for surgical assistance using extended image plane techniques and experimentally validated the ability of the robot to autonomously retrieve surgical instruments from the surgeon [115]. Also, Azizian *et al.* gave a comprehensive review regarding the applications of such vision assisted medical robots [114].

- (b) *Parallel Manipulators*: The structure of these particular type of manipulators can be described as closed-loop kinematic mechanisms [131], whereby the end effector is linked to the base by several independent actuator driven kinematic chains leading to a system capable of generating n -DoF motion. A thorough survey reveals that research on visual servoing of robotic manipulators in parallel configuration has been extremely limited. The research available on the topic, seems to have preferred the **IBVS** scheme over the **PBVS**, since the latter is known to be prone to camera calibration errors, has poor robustness of pose estimation and requires an accurate 3D target model, which in turn diminishes the positioning accuracy of the system [132]. It is also observed that the prior research seems to have diverged into two different approaches. In one approach, a group of researchers used an eye-on-hand visual servoing configuration to control the end-effector motion of parallel manipulators by observing the position of its legs. It was shown by experiments and simulations that this approach is suitable for high-speed tasks [133]–[135]. In another approach, Qi *et al.* proposed [132], [136], [137] the use of parallel robot based pointing systems for position regulation, utilizing an eye-in-hand **IBVS** scheme. Simulation studies of the proposed system demonstrated that such systems are robust and can handle large initial feature errors. However, no experimental validation of these simulations were provided.

From the above discussions, it is quite obvious that vision based feedback control is playing a major part in the development of industrial automation. However, it should also be noted there is ample scope for development in certain areas involving applications of parallel manipulators like the Stewart Platform (**SP**).

1.5 Motivation

It is quite evident from the literature survey in *Section 1.4* that a vast body of work, both experimental and numerical, is available for electrohydraulic control and vision based feedback control of robotic manipulators. However, certain gaps can be identified in the literature associated with **EHAS**, which may be addressed. A summary of these gaps is provided below.

1. Most of the studies involving control of **EHAS**, consider servo-valve driven systems. However, as mentioned in *Section 1.4.1.1*, achieving servoclass performance with **PV** driven **EHAS** has emerged as a research trend in the last decade. Hence, the possibility of utilizing suitable algorithms to ensure efficient control of rugged **PV** driven **EHAS**, comparable in performance to a servo-grade **EHAS**, is still being explored.

2. Recurrent Neural Networks (**RNN**), which are known for their dynamic mapping capabilities, have been used in only a few investigations, primarily for force-tracking problems of servo-valve driven **EHAS**. Therefore, the possibility of using **RNN** in a synergistic combination with known robust controllers like Sliding Mode Controller (**SMC**) for addressing the position-tracking problem of a rugged **EHAS**, may also be explored.
3. It is also evident from the literature that most studies on control of real-time **EHAS** is aided by position transducers, which need to be attached to the actuator, in order to form a closed-loop system. However, the advantages offered by a visual sensor in association with a robust feedback control scheme, may also be exploited for the real-time position-tracking problem of a single degree-of-freedom **EHAS**.
4. It is also evident from *Section 1.4.2.2* that studies on visual servoing of parallel manipulators have been extremely limited. Most of these studies are numerical in nature and proper experimental evidence to support the simulation results is absent. Hence, the possibility of extending the advantages of vision based sensing to a higher degree-of-freedom Electrohydraulic System, in particular a 6-**DoF** Parallel Manipulator, using visual servoing algorithms may be explored.

Motivated by these open problems in electrohydraulic control and visual servoing, the present thesis work aims at addressing these issues in order to develop certain real-time applications of **EHAS** driven systems.

1.6 Objectives of the Present Work

The overall objective of the present thesis work is to address the identified gaps which are highlighted in *Section 1.5*. This work focuses on developing real-time control solutions for applications involving **PV** driven **EHAS**.

The first and foremost objective of the present work is to develop a closed-loop control scheme based on a synergistic combination of **RNN** and **SMC** to address the real-time position control issue of **PV** driven linear **EHAS** with emphasis on the low-velocity regime. The control scheme may also consider certain Lyapunov based adaptive laws to update the controller parameters, in order to tackle the disturbances undergoing continuous variation.

In order to investigate the effectiveness of a vision based feedback control approach compared to a conventional position sensor, a real-time closed loop position tracking algorithm is to be developed for a **PV** driven linear **PV** utilizing **SMC** based control schemes with adaptive laws.

The domain of the vision based feedback control is extended from a linear **EHAS** to a 6-**DoF** system by laying emphasis on the development of a real-time target motion tracking and replicating system utilizing an Electrohydraulic Parallel Manipulator (or Stewart Platform) with the aid of **IBVS** approach for a monocular eye-in-hand configuration of the end-effector of the manipulator. The parallel manipulator to be used for this study is constructed with the aid of six **PV** driven **EHAS** arms or kinematic chains, thereby linking the end effector to the static base of the manipulator.

1.7 Contributions of the Present Work

Based on the objectives stated in the previous section, some novel control methodologies and applications for electrohydraulic systems are proposed in the present work. The major contributions of the present work are described below.

1. A real-time adaptive feedback position control scheme based on the synergistic combination of a recurrent neural network with hermite polynomial based activation functions and an integral sliding mode controller has been developed for a rugged linear **EHAS**. The adaptive scheme has been formulated based on the Lyapunov criterion. It has also been augmented with a tan-hyperbolic bias controller to tackle the discontinuous nonlinearities of the system [138].
2. A marker-enabled visual feedback based closed-loop position control scheme has been developed for a 1-**DoF** electrohydraulic system with a single-rod double-acting cylinder and an industry grade **PV**. A monocular **CCD** camera is used for sensing the actuator rod position. The control algorithm utilizes an input linearized system model and is based on the combination of a first-order sliding mode controller, a tan-hyperbolic deadband controller and Lyapunov based adaptive laws for the controller parameters [139].
3. A real-time object motion sensing and trajectory replicating system based on the image-based visual servoing capabilities of an eye-in-hand electrohydraulic parallel manipulator has been developed. The tracking system consists of a monocular **CCD** camera affixed to the top platform of a 6-**DoF** parallel manipulator constructed with industry grade double-acting single rod hydraulic actuators and **PV**. The control algorithm is an amalgamation of three principles, viz. the machine vision based object tracking, image based visual servoing and actuator-level feedback control utilizing the inverse kinematics of the manipulator [140].

It is obvious that the field of electrohydraulic control and its applications is quite vast and numerous avenues for research exist within it. The present research work is aimed at addressing few of these avenues in order to develop real-time applications of **EHAS**.

1.8 Thesis Layout

The work presented in this thesis mainly encompass the contributions mentioned in the previous section. The organisation of the rest of the thesis and the interrelation amongst the chapters are mentioned below.

Chapter 2 discusses the design and implementation of an adaptive neuro-sliding mode based position control scheme designed for the linear **EHAS** with a conventional position sensor. The proposed control architecture involving a recurrent neural network **RNN** embedded with orthonormal Hermite polynomial based activation functions, an integral sliding mode control **SMC** technique and Lyapunov based adaptation laws have been elaborated in this chapter.

A stepwise implementation of the proposed controller on a laboratory scale linear **EHAS** along with the specifics of the setup, has also been included. The results obtained from the experimental studies have also been furnished following a logical progression.

Chapter 3 presents the design of an adaptive position controller for a linear **EHAS** based on visual feedback of the actuator position from an eye-on-hand monocular camera. The methodology involved in developing the proposed control scheme composed of a first-order sliding mode controller, a tan-hyperbolic deadband controller and Lyapunov based adaptation laws for several controller parameters has been explained in this chapter.

A detailed description of the laboratory scale vision enabled **EHAS** test setup and the implementation of the aforementioned control methodology on the setup has been provided. A comparative study of the proposed controller with a conventional control scheme in terms of regulation and tracking performance has been done to assess its feasibility.

Chapter 4 describes the development of a novel real-time object motion sensing and trajectory replicating system based on visual servoing of an eye-in-hand electrohydraulic parallel manipulator. The working principles of the parallel manipulator based servoing system and the corresponding laboratory scale setup have been expounded in this chapter.

Apart from discussing the laboratory scale setup, the multi-domain control methodology involving five separate but interconnected stages, required for the real-time operation of the motion-sensing and replicating system has been explained in complete detail. The experimental results for 1D (Horizontal and Vertical) and 2D (Diagonal) target trajectories are provided and assessed based on several performance parameters.

Chapter 5 concludes the thesis with a summary of the developments achieved during this work. It also discusses future avenues for further research to be conducted in this field of study.

Recurrent Neuro-Sliding Mode based Position Control of EHAS

This chapter discusses one of the contributions of the present work as mentioned in the introductory chapter, whereby a feedback position control scheme involving a combination of Recurrent Neural Network (**RNN**) and an Integral Sliding Mode Control (**ISM**C) has been developed for real-time application, on a linear **EHAS** setup. The chapter presents the philosophy behind the utilization of a particular **RNN** structure by taking cues from the available literature and associating the same with the position control problem of an **EHAS**. The necessity for developing adaptive rules for the control variables has also been introduced here and later analysed by comparing adaptive and non-adaptive controller performances. Subsequently, a systematic procedure for developing this control structure is also given in the chapter, followed by a detailed account of the closed-loop performance of the system with the controller in loop for various real-time scenarios for a representative **EHAS** test setup.

2.1 Background

Artificial Neural networks are widely employed in analysis and control of non-linear dynamic systems, owing to their model-free architecture and approximation abilities based on self-learning methodologies [78]. Although, the network structures of **ANNs** are quite complex at times, the basic networks can be broadly classified into Feed-Forward Neural Network (**FFNN**) and Recurrent Neural Network (**RNN**) [78]. The **FFNNs** are known to approximate any continuous function closely. However, they generally cannot comprehend the dynamic mapping

often associated with real-time systems due to their static nature [99], [141]. The **RNNs**, on the other hand, can perform seamless dynamic mapping owing to the presence of recurrent neurons capable of capturing the dynamic response of a system with the aid of internal or external or dual feedbacks [98], [99], [141], [142]. Researchers have often taken advantage of this feature of the **RNNs** to address various issues related to nonlinear dynamic systems, such as motion control of the X-Y table in a computer numerical control (**CNC**) machine driven by permanent magnet synchronous motors [99], prediction of cylinder chamber pressures of a variable speed-pump controlled hydraulic system [143] and soft sensor validation from indoor air quality (**IAQ**) measurements of subway ventilation control systems [144].

Although the dynamic mapping capability of the **RNNs** can take care of the modeling uncertainties in a system, most dynamic systems are often difficult to control in real-time owing to the presence of system nonlinearities, parametric variations and external disturbances. As mentioned earlier in *Section 1.4.1*, research has revealed that integral sliding mode control (**ISMC**) schemes are one of the most robust nonlinear control methodologies, which are insensitive to external disturbances and variation of the system parameters since it applies sliding mode across the entire phase plane and eliminates the reaching phase [56], [145], [146]. Therefore, in recent years, researchers from many fields have gravitated towards designing hybrid control methodologies combining the characteristic features of **RNN** and **ISMC** into a nonlinear control technique applicable to real-time dynamic systems. For instance, Zhang *et al.* developed an adaptive sliding mode control technique based on local **RNNs** to address the trajectory tracking control problem of underwater robots [147]. In order to restore movements to paralysed limbs, Ajoudani *et al.* designed a neuro-sliding mode controller for the functional electrical stimulation system, where a form of **RNN** was used to model the system uncertainties [148]. Fei *et al.* proposed a double loop recurrent neural network (**DLRNN**) based adaptive sliding mode system for controlling the vibration dynamics of the proof mass in a Z-axis microelectromechanical system (**MEMS**) gyroscope [142]. In order to improve the current tracking control for three phase active power filters (**APF**), a fractional-order sliding mode control scheme using a **RNN** approximator was investigated by Fei *et al.* [149]. Later, a double hidden layer recurrent neural network (**DHLRNN**) based adaptive global sliding-mode controller was proposed by Chu *et al.* for a class of dynamic systems exemplified by the use of an **APF** [98].

Furthermore, research has indicated that the dynamic approximation capability of **RNNs** can be enhanced by the choice of proper activation functions for the hidden neurons [150], [151]. Although, it is a general practice to assign the same

activation functions to all the hidden neurons, typically a sigmoidal or radial basis function, the notion of using different activation functions for each hidden neurons have been proposed by researchers now and again [146], [151]. Some studies have shown that different orthonormal Hermite polynomial basis functions (**OHPBFs**) employed for each hidden neuron may serve as a better alternative to traditional functions, since the Hermite polynomials are defined over the interval $(-\infty, \infty)$ which provides an unrestricted input space, thus improving the representational capability of the network [146], [150], [151]. Consequently, the weighted sum of the **OHPBFs** series leads to a superior function approximation ability of the network, which may be regarded as a Recurrent Hermite neural network (**RHNN**). Despite these special features, limited literature is available pertaining to the use of a hybrid control strategy involving **RHNN** and **SMC** for real-time dynamic systems. For instance, few studies related to trajectory tracking by nonlinear thrust active magnetic bearing (**TAMB**) system utilize non-singular terminal sliding mode control (**NTSMC**) and adaptive complementary sliding mode control (**ACSMC**) in conjunction with a **RHNN** [146], [152]. In these studies, the **RHNN** works either as a lumped uncertainty estimator or a dynamic estimator for some functions of the **TAMB**.

A similar real-time position control problem arises for **EHAS** as previously mentioned in *Section 1.2*. This study was undertaken to design and implement a real-time adaptive position control solution utilizing a synergistic combination of **ISMC**, **OHPBF**, **RNN** and a bias controller, for an **EHAS** employing a double-acting single rod actuator (**SRA**) and a proportional valve (**PV**), similar to those shown in Figs.1.1 and 1.3 in the introductory chapter. The fact that the **RHNN** has been used with an **ISMC** scheme in a manner so as to dynamically approximate not just the uncertainties but also the continuous (soft) nonlinearities of the **EHAS** brings novelty to this study. Also, the effect of hard nonlinearities in the system is dealt with the help of a tan-hyperbolic bias controller [9], [139]. Moreover, the use of Lyapunov function based adaptive laws ensures an online dynamic update of the controller parameters such as the connective weights of the **RHNN** and the saturation values of the bias controller, thus tackling the disturbances undergoing continuous variation [99], [145].

The rest of the chapter has been organised in the stated manner. The formulation of the adaptive **RHNN-ISMC** based control scheme has been described in detail in *Section 2.2*. The implementation of the controller on a laboratory scale **EHAS** along with the specifications of the setup have been explained in *Section 2.3*. The results obtained from the real-time experiments performed by applying the proposed methodology on the laboratory setup and the central findings of this investigation has been described in *Section 2.4*.

2.2 Adaptive RHNN-Bias-Integral SMC

An adaptive control has been formulated here for an n^{th} order, single-input-single-output (**SISO**), nonlinear system

$$\mathbf{x}^{(n)} = g(\mathbf{x}(t), v(t), t) + d(\mathbf{x}(t), t) \quad (2.1a)$$

with the state vector,

$$\mathbf{x}(t) = [x_1 \quad x_2 = x_1^{(1)} \cdots x_i = x_1^{(i-1)} \cdots x_n = x_1^{(n-1)}]^T \quad (2.1b)$$

the scalar output,

$$y(t) = x_1(t) \quad (2.1c)$$

and a scalar input $v(t)$. Here, g is a partially unknown bounded nonlinear function and d is an unknown but bounded (with unknown bound) disturbance. The superscripts in (2.1a) and (2.1b) indicate the order of derivative of the corresponding state with respect to time t . It is further surmised that the partial knowledge of the function g allows one to construct an input u such that the system dynamics expressed by (2.1a) can be represented in an input-linearized form as,

$$\mathbf{x}^{(n)} = f(\mathbf{x}(t), t) + bu(\mathbf{x}(t), t) + d(\mathbf{x}(t), t) \quad (2.2)$$

The control objective is to design a u such that the system state vector $\mathbf{x}(t)$ follows the desired state vector, say $\mathbf{x}^*(t)$ in presence of uncertainties and disturbances, or in other words, to ensure that the error vector, say $\mathbf{e}(t)$ approaches zero asymptotically. In mathematical terms, the control objective can be stated as,

$$\mathbf{e}(t) = \mathbf{x}^*(t) - \mathbf{x}(t) \rightarrow 0 \quad \text{as } t \rightarrow \infty \quad (2.3)$$

Prior to designing the variable u , we make the following assumptions, based on [145]:

Assumption 1: External disturbance $d(\mathbf{x}(t), t)$ is bound by a constant D whose value need not be known, i.e.,

$$|d(\mathbf{x}(t), t)| \leq D \quad (2.4a)$$

Assumption 2: The state vector $\mathbf{x}^*(t)$ is an n^{th} order continuous, differentiable, known and bounded function such that:

$$\|\mathbf{x}^*(t)\| < c \quad (2.4b)$$

2.2.1 Recurrent Hermite Neural Network and the Universal Approximation Theory

In the present work, a **RNN** [78] with multiple inputs and a single output is considered as the prime component of control input u owing to its universal approximation property [99], [141], [152]. The hidden neurons in this network use orthogonal Hermite polynomial basis functions (**OHPBFs**) as activation functions to increase the search space significantly [152]. This Recurrent Hermite Neural Network (**RHNN**) shown in Fig. 2.1 includes an input layer (*Layer 1*) with $(n+1)$ input neurons, a hidden layer (*Layer 2*) with q hidden neurons and an output layer (*Layer 3*) with a single output neuron. The z^{-1} term in the hidden layer neurons represent a time delay.

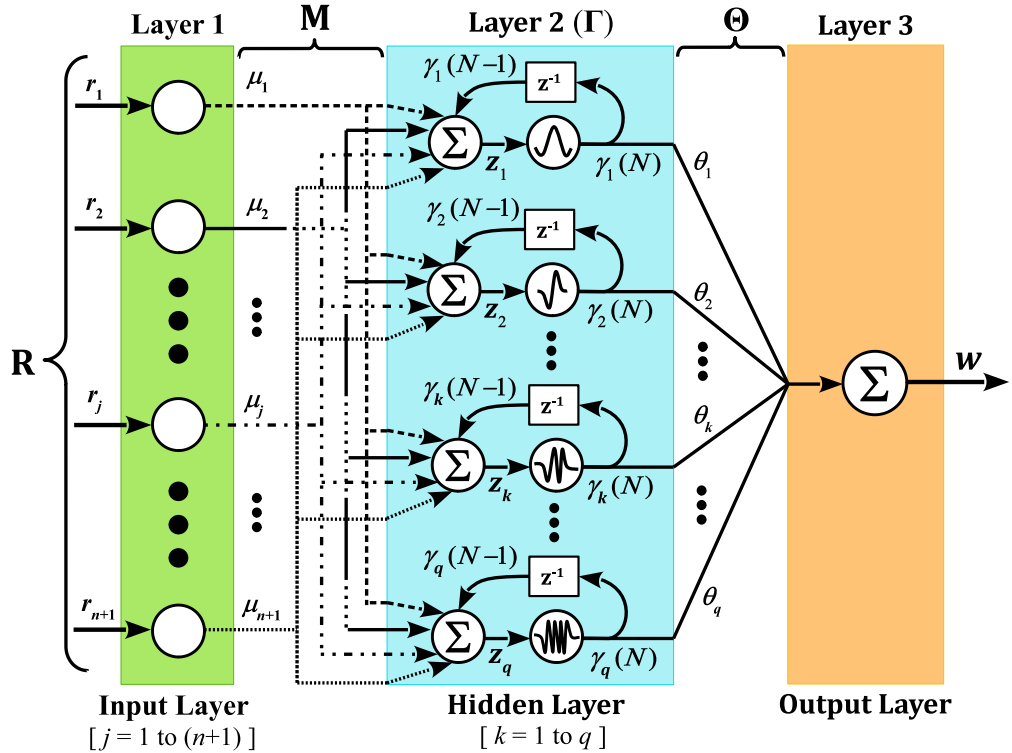


Figure 2.1: Network Structure of the Recurrent Hermite Neural Network.

1. *Layer 3 (Output Layer)*: The output $w(\mathbf{R})$ of the **RHNN** [152] can be expressed as,

$$w(\mathbf{R}) = \Theta^T \Gamma(N) \quad (2.5a)$$

where, N denotes the iteration number, Θ and $\Gamma(N)$ are the weight vector and hidden layer output vector, respectively. Mathematically, they can be expressed as,

$$\Theta = [\theta_1 \quad \theta_2 \quad \cdots \quad \theta_k \quad \cdots \quad \theta_q] \quad (2.5b)$$

$$\mathbf{\Gamma}(N) = [\gamma_1(N) \quad \gamma_2(N) \quad \cdots \quad \gamma_k(N) \quad \cdots \quad \gamma_q(N)] \quad (2.5c)$$

Here, θ_k represents the weight between the k^{th} hidden layer neuron and the output neuron. Also, γ_k represents the output of the k^{th} hidden layer neuron.

2. *Layer 2 (Hidden Layer)*: The output of the k^{th} hidden layer neuron can be expressed as [152],

$$\gamma_k(N) = h_k(z_k(N)) = e^{-z_k^2(N)/2} H_k(z_k(N)) / \sqrt{2^k k!} \sqrt{\pi} \quad (2.5d)$$

In the above equations, h_k represents the **OHPBFs** which have been considered as activation function for each hidden neuron. Moreover, H_k is a k^{th} order **OHP** corresponding to the k^{th} hidden neuron, which can be represented in the form based on [150], [152],

$$\begin{aligned} H_1(z_k) &= 1, \quad H_2(z_k) = 2z_k, \cdots, \\ H_q(z_k) &= 2z_k H_{q-1}(z_k) - 2(q-1)H_{q-2}(z_k), \quad q \geq 3 \end{aligned} \quad (2.5e)$$

3. *Layer 1 (Input Layer)*: In (2.5e), z_k represents the input to k^{th} hidden layer neuron, which can be expressed as [152],

$$z_k = \mathbf{M}^T \mathbf{R} + \psi \gamma_k(N-1) \quad (2.5f)$$

where, ψ is a fixed recurrent gain and $\mathbf{M}$ is the constant value weight vector between input and hidden layer represented by,

$$\mathbf{M} = [\mu_1 \quad \mu_2 \quad \cdots \quad \mu_j \quad \cdots \quad \mu_{n+1}]^T \quad (2.5g)$$

with μ_j as the weight between j^{th} input neuron and any one of the m hidden neurons. The input vector $\mathbf{R}$ is expressed in the form,

$$\mathbf{R} = [r_1 \quad r_2 \quad \cdots \quad r_j \quad \cdots \quad r_{n+1}]^T \quad (2.5h)$$

where, r_j represents the j^{th} input signal.

Based on this background, we may state the following theorem.

Theorem 1 *By universal approximation theory, there exists an optimal RHNN function estimator in the form of (2.5a) such that,*

$$P = w^* + \varepsilon = \mathbf{\Theta}^{*T} \mathbf{\Gamma} + \varepsilon \quad (2.6a)$$

Here, $\mathbf{\Theta}^{*T}$ is the optimal weighting vector that achieves a satisfactory approxima-

tion with a minimum reconstruction error ε such that $|\varepsilon| < \chi$, where χ is a small positive number. Hence, for any given real continuous function g on the compact set $U \subset \mathbf{R}$ and arbitrary $\varepsilon > 0$, there exists an RHNN, $w^* = \Theta^{*T} \mathbf{\Gamma}$ in the form of (2.5a) such that,

$$\sup_{\mathbf{R} \in U} |w^*(\mathbf{R}) - g(\mathbf{R})| < \varepsilon \quad (2.6b)$$

This theorem states that **RHNN** in the form (2.5a) has universal approximation property and can be used to approximate any real, continuous function to any degree of accuracy [141], [151], [152].

2.2.2 Control Methodology

2.2.2.1 Control Objective

The n^{th} order SISO nonlinear system represented by (2.1a)-(2.1c) can be controlled, only if (2.3) is true. Based on (2.3), the linearized system in (2.2) and the desired state vector $\mathbf{x}^*(t)$, an error polynomial is formulated. The control objective of the proposed methodology is to nullify this polynomial, which may be represented as,

$$e^{(n)} + \lambda_{n-1}e^{(n-1)} + \dots + \lambda_1e^{(1)} + \lambda_0e = 0 \quad (2.7a)$$

where, $\lambda_0, \lambda_1, \dots, \lambda_{n-1}$ are the Hurwitz coefficients. By using (2.2) and (2.3) with (2.7a), the ideal control law is formulated and represented as,

$$\begin{aligned} u^*(t) = & b^{-1}[-f(\mathbf{x}(t), t) - d(\mathbf{x}(t), t) + \mathbf{x}^{*(n)}] \\ & + b^{-1}[\lambda_{n-1}e^{(n-1)} + \dots + \lambda_1e^{(1)} + \lambda_0e] \end{aligned} \quad (2.7b)$$

which satisfies the control objective in (2.7a), provided the functions $f(\mathbf{x}(t), t)$, $d(\mathbf{x}(t), t)$ and the constant b are completely known. However, in practical scenarios, these quantities remain unknown. Hence, the ideal control law in (2.7b) cannot be constructed.

2.2.2.2 Proposed Control Law

In order to solve the problem regarding the construction of the control law, a **RHNN** based control structure for the control variable $u^*(t)$ is proposed in this paper owing to its universal approximation property as stated in *Theorem 1*. This proposed control structure is synergized with an integral sliding mode technique owing to their robust nature throughout the entire response starting at the initial time instant [56], [145]. The integral sliding surface acts as the input ($\mathbf{R}$) to the

RHNN hidden neurons and is defined by the scalar function,

$$s(t) = -x^{(n-1)}(t) + \int_0^t \{x^{*(n)}(\tau) + \lambda_{n-1}e^{(n-1)}(\tau) + \dots + \lambda_1e^{(1)}(\tau) + \lambda_0e(\tau)\}d\tau \quad (2.8)$$

Based on *Theorem 1*, there exists a Θ^* , such that,

$$u^*(t) = u_{RHNN}^*(s(t), \Theta^*) + \Delta \quad (2.9a)$$

where, u_{RHNN}^* is a **RHNN** of the form shown in (2.5a) and Δ stands for the approximation error. This approximation error (Δ) is treated as the compensation controller (u_c) in this paper. Thus, the proposed control law in (2.9a) is recast as,

$$u^*(t) = u_{NNSMC} = u_{RHNN}^*(s(t), \Theta^*) + u_c(s(t)) \quad (2.9b)$$

The control law in (2.9b) is further enhanced by the addition of a bias controller (u_b) to overcome the specific nonlinearities encountered in an electrohydraulic actuation systems [9]. Hence, the final form of the proposed control law can be expressed as,

$$\begin{aligned} u^*(t) &= u_{NNSMC} + u_b(t) \\ &= u_{RHNN}^*(s(t), \Theta^*) + u_c(s(t)) + u_b(t) \end{aligned} \quad (2.9c)$$

It should be noted here that only estimates of the control variable are determined during practical scenarios, not the ideal values. In general, the hat symbol ($\hat{\cdot}$) over any variable represents its estimate. The estimate of the proposed control law in (2.9c) is represented as,

$$\hat{u}(t) = \hat{u}_{RHNN}(s(t), \hat{\Theta}) + \hat{u}_c(s(t)) + \hat{u}_b(t) \quad (2.9d)$$

where, $\hat{u}_{RHNN}$, $\hat{u}_c$ and $\hat{u}_b$ are the estimates of the **RHNN**, compensation and bias controllers, respectively. The role of each of these components of u is discussed next.

2.2.3 Controller Components

2.2.3.1 RHNN-Sliding Mode Controller

The synergistic **RHNN** - integral **SMC** architecture handles the continuous nonlinearities and system uncertainties. The sliding variable ($s(t)$) in (2.8) forms the effective input for the **RHNN** structure shown in Fig. 2.1. The input to this

RHNN controller can be constructed based on (2.8), (2.5g), (2.5h) as,

$$\begin{aligned} r_j &= e^{(n-j)}; \quad \text{for } j = 1, \dots, n \\ &= \int_0^t e(\tau) d\tau; \quad \text{for } j = (n+1) \end{aligned} \quad (2.10a)$$

$$\begin{aligned} \mu_j &= 1; \quad \text{for } j = 1 \\ &= \lambda_{n-j+1}; \quad \text{for } j = 2, 3, \dots, (n+1) \end{aligned} \quad (2.10b)$$

The output of this **RHNN** structure can be written as,

$$\hat{u}_{RHNN}(s(t), \hat{\Theta}) = w(\mathbf{R}) = \hat{\Theta}^T \mathbf{\Gamma} \quad (2.10c)$$

2.2.3.2 Compensation Controller

The compensation controller takes care of the unknown approximation error Δ by continuously compensating for it. The approximation error Δ is assumed to be smaller than a small positive constant E i.e., $|\Delta| \leq E$. It is assumed that this bound is unknown and an estimate of E , denoted by $\hat{E}$, is used to model it [152] as a switching control about the sliding surface, which is expressed as [145],

$$\hat{u}_c(t) = \hat{E} \text{sgn}(s) \quad (2.11)$$

2.2.3.3 Bias Controller

The bias controller deals with the discontinuous nonlinearities like stiction in the cylinders (refer to Fig. 1.4) and deadband present in the proportional valve. The bias controller – usually designed as a discontinuous step function, has been approximated in the present work as a continuous hyperbolic tangent because of its continuous sigmoidal nature [9]. This can be expressed mathematically as ,

$$\hat{u}_b(t) = \sum_{l=1}^2 \hat{u}_{bl} \xi_{bl} \quad (2.12a)$$

$$\xi_{bl} = \exp\{(-1)^{l+1} \alpha_b e\} / [\exp(\alpha_b e) + \exp(-\alpha_b e)] \quad (2.12b)$$

Here, $\hat{u}_{bl}$ are the estimates of the bias saturation for large positive and negative errors respectively, along with a positive parameter α_b .

2.2.3.4 Adaptation Laws

The control law in (2.9d) is combined with the system equation in (2.2) to give,

$$\begin{aligned} \mathbf{x}^{(n)} &= f(\mathbf{x}(t), t) + d(\mathbf{x}(t), t) + \\ &\quad b[\hat{u}_{RHNN}(s(t), \hat{\Theta}) + \hat{u}_c(s(t)) + \hat{u}_b(t)] \end{aligned} \quad (2.13a)$$

Now, by differentiating (2.8) and using (2.3), (2.7b), (2.13a), we get,

$$\begin{aligned} \dot{s}(t) &= e^{(n)}(t) + \lambda_{n-1}e^{(n-1)}(t) + \dots + \lambda_1e^{(1)}(t) + \lambda_0e(t) \\ &= b[u^*(t) - \hat{u}_{RHNN}(s(t), \hat{\Theta}) - \hat{u}_c(s(t)) - \hat{u}_b(t)] \end{aligned} \quad (2.13b)$$

Then by considering the following equation,

$$\tilde{u}_{RHNN} = u^* - \hat{u}_{RHNN} \quad (2.13c)$$

where, the symbol $(\sim)$ over the variable signifies the difference between its ideal and estimated values, (2.9a) is recast as,

$$\begin{aligned} u^*(t) &= u_{RHNN}^*(s(t), \Theta^*) - \tilde{u}_{RHNN}(s(t), \hat{\Theta}) + \Delta \\ &= \Theta^{*T} \Gamma - \hat{\Theta}^T \Gamma + \Delta = \tilde{\Theta}^T \Gamma + \Delta \end{aligned} \quad (2.13d)$$

with $\tilde{\Theta} = \Theta^* - \hat{\Theta}$. Consequently from (2.13b) and (2.13d), we have,

$$\dot{s}(t) = b[\tilde{\Theta}^T \Gamma + \Delta - \hat{u}_c(s(t)) - \hat{u}_b(t)] \quad (2.13e)$$

Based on this, the following theorem can be stated.

Theorem 2 *Considering a class of nonlinear systems as specified in (2.2) and the control law proposed in (2.9d), the closed loop system signals are bounded and the tracking error converges to zero asymptotically, if the following adaptation laws are incorporated:*

$$\dot{\hat{\Theta}} = \eta_\theta s \Gamma \quad (2.14a)$$

$$\dot{\hat{E}} = \eta_E |s| \quad (2.14b)$$

$$\dot{\hat{u}}_{bl} = \eta_b s \xi_{bl} \quad \text{for } l = 1, 2 \quad (2.14c)$$

where, $\eta_\theta, \eta_E, \eta_b$ are positive learning rates.

Proof of Theorem 2: Let us consider the following candidate Lyapunov function for the system [9], [145]:

$$W = s^2/2 + b[\tilde{\Theta}^T \tilde{\Theta}/\eta_\theta + \tilde{E}^2/\eta_E + \hat{u}_{b1}^2/\eta_b + \hat{u}_{b2}^2/\eta_b]/2 \quad (2.15a)$$

where, $\tilde{E} = E - \hat{E}$. Taking the time derivative of (2.15a) along with (2.2) and using (2.13e) and (2.14a)-(2.14c) leads to,

$$\begin{aligned}\dot{W} &= s\dot{s} + b[\tilde{\Theta}^T \dot{\tilde{\Theta}}/\eta_\theta + \tilde{E}\dot{\tilde{E}}/\eta_E + \hat{u}_{b1}\dot{\hat{u}}_{b1}/\eta_b + \hat{u}_{b2}\dot{\hat{u}}_{b2}/\eta_b] \\ &= b\tilde{\Theta}^T (s\Gamma - \dot{\tilde{\Theta}}/\eta_\theta) + bs(\Delta - u_c) - b\tilde{E}\dot{\tilde{E}}/\eta_E + \\ &\quad b\hat{u}_{b1}(\dot{\hat{u}}_{b1}/\eta_b - s\xi_{b1}) + b\hat{u}_{b2}(\dot{\hat{u}}_{b2}/\eta_b - s\xi_{b2}) \\ &= (bs\Delta - bE|s|) \leq -b|s|(E - |\Delta|) \leq 0\end{aligned}\tag{2.15b}$$

Equation (2.15b) suggests that $\dot{W}$ is negative semi-definite, i.e.,

$$W(s(t), \tilde{E}(t), \tilde{\Theta}, \hat{u}_{b1}, \hat{u}_{b2}) \leq W(s(0), \tilde{E}(0), \tilde{\Theta}, \hat{u}_{b1}, \hat{u}_{b2})\tag{2.15c}$$

which implies that $s(t), \tilde{E}(t), \tilde{\Theta}, \hat{u}_{b1}, \hat{u}_{b2}$ are bounded. Now, in order to prove that,

$$\lim_{t \rightarrow \infty} |\mathbf{e}(t)| = 0\tag{2.15d}$$

it is necessary to show that

$$\lim_{t \rightarrow \infty} |s(t)| = 0\tag{2.15e}$$

In order to proceed with such a proof, a new variable σ is defined such that

$$\sigma(t) = b|s|(E - |\Delta|) \leq -\dot{W}\tag{2.15f}$$

and further integrating it with respect to time yields,

$$\int_0^t \sigma(\tau) d\tau \leq \left\{ \begin{array}{l} W(s(0), \tilde{E}(0), \tilde{\Theta}, \hat{u}_{b1}, \hat{u}_{b2}) \\ -W(s(t), \tilde{E}(t), \tilde{\Theta}, \hat{u}_{b1}, \hat{u}_{b2}) \end{array} \right\}\tag{2.15g}$$

Here, $W(s(0), \tilde{E}(0), \tilde{\Theta}, \hat{u}_{b1}, \hat{u}_{b2})$ is absolutely bounded and $W(s(t), \tilde{E}(t), \tilde{\Theta}, \hat{u}_{b1}, \hat{u}_{b2})$ is bounded and non-increasing. Hence, the following statement may be articulated

$$\lim_{t \rightarrow \infty} \int_0^t \sigma(\tau) d\tau < \infty\tag{2.15h}$$

Also, $\dot{\sigma}(t)$ is bounded. So by using *Barbalat's lemma* [99], [152], it can be shown that,

$$\lim_{t \rightarrow \infty} \sigma(t) = 0\tag{2.15i}$$

Therefore, we may say that $\lim_{t \rightarrow \infty} |s(t)| = 0$, which further implies that $\lim_{t \rightarrow \infty} |e(t)| = 0$. This ensures that the designed controller guarantees the asymptotic stability of the closed-loop system. This control architecture has been depicted in the next section along with a description of the **EHAS** test bench, where it is applied.

2.3 Real-Time Control Implementation

The control scheme designed in the previous section is implemented on a rugged laboratory scale **EHAS** test bench comprising an industry grade single-rod actuator and proportional valve with a $\sim 10\%$ deadband for evaluating its position tracking performance.

2.3.1 Electrohydraulic Actuation System Test Setup

The complete real-time **EHAS** test setup is shown in Fig. 2.2 and is composed of three distinct sections *viz.* the Hydraulic section, the Electrical/Electronics section and the Interface, which are described below in detail.

1. *Hydraulic section:* This part of the setup is used to operate a double-acting single-rod hydraulic actuator (**SRA**) with the aid of a proportional valve (**PV**) against a compression spring (**S**) which acts as a variable load, as shown in the photographic view of the setup shown in Fig. 2.2. A highly pressurised hydraulic oil (Indian Oil HLP-68) is used as the working agent for the hydraulic circuit and is supplied from a hydraulic power pack as shown in the photographic view at the bottom right of Fig. 2.2. The primary component of the power pack is a pressure-compensated axial piston pump (**P**) [153] connected to a 200L hydraulic oil reservoir/tank (**T**) at its inlet and driven by a 37kW 3-phase induction motor (**IM**). The pressurized oil flows from the pump **P** outlet via a non-return valve (**NRV**) to a high-pressure in-line filter (**F**). Additionally, a pressure relief valve (**PRV**) is also installed with the power pack for safety purposes. The outlet of the filter **F** is connected to the pump port of the **PV** and the tank port of the **PV** is connected to the oil reservoir/tank via hosepipes. The movement of the spool-land system of the **PV** connects one of the metered ports (A or B) of the valve to the incoming flow of the pressurized oil (Q_1) from the pump **P** at pressure P_P and directs it towards one of the cylinder chambers of the **SRA** (Ch_1 or Ch_2) with asymmetrical areas (A_{a1} or A_{a2}). The flow coming from the other cylinder chamber (Q_2) is directed towards the tank **T** at pressure P_T . This results in the extension or retraction movement of the cylinder piston rod of the **SRA**.
2. *Electrical / Electronics section:* The instantaneous position of the **SRA** piston rod ($\mathbf{x}(t)$) is measured with the aid of a position transducer (**LVDT**) in the form of an analog voltage signal. This position signal is fed to the Real-time System (**RTS**) used for data acquisition and control, which in

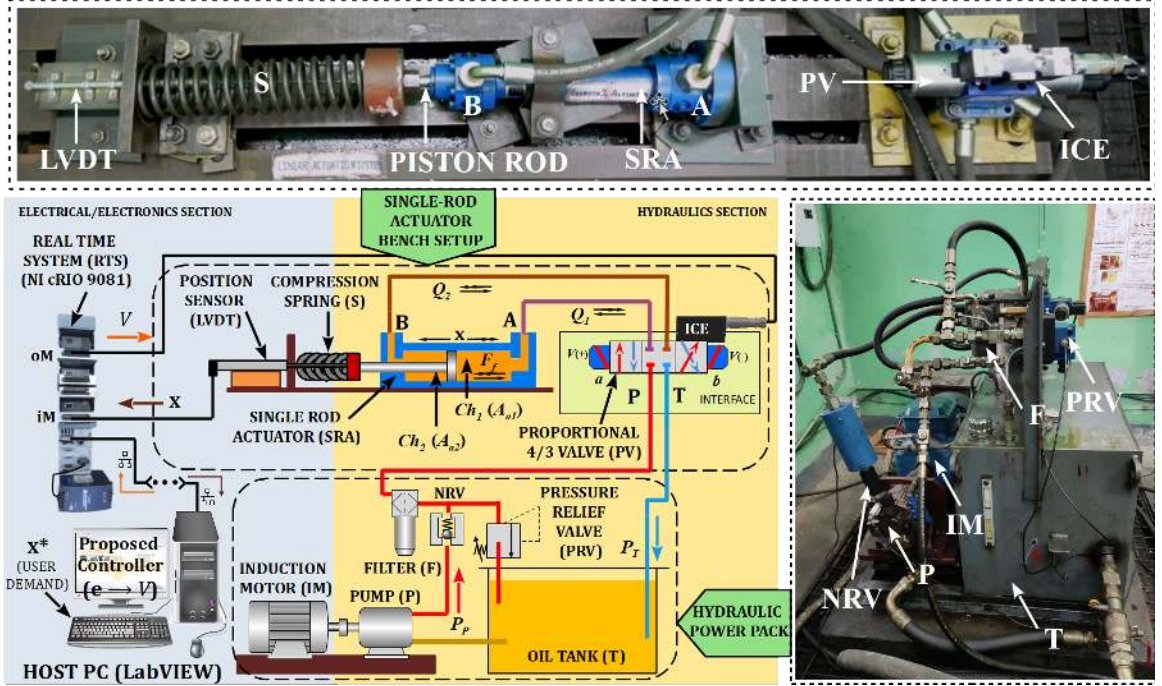


Figure 2.2: Laboratory scale electrohydraulic actuation system (EHAS) test bench setup.

turn converts it into a digital form with the aid of an A/D converter module (iM) attached to the RTS. The RTS is connected via an Ethernet cable to a HOST PC for the LabVIEW program environment, within which the entire control architecture (shown in Fig. 2.3) has been developed. The digital position signal ($x(t)$) of the SRA is compared to the user-defined demand signal ($x^*(t)$) within a LabVIEW program to compute the position error ($e(t)$), from which the proposed control scheme calculates the value of the instantaneous control variable ($\hat{u}(t)$) required to reduce this error in position. Based on a model of the EHAS (discussed in the next subsection), the control voltage signal ($V(t)$) is subsequently calculated from $\hat{u}(t)$. This control voltage ($V(t)$) is then relayed back in analog form to the PV with the support of a D/A converter module (oM) of the RTS, thus creating a closed loop system as shown in Fig. 2.2.

3. *Interface:* The solenoid operated 4port/3way PV acts as the interface between the Hydraulic and the Electrical sections with the aid of an integrated control electronics (ICE) card. The analog control voltage ($V(t)$) from the RTS is received and amplified by the ICE card integrated with the PV, which is then applied to the valve solenoids (a and b). These solenoids, in turn drive the spool-land system of the PV by the appropriate amount so as to pass a specific quantity of high pressurised oil to one of the cylinder

chambers of the **SRA**, depending on the necessary action. The deadband in the **PV** refers to the range of control voltage ($V(t)$) for which no recorded movement of the piston rod of the **SRA** occurs.

The operation of the entire test setup described above takes place in real-time within milliseconds. The instrument specifications of this **EHAS** setup are given in Table 2.1.

Table 2.1: Component specifications of the laboratory scale Single Actuator Test Bench in Fig. 2.2

Component	Brand	Specifications
Double-acting single-rod actuators (SRA)	Rexroth	CD250D40/20-200A1X/01-CGDHT; Full Stroke: 0.200m; Bore diameter: 0.040m; Rod diameter: 0.020m
Proportional valve (PV)	Rexroth	4WREE-10-E50-23/G24K31/A1V with Integrated Control Electronics (ICE), Command Signal: $\pm 10V$
Linear variable differential transformer (LVDT)	HBM	Plunger type; 0-0.200m range with 0.5-10V analog output; Nominal Sensitivity: 80mV/V
Compression spring (S)	Local	20kN/m Stiffness
Induction Motor (IM)	ABB	3GBA201044-ADEIN; 3-phase; 2-pole; 37 kW; 2945 rpm
Axial-Piston Pump (P)	Rexroth	A10VSO/32 ; 280 bar; 60 lpm
Inline Filter (F)	Stauff	SF045B/TB/B; Working Pressure: 420 bar (max)
Pressure-relief Valve (PRV)	Rexroth	DB10-1-41/315-IN001; Max. Pressure - 315 bar
Hydraulic Oil Tank (T)	Special Technique	Dimensions: 1.0m x 0.6m x 0.5m; Capacity: 200L
Real-time system (RTS)	National Instruments	NI-cRIO 9081 with 1.06GHz Intel Celeron U3405 & Spartan-6 LX75
Input-module (iM)	National Instruments	NI 9215 (4 Channel; $\pm 10V$ 16-bit simultaneous analog input)
Output-module (oM)	National Instruments	NI 9263 (4 Channel; $\pm 10V$ 16-bit simultaneous analog output)

2.3.2 Proposed Controller Implementation for EHAS Test Setup

The **EHAS** test setup shown in Fig. 2.2 can be considered as a second order nonlinear system ($n = 2$), where the motion of the cylinder piston may be expressed by a modified form of the motion dynamic equation (1.1), from an initial position, say x_0 , against the compression spring **S** of stiffness k with pre-compression δ in the form,

$$m\ddot{x} = P_1 A_{a1} - P_2 A_{a2} - k(x - x_0 + \delta) - F_f \quad (2.16)$$

Here, m represents the piston mass; P_1, A_{a1} are the cap end chamber pressure and flow area; P_2, A_{a2} are the rod end chamber pressure and flow area and F_f stands for the simplified model of the cylinder friction [8], [9]. Keeping (2.16) in mind, the proposed control law in *Section 2.2.2* can be implemented in real-time by following the steps enumerated below.

Step 1. *Formation of the control variable ($\hat{u}$):* The control variable described in (2.9d) is formed by the summation of three components viz. the RHNN controller ($\hat{u}_{RHNN}$), the compensation controller ($\hat{u}_c$) and the bias controller ($\hat{u}_b$), which are structured in the following manner for the **EHAS** setup.

1. *RHNN component:* The neuron structure of the Input Layer (r_j) and the connective weights (μ_j) of the ($\hat{u}_{RHNN}$) component are modelled based on the order of the system (n) as expressed by (2.10a)-(2.10b). Since, the system under investigation is a second order system ($n = 2$) based on (2.16), the number of input neurons (j) is 3, whereby, $r_1 = e^{(1)}$, $r_2 = e$ and $r_3 = \int_0^t e(\tau) d\tau$ from (2.10a). The connective weights of these three input neurons to the hidden neurons are $\mu_1 = 1$, $\mu_2 = \lambda_1$ and $\mu_3 = \lambda_0$, respectively from (2.10b). It is also to be noted that the underlying structure of the hidden layer remains unaltered as mentioned in *Section 2.2.1*. However, the number of hidden neurons (q) in the structure may vary for different systems. For the purposes of this study on the **EHAS** test setup, the value of q is chosen as 5, as a trade-off between computational time and determination of higher order dynamics. Also, the parameters ψ , λ_1 and λ_0 used in the hidden layer input, needs to be determined experimentally.
2. *Compensation and Bias components:* The controller components $\hat{u}_c$ and $\hat{u}_b$ are primarily functions of the error signal ($e(t)$) or the sign of the sliding variable ($s(t)$) as mentioned earlier. Hence, it is obvious that they do not structurally change based on the order (n) of the system being investigated and are expressed by (2.11), (2.12a) and (2.12b).

3. *Adaptive Laws*: The adaptive laws formulated in (2.14a), (2.14b) and (2.14c) are functions of sliding variable ($s(t)$) which is formulated from (2.8) for $n = 2$. Although the structure of these laws remain unaltered, the learning rates ($\eta_\theta, \eta_E, \eta_b$) needs to be determined experimentally.

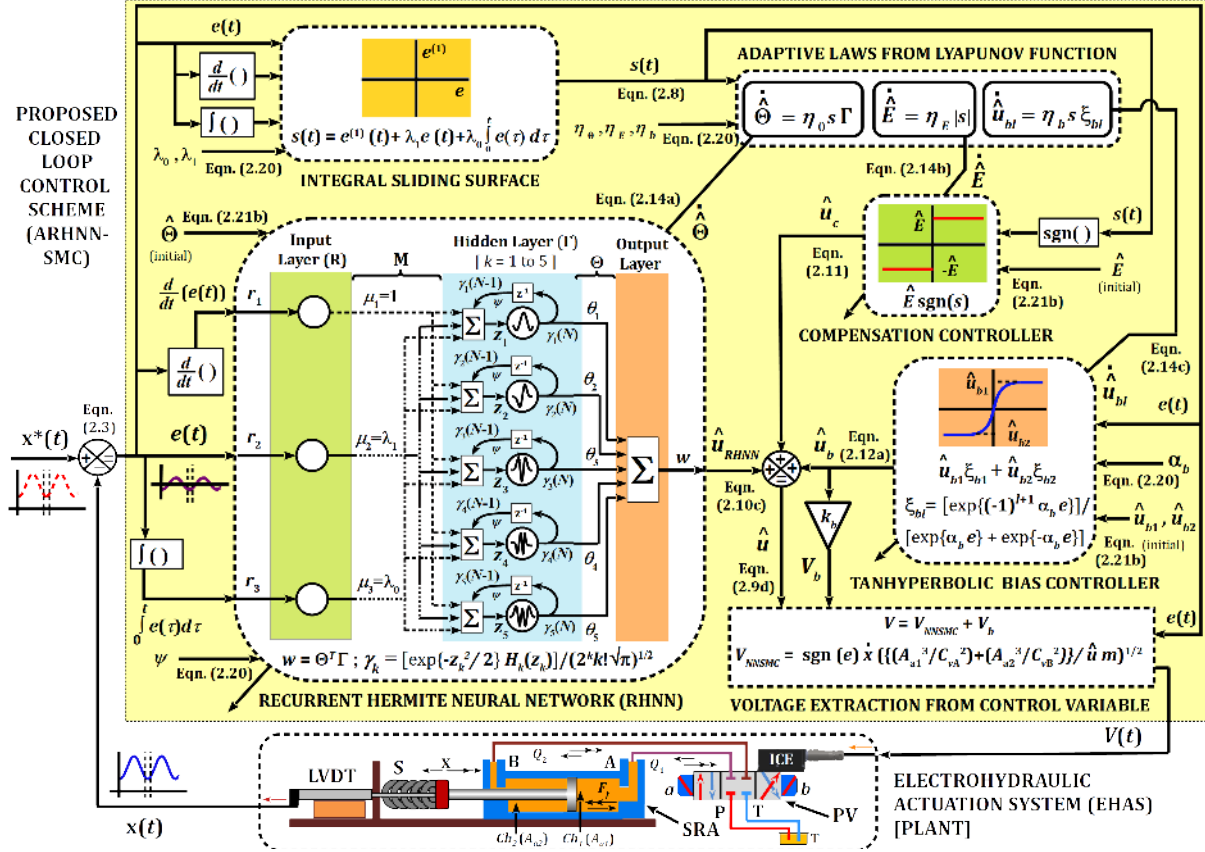


Figure 2.3: Schematic illustration of the proposed closed loop control architecture with the **EHAS** test setup

Step 2. Voltage Extraction from control variable ($\hat{u}$): Based on the control variable ($\hat{u}$) described in **Step 1**, the control voltage to the **PV** may be extracted from the system equations in the following manner.

1. *Actuator chamber pressures*: Assuming that the hydraulic oil being used is incompressible, the chamber pressures P_1 and P_2 in (2.16) can be extracted from the pressure-discharge relation of the proportional valve in the form [8], [9] of (1.4a)-(1.4b) as,

$$\begin{aligned} Q_1 &= A_{a1}\nu = C_{vA}(V - V_b(t))\sqrt{(P_P - P_1)} \quad \text{for } V \geq V_b \\ &= C_{vA}(V + V_b(t))\sqrt{(P_1 - P_T)} \quad \text{for } V < V_b \end{aligned} \quad (2.17a)$$

$$\begin{aligned} Q_2 &= A_{a2}\nu = C_{vB}(V - V_b(t))\sqrt{(P_2 - P_T)} \quad \text{for } V \geq V_b \\ &= C_{vB}(V + V_b(t))\sqrt{(P_P - P_2)} \quad \text{for } V < V_b \end{aligned} \quad (2.17b)$$

where, ν is the piston velocity; C_{vA} , C_{vB} are the valve coefficients of metered ports (A and B) and P_P , P_T are the pump and tank pressures, respectively. As mentioned in *Chapter 1*, the values of the valve coefficients are unequal based on the area asymmetry of the SRA. In this case, it is considered that the port areas have a ratio of 2:1, which is equal to the ratio C_{vA}/C_{vB} . This is used to derive the value of the coefficients from the valve catalogue [154] as, $C_{vA} = 2C_{vB} = 2.19 \times 10^{-7} \text{ m}^3/(\text{s.V}\sqrt{\text{Pa}})$.

2. *Bias Voltage*: In (2.17a)-(2.17b), $V_b(t)$ represents the bias voltage used to compensate for the discontinuous nonlinearities induced by the static friction in the cylinder and the valve dead band. It may be formulated from the bias controller $\hat{u}_b(t)$ in (2.12a) as,

$$V_b(t) = k_b \hat{u}_b(t); \quad k_b = 1 \text{ V.s}^2/\text{m} \quad (2.18)$$

3. *Input Linearization*: Based on (2.16), (2.17a) and (2.17b), an input linearized second order system dynamics model of the form represented in (2.2), can be written in the form of (1.6a)-(1.6b) as,

$$\ddot{x} = -\dot{x} + u + d \quad (2.19a)$$

where, the input linearized control u is given by,

$$u = -\text{sgn}(V)(\dot{x}^2/m)\{(A_{a1}^3/C_{vA}^2) + (A_{a2}^3/C_{vB}^2)\}/\{V - \text{sgn}(V)V_b\}^2 \quad (2.19b)$$

and the disturbance d has the form,

$$\begin{aligned} d &= \{(k(x_0 - \delta) + P_P A_{a1} - P_T A_{a2} - F_f - kx)/m\} + \dot{x} \\ &\quad \text{for } V \geq V_b \\ &= \{(k(x_0 - \delta) + P_P A_{a2} - P_T A_{a1} - F_f - kx)/m\} + \dot{x} \\ &\quad \text{for } V < V_b \end{aligned} \quad (2.19c)$$

4. *Control Voltage*: The variable V in (2.19b) represents the control voltage and can be expressed mathematically as,

$$V = V_{NNSMC} + V_b \quad (2.19d)$$

where, V_{NNSMC} can be derived from (2.19b) and (2.19d) in the following form,

$$V_{NNSMC} = \text{sgn}(e)\dot{x}\{(A_{a1}^3/C_{vA}^2) + (A_{a2}^3/C_{vB}^2)\}/um)^{1/2} \quad (2.19e)$$

Hence, the equations (2.19d) and (2.19e) together constitute the control voltage ($V(t)$), extracted from the control variable ($\hat{u}$) and relayed to the **PV** at every time step.

The execution of the proposed control scheme detailed in **Step 1** and the extracted control voltage ($V(t)$) in **Step 2** has been illustrated in Fig. 2.3, in the form of a schematic diagram encompassing the entire control architecture along with the diagrammatic view the **EHAS** test setup with relevant equation numbers. The entire proposed control scheme in Fig. 2.3 is developed and executed in the LabVIEW environment of the **HOST PC** of the **EHAS** setup.

2.4 Results and Discussion

2.4.1 Selection of fixed parameters

The fixed parameter vector of the system can be defined as,

$$\mathbf{f} = [q \quad \psi \quad \lambda_0 \quad \lambda_1 \quad \alpha_b \quad \eta_\theta \quad \eta_b \quad \eta_E]^T \quad (2.20)$$

The values of the vector components are chosen based on real-time experiments, conducted with a sinusoidal tracking demand signal $\mathbf{x}^*(t)$ of frequency 0.5Hz and 0.02m peak-to-peak amplitude. The number of hidden neurons (q) is chosen as 5, as a trade-off between computational burden and determination of higher order dynamics, based on several real-time experimental trials involving 3, 5 and 7 hidden neurons. The studies for the selection of ψ , λ_0 and λ_1 are conducted by disabling the bias controller and the adaptive parameters. In order to proceed with the study, the parameter settling time T_s is defined as the time required for the system response $\mathbf{x}(t)$ to reach within 1% error band.

In Fig. 2.4(a), the studies conducted for selection of the recurrent gain ψ in (2.5f) with λ_0 and λ_1 as unity, are shown. It can be clearly observed that both the settling time and the average error per cycle is least for $\psi = 1.5$. With this value of ψ , another study is presented in Fig. 2.4(b) for determining the coefficients λ_0 and λ_1 with the same demand. It can be seen that the coefficients λ_0 and λ_1 can be chosen as 2 and 3, respectively, corresponding to the lowest absolute value of the average error over a cycle. The learning rates in (2.20) are determined by considering the chosen values of ψ , λ_0 and λ_1 . Fig. 2.4(c) shows the variation of settling time for different values of η_θ , η_E and η_b . In order to determine the nominal setting for η_b , the non-adaptive index α_b is chosen as 10. Corresponding to minimum settling time, η_θ is chosen as 2500, η_E as 15 and η_b as 25.

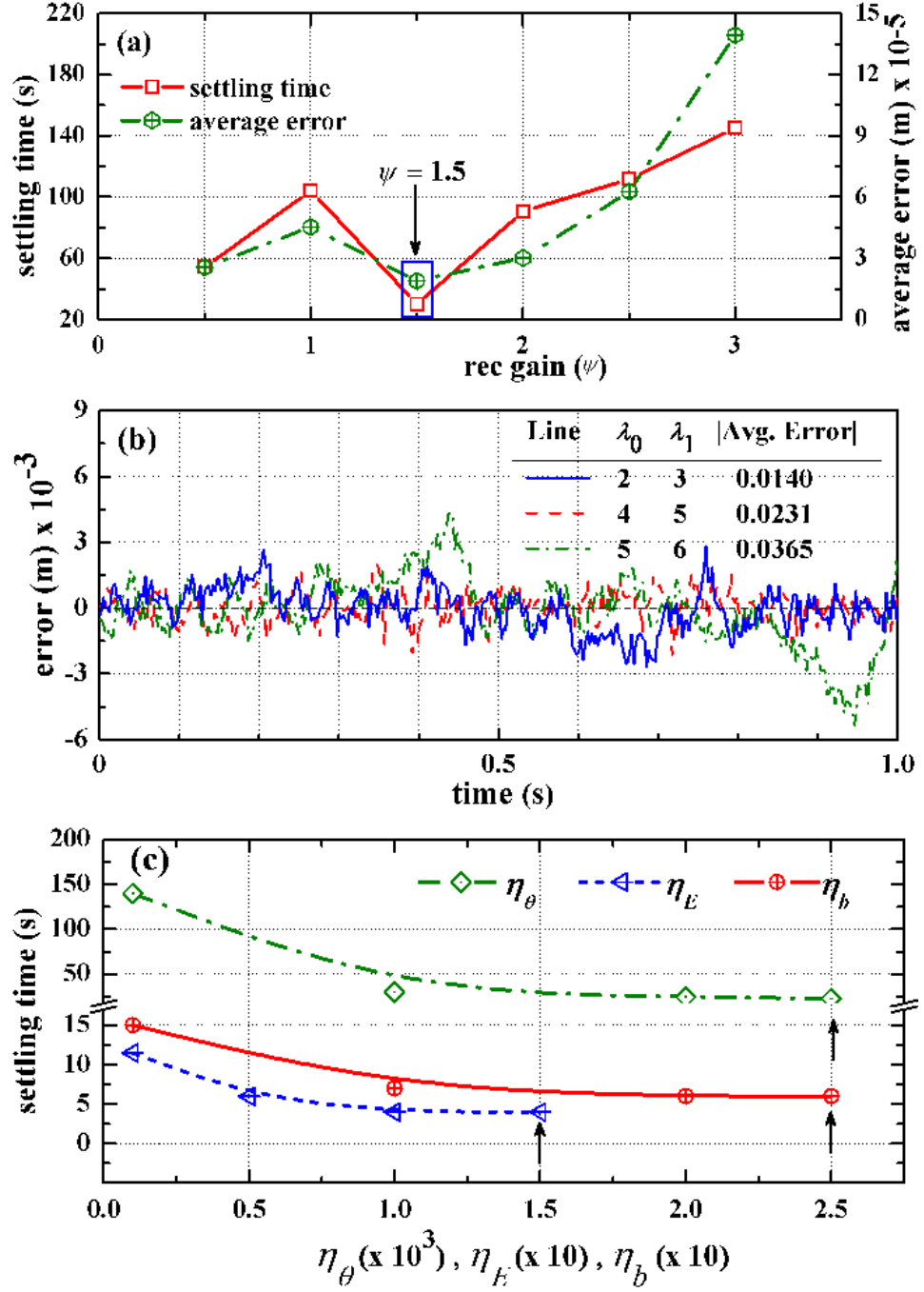


Figure 2.4: Results of experimental studies for selection of fixed parameters: (a) Recurrent Gain (ψ) (b) Hurwitz coefficients (λ_0 and λ_1) (c) Learning Rates (η_θ , η_E and η_b).

2.4.2 Nominal settings of free parameters

The free or adaptive parameter vector of the system can be defined as,

$$\mathbf{a} = [\hat{\theta}_1 \quad \hat{\theta}_2 \quad \cdots \quad \hat{\theta}_{q=5} \quad \hat{E} \quad \hat{u}_{b1} \quad \hat{u}_{b2}]^T \quad (2.21a)$$

the nominal settings for which have been chosen as,

$$\mathbf{a} = [0.09 \quad 0.15 \quad 0.08 \quad 0.05 \quad 0.08 \quad 1 \quad 1 \quad -1]^T \quad (2.21b)$$

The values of the weight vector components $(\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_5)$ has been chosen after several trials with the real-time system. As mentioned in Section 2.2.3.2, the approximation error Δ , introduced due to the presence of uncertainties in the electrohydraulic system, is smaller than a small positive constant E . Hence, a typical small positive constant value of unity is chosen as the nominal or initial value of the adaptive parameter $\hat{E}$. The estimates of the bias saturation values are considered as 1 and -1 respectively, keeping in mind the $\sim 10\%$ dead band in proportional valve with control range of ± 10 V as specified in Table 2.1.

2.4.3 Frequency Response

The frequency response analysis of the proposed controller is carried out with sinusoidal tracking demand signals of 0.02m peak-to-peak amplitude and within the frequency range of 0.1Hz to 10Hz. The magnitude gain and phase plots of the frequency response are shown in Fig.2.5, where the circular markers represent the gain magnitude values and the diamond markers represent the phase shift values for the corresponding frequencies. It is evident from the plot that the -3dB magnitude lies around the frequency of 6.0Hz.

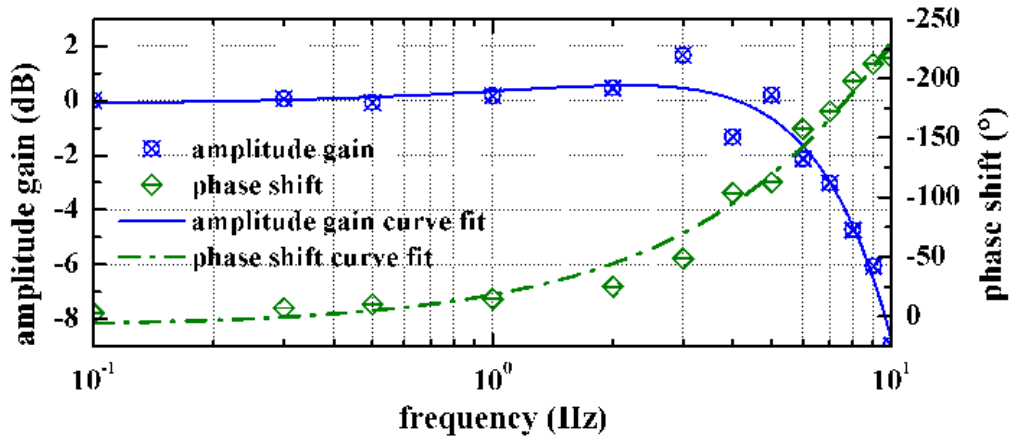


Figure 2.5: Frequency response of the proposed control scheme for sinusoidal demands of 0.02m amplitude.

2.4.4 Evaluation of the Controller Adaptation Scheme

The performance of the adaptive laws as represented by (2.14a)-(2.14c), for the proposed controller can be evaluated based on the three performance indices or metrics, namely the Integral Absolute Error (**IAE**), Integral of Time-Weighted Absolute Error (**ITAE**) and Control Effort (**CE**) [9], [77], which are defined in their discretized form as,

$$\mathbf{IAE} = \sum_{i=0}^{N_s} |e(i)| \tau \quad (2.22a)$$

$$\mathbf{ITAE} = \sum_{i=0}^{N_s} (i\tau) |e(i)| \tau \quad (2.22b)$$

$$\mathbf{CE} = \sum_{i=0}^{N_s} V(i)^2 \tau \quad (2.22c)$$

where, N_s denotes the total number of samples and τ is the sample time. However, a normalised form of the equations in (2.22) is required to obtain frequency and amplitude independent performance values for the all the position demands used in this study. The normalisation is carried out by using the time period (T) of the demand signal, the maximum stroke of the actuator (s_{max}) and the square of the available voltage (V_{max}). These normalised metrics are named as the Normalised Discrete Absolute Error (**NDAE**), the Normalised Discrete Time-weighted Absolute Error (**NDTAE**) and the Normalised Control Effort (**NCE**). These are defined mathematically as,

$$\mathbf{NDAE} = \frac{1}{s_{max}T} \sum_{i=0}^{N_s} |e(i)| \tau \quad (2.23a)$$

$$\mathbf{NDTAE} = \frac{1}{s_{max}T^2} \sum_{i=0}^{N_s} (i\tau) |e(i)| \tau \quad (2.23b)$$

$$\mathbf{NCE} = \frac{1}{V_{max}^2 T} \sum_{i=0}^{N_s} V(i)^2 \tau \quad (2.23c)$$

The normalisation is carried out by considering the maximum available stroke of the **SRA**(s_{max}) as 0.20m and the maximum voltage to the **PV**(V_{max}) as $\pm 10V$, based on the system specifications in Table 2.1. These metrics are evaluated for 0.02m sinusoidal, trapezoidal and square demand signals of frequencies 0.1, 0.3 and 0.5Hz as presented in Figs. 2.6, 2.7 and 2.8, respectively, with two control configurations: the fixed **RHNN-SMC** with fixed bias (**RHNNSMC-FB**) and the adaptive **RHNN-SMC** with adaptive bias (**ARHNNSMC-AB**).

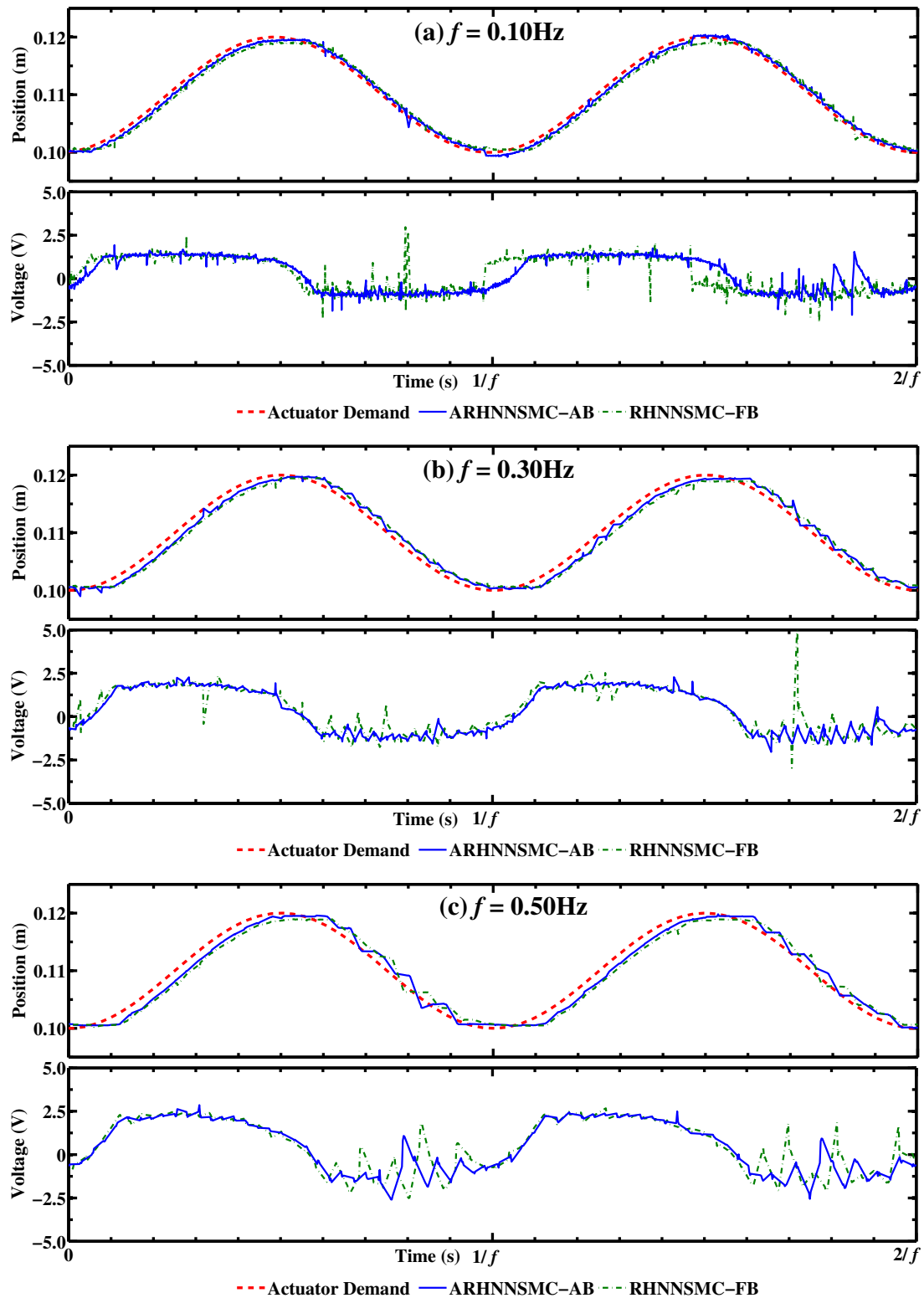


Figure 2.6: Comparison of System responses for Adaptive (**ARHNN-SMC-AB**) and Fixed (**RHNN-SMC-FB**) Controllers with Sinusoidal demand signals of amplitude 0.02m and fundamental frequencies of (a) 0.1Hz (b) 0.3Hz (c) 0.5Hz

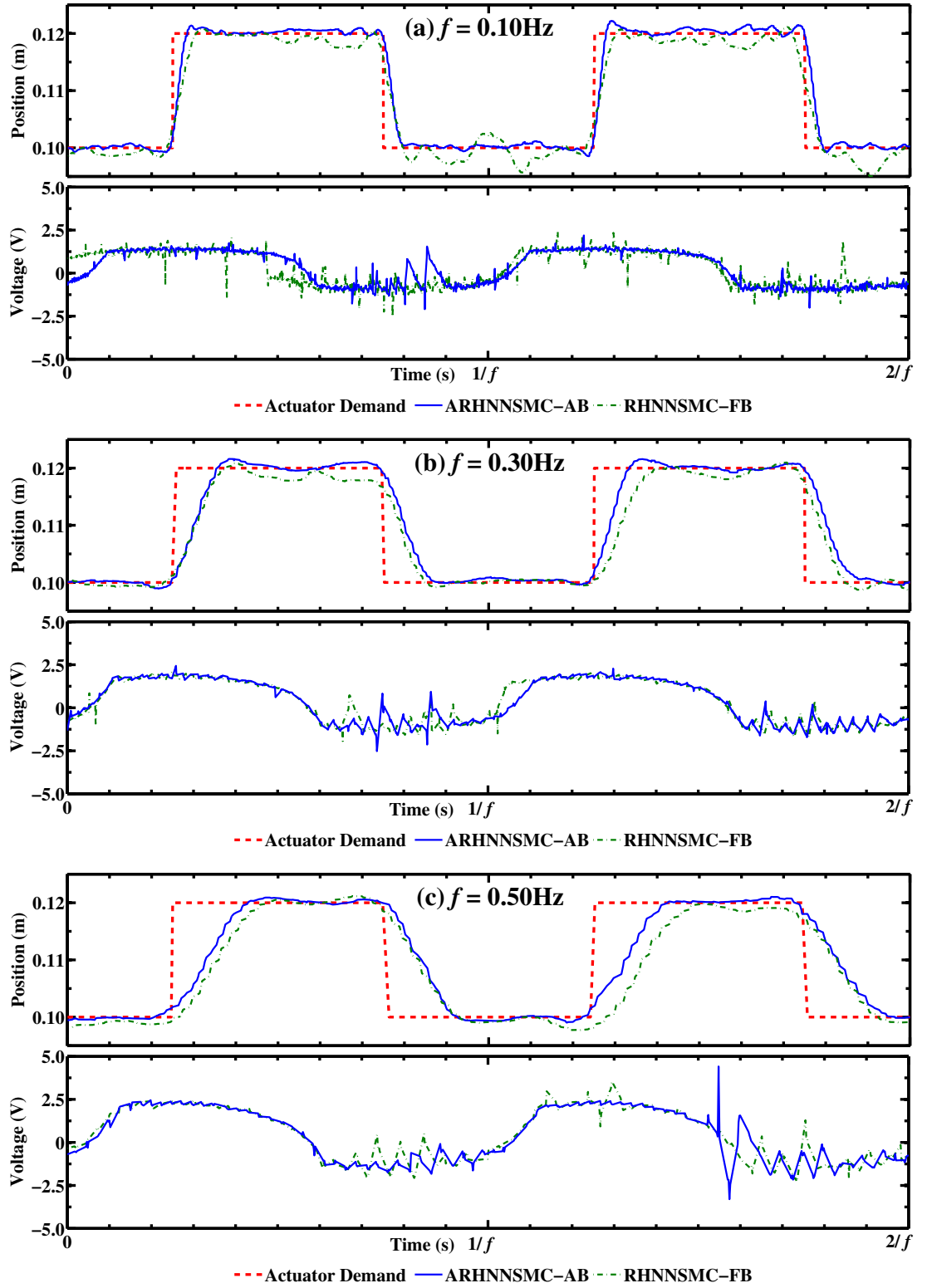


Figure 2.7: Comparison of System responses for Adaptive (ARHNSMC-AB) and Fixed (RHNSMC-FB) Controllers with Square wave demand signals of amplitude 0.02m and fundamental frequencies of (a) 0.1Hz (b) 0.3Hz (c) 0.5Hz

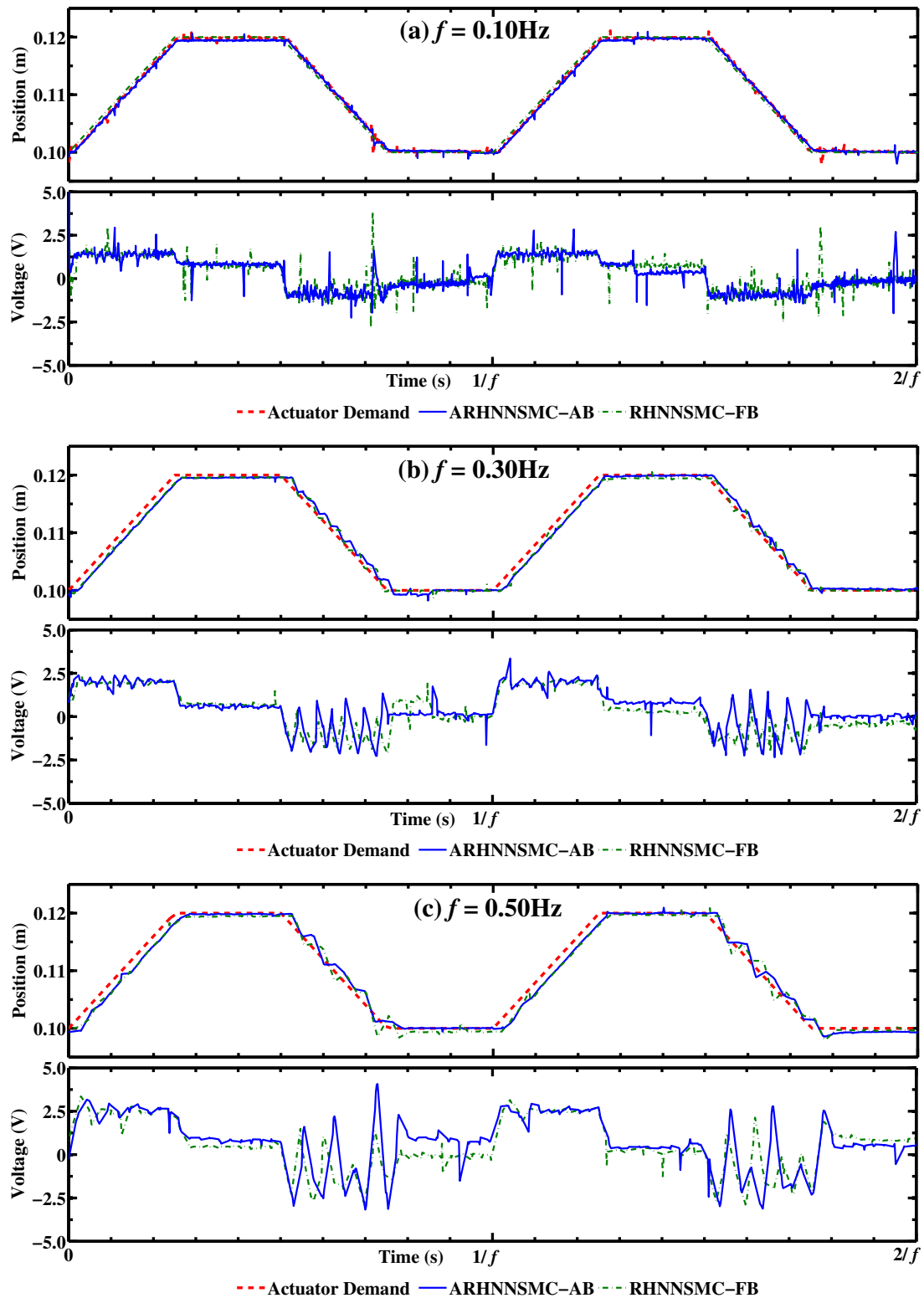


Figure 2.8: Comparison of System responses for Adaptive (ARHNSMC-AB) and Fixed (RHNSMC-FB) Controllers with Trapezoidal demand signals of amplitude 0.02m and fundamental frequencies of (a) 0.1Hz (b) 0.3Hz (c) 0.5Hz

Table 2.2: Comparison of performance indices for fixed (**RHNNSMC-FB**) and adaptive (**ARHNNSMC-AB**) controllers in normalised forms

Demand Type	Freq (Hz)	Controllers	
		RHNN-SMC-FB (Fixed)	ARHNN-SMC-AB (Adaptive)
NDAE values			
Sinusoid (0.02m Amplitude)	0.1	0.755	0.465
	0.3	1.005	0.675
	0.5	1.025	0.65
Trapezoid (0.02m Amplitude)	0.1	0.51	0.38
	0.3	0.63	0.495
	0.5	1.175	0.775
Square (0.02m Amplitude)	0.1	0.02	0.01
	0.3	0.0255	0.0225
	0.5	0.0425	0.035
NDTAE values			
Sinusoid (0.02m Amplitude)	0.1	0.72	0.46
	0.3	15.39	11.475
	0.5	16.625	12.875
Trapezoid (0.02m Amplitude)	0.1	0.431	0.357
	0.3	1.917	1.706
	0.5	6.35	5.275
Square (0.02m Amplitude)	0.1	0.039	0.024
	0.3	0.081	0.072
	0.5	0.1375	0.1125
NCE values			
Sinusoid (0.02m Amplitude)	0.1	0.00159	0.00153
	0.3	0.00264	0.00252
	0.5	0.00345	0.00355
Trapezoid (0.02m Amplitude)	0.1	0.00164	0.00159
	0.3	0.00267	0.00261
	0.5	0.00515	0.00425
Square (0.02m Amplitude)	0.1	0.00688	0.00665
	0.3	0.0131	0.0123
	0.5	0.021	0.018

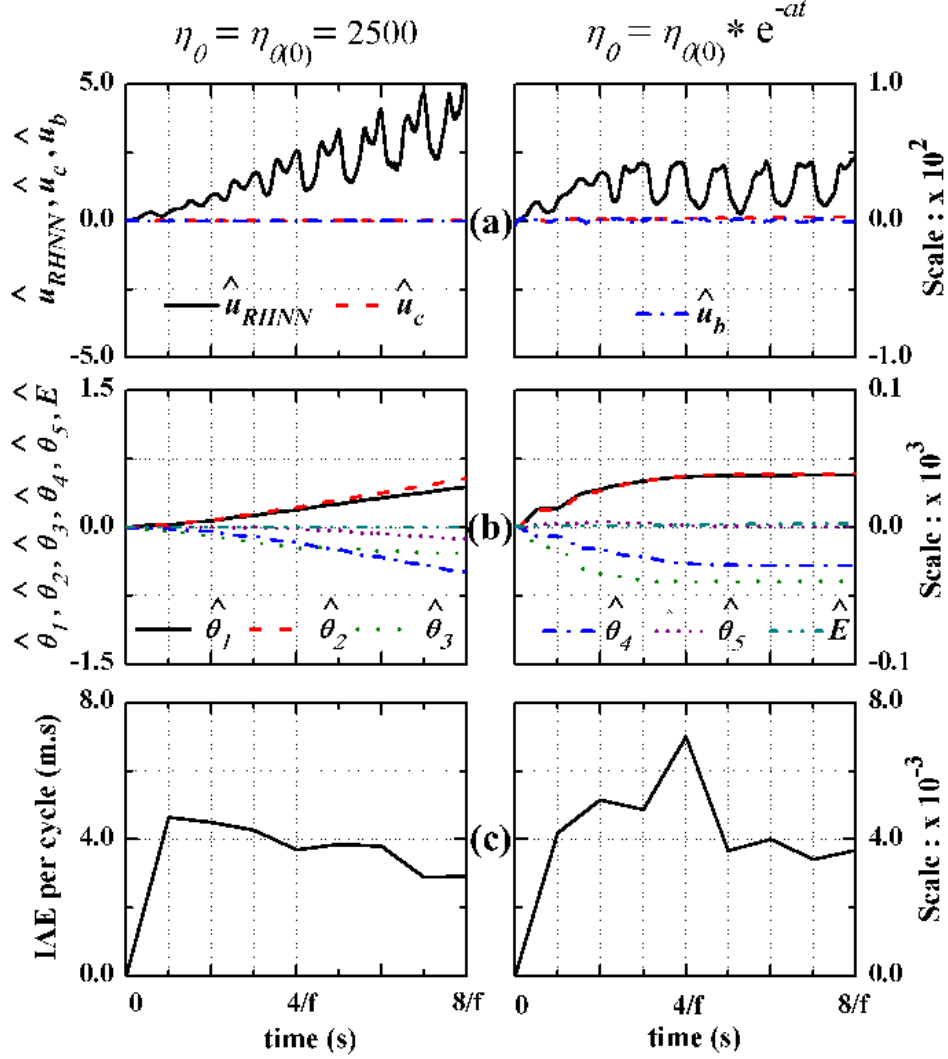


Figure 2.9: Comparison of (a) the components of the manipulated variable $\hat{u}$ ($\hat{u}_{RHNN}$, $\hat{u}_c$ and $\hat{u}_b$) (b) the RHNN weight vector components ($\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3, \hat{\theta}_4, \hat{\theta}_5$) & the uncertainty bound $\hat{E}$ (c) IAE values per cycle for a constant and an exponentially decaying variation of the learning rate η_θ based on a sinusoidal reference signal of 0.02m amplitude and 0.5Hz frequency.

The normalised performance indices for all the demand signals mentioned above for both the controllers are tabulated in Table 2.2. It can be noticed from this table that in most cases the adaptive strategy (**ARHNN-SMC-AB**) shows promising results compared to the fixed scheme (**RHNNSMC-FB**). In Figs. 2.6, 2.7 and 2.8, the voltage oscillations seem to reduce with increasing frequency. This can be attributed to the presence of significant Stribeck friction for lower piston velocities which subsequently lead to higher tracking error and control effort. During the retraction phase of the 0.5Hz signals, certain low magnitude trajectory oscillations are observed, which may be attributed to switching about the sliding surface by the $\hat{u}_c$ controller.

Fig. 2.9 presents the variation of the control variable ($\hat{u}$), components, i.e., $\hat{u}_{RHNN}$, $\hat{u}_c$ and $\hat{u}_b$, for a constant learning rate η_θ determined from Fig. 2.4(c) and an exponentially decaying η_θ from the same initial value as the constant rate system, corresponding to the sinusoidal signals represented in Fig. 2.6. It is evident from this figure that $\hat{u}_{RHNN}$ dominates the formation of $\hat{u}$ and hence, pacifies the switching effect of $\hat{u}_c$ to a large extent, which renders these oscillations insignificant for the **EHAS** system under investigation. It is also noticeable from Fig. 2.9(a) that the magnitude of $\hat{u}_{RHNN}$ tends to vary within a band for the exponentially decaying case. Similarly, it is clear from the comparison of the weight vector (Θ) components and the uncertainty estimate ($\hat{E}$) shown in Fig. 2.9(b), that the weights tend to saturate after a few cycles for the exponentially decaying case. From Fig. 2.9(c) it is evident that the **IAE** values per cycle tend to lie around similar numerical values for both the cases, after the saturation of the weight vector (Θ) components and controller ($\hat{u}$) components. This seems to suggest that either of the learning rate cases may work out similarly.

2.4.5 Performance Evaluation of the Proposed Controller

The performance of the proposed controller (**ARHNNSMC-AB**) is evaluated in comparison with three established control configurations:

1. *Proportional-Integral Controller (PI)*: The most widely used industrial control scheme with fixed gains $K_P=175$ V/m and $K_I=90$ V/(m-s) [139].
2. *GA-Optimized Fuzzy-Feedforward-Bias Controller (GA-OFFB)*: The combination of a feedforward model and a fuzzy-based feedback control scheme with binary-coded genetic algorithm (**BCGA**) based parameter optimization [48].
3. *Adaptive Fuzzy-Sliding Mode Controller with Adaptive Bias (AFSMC-AB)*: An integral sliding mode based nonlinear control scheme with fuzzy rule-base and adaptive laws for the free parameters of the system [9].

2.4.5.1 Temporal Analysis

In this study, the system is subjected to sinusoidal, trapezoidal and square type demand signals of 40mm amplitude and signal frequencies of 0.1 Hz, 0.5Hz and 1.0 Hz for each of the four control configurations mentioned, as displayed in Figs. 2.10, 2.11 and 2.12, respectively. A closer inspection of these figures suggests that the proposed scheme has a similar behavior to that of the **AFSMC-AB** [9] for all the three demand types under investigation. However,

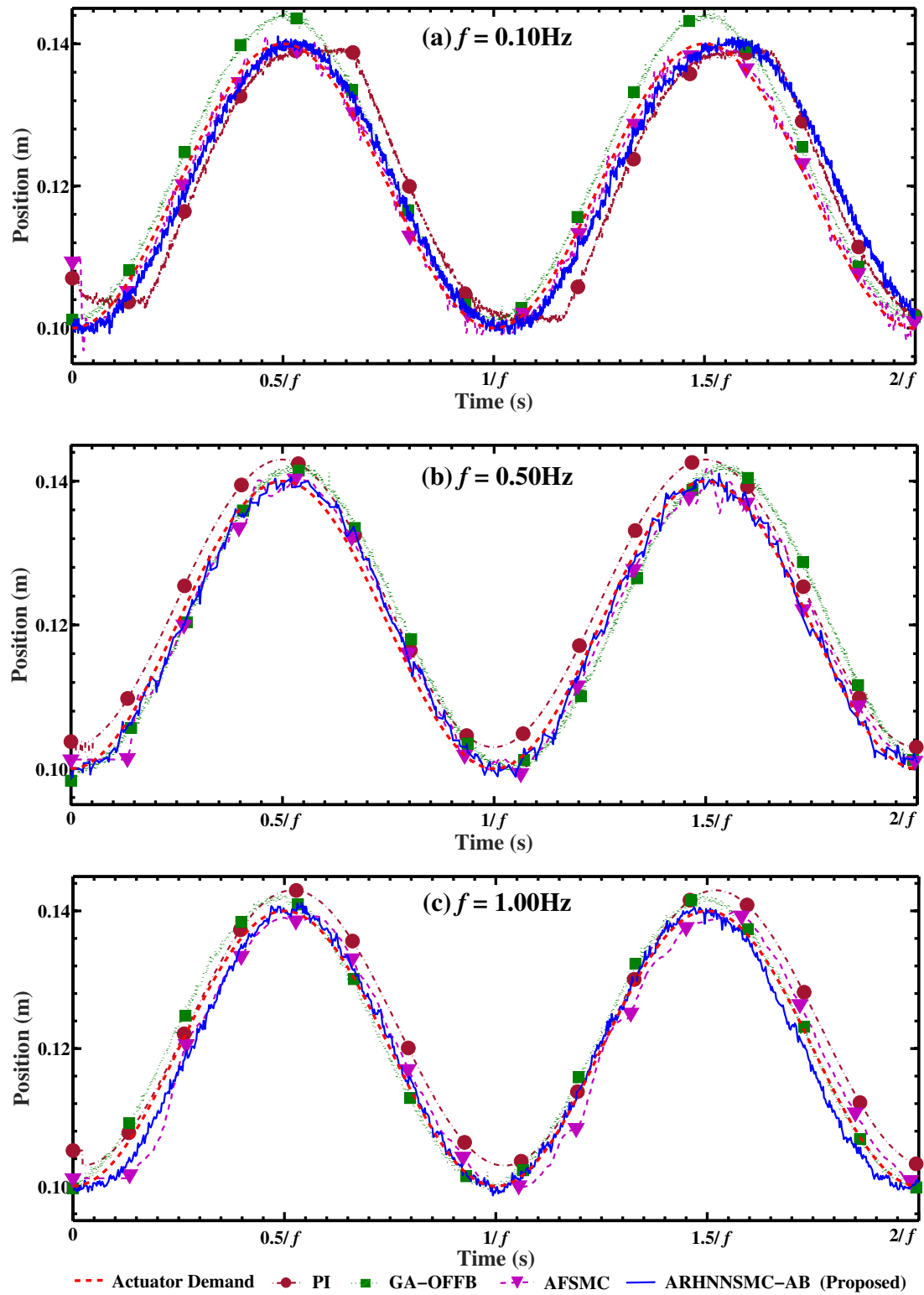


Figure 2.10: Comparison of System responses of various controllers for Sinusoidal demand signals of amplitude 0.04m and fundamental frequencies of (a) 0.1Hz (b) 0.5Hz (c) 1.0Hz

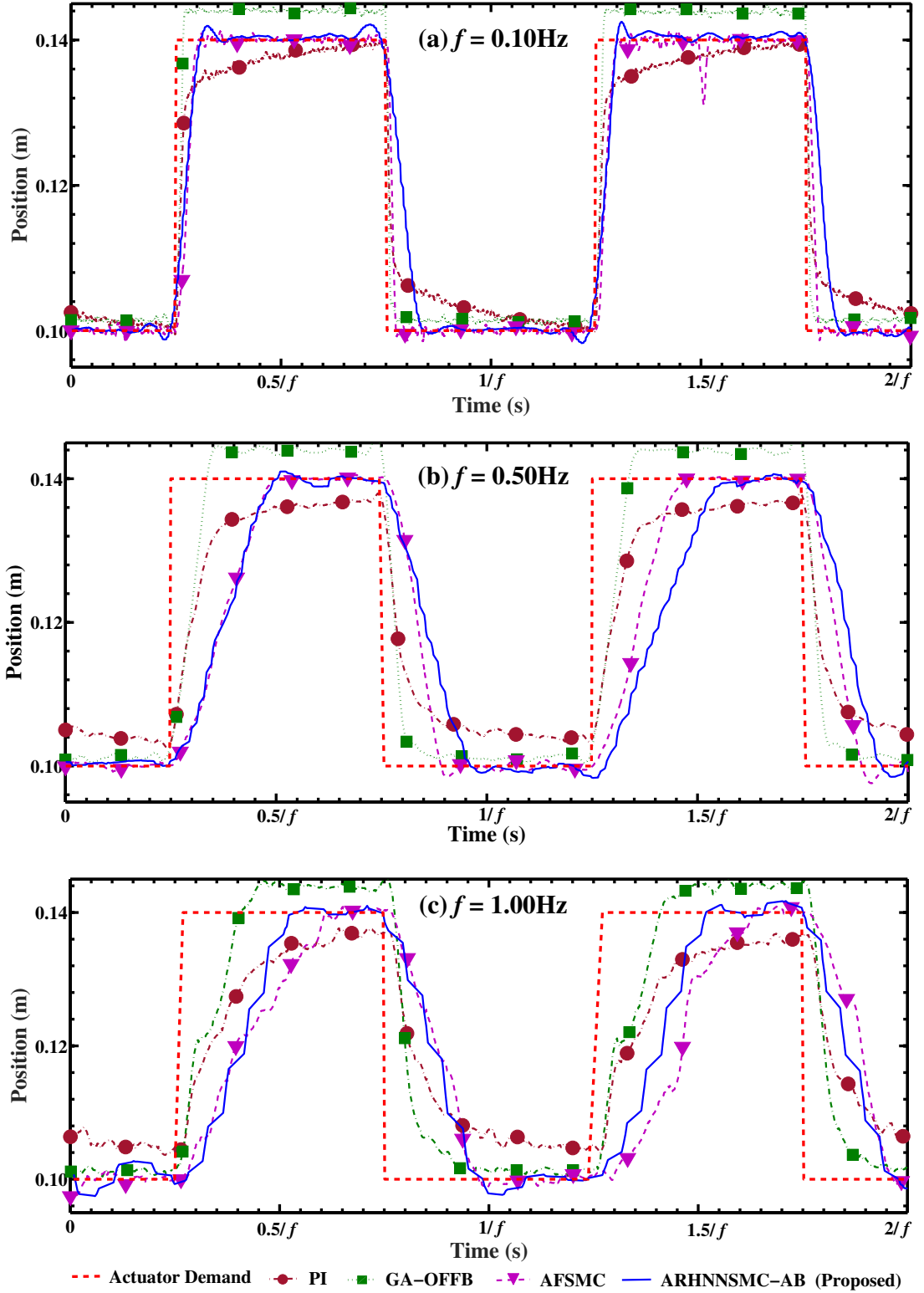


Figure 2.11: Comparison of System responses of various controllers for Square wave demand signals of amplitude 0.04m and fundamental frequencies of (a) 0.1Hz (b) 0.5Hz (c) 1.0Hz

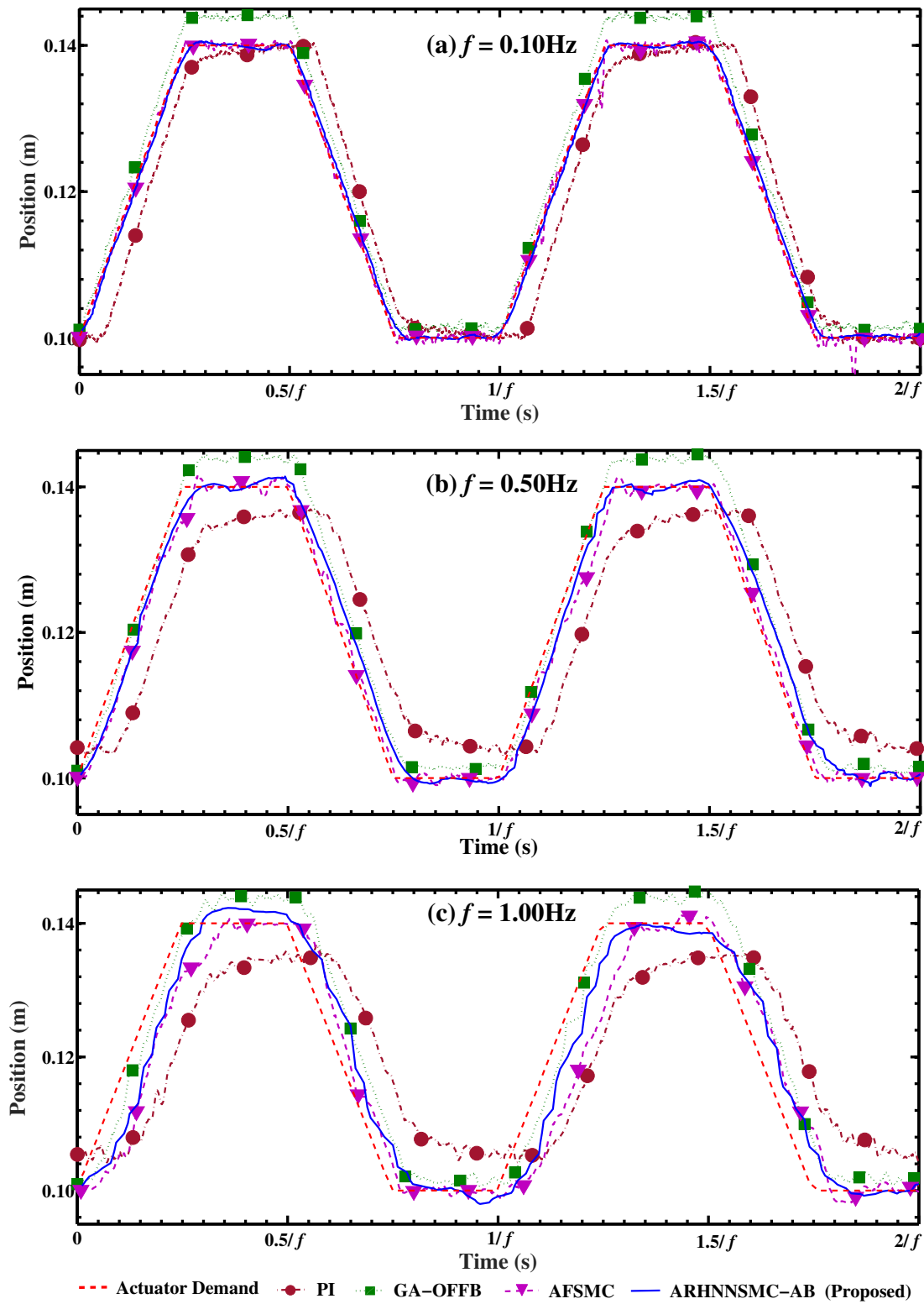


Figure 2.12: Comparison of System responses of various controllers for Trapezoidal demand signals of amplitude 0.04m and fundamental frequencies of (a) 0.1Hz (b) 0.5Hz (c) 1.0Hz

Table 2.3: Comparison of normalized performance indices for four control configurations and three demand types as depicted in Figs. 2.10, 2.11 and 2.12

Demand Type	Freq (Hz)	Controllers (Performance Indices)			
		PI	GA-OFFB	AFSMC-AB	ARHNN-SMC-AB (Proposed)
NDAE values					
Sinusoid (0.04m Amplitude)	0.1	0.127	0.061	0.303	0.058
	0.5	0.078	0.06	0.29	0.033
	1.0	0.395	0.28	0.155	0.13
Trapezoid (0.04m Amplitude)	0.1	0.123	0.069	0.819	0.066
	0.5	0.245	0.08	0.848	0.07
	1.0	0.255	0.09	0.305	0.075
Square (0.04m Amplitude)	0.1	0.098	0.073	0.381	0.066
	0.5	0.205	0.113	0.753	0.08
	1.0	0.325	0.175	0.87	0.08
NDTAE values					
Sinusoid (0.04m Amplitude)	0.1	0.129	0.059	0.341	0.056
	0.5	0.078	0.058	0.352	0.036
	1.0	0.395	0.29	0.19	0.14
Trapezoid (0.04m Amplitude)	0.1	0.357	0.214	2.502	0.206
	0.5	0.985	0.32	3.44	0.286
	1.0	1.79	0.62	2.065	0.52
Square (0.04m Amplitude)	0.1	0.198	0.144	0.857	0.139
	0.5	0.82	0.446	3.066	0.314
	1.0	1.645	0.88	4.33	0.385
NCE values					
Sinusoid (0.04m Amplitude)	0.1	0.054	0.111	0.081	0.006
	0.5	0.089	0.131	0.259	0.016
	1.0	0.072	0.110	0.163	0.068
Trapezoid (0.04m Amplitude)	0.1	0.063	0.029	0.069	0.003
	0.5	0.141	0.115	0.215	0.012
	1.0	0.188	0.181	0.377	0.030
Square (0.04m Amplitude)	0.1	0.065	0.073	0.069	0.003
	0.5	0.171	0.352	0.180	0.015
	1.0	0.352	0.748	0.438	0.029

the comparison of normalised performance indices presented in Table 2.3 clearly point to the fact that the proposed scheme has much better performance values amongst all the control configurations. Also, it can be noticed that the **NDAE** and **NDTAE** values of the **GA-OFFB** [48] and the proposed scheme are quite close in a few cases, however, the **NCE** values suggest that the proposed controller requires much less effort to achieve a similar performance. This feature may prove useful for multi-actuator systems.

2.4.5.2 Spectral Analysis

The Fast Fourier Transform (**FFT**) has been used to conduct a spectral analysis for each controller corresponding to the three sinusoidal demands shown in Fig. 2.10. The sampling rate of the recorded responses is about 75Hz. A combined spectral analysis plot for all three fundamental frequencies under investigation (0.1Hz, 0.5Hz and 1Hz) is shown in Fig. 2.13 by truncating the higher order frequencies. The spectral analysis is performed to ascertain the degree of variation amongst the responses.

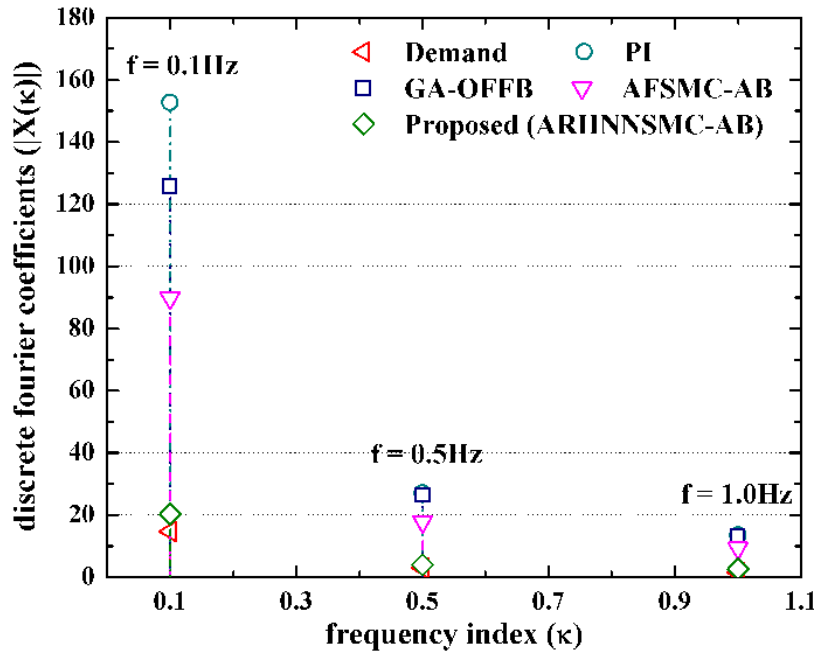


Figure 2.13: Spectral analysis using FFT of controller responses for 0.1Hz, 0.5Hz and 1Hz sinusoidal demand frequencies as shown in Fig. 2.10(a)

In order to represent this variation in response magnitude of each controller against the demand magnitude, a new performance ratio, Relative Error in Magnitude (**REM**), is defined in the form,

$$\mathbf{REM} = [(X_d(\kappa) - X_r(\kappa))/(X_d(\kappa))]_f \quad (2.24)$$

In (2.24), $X_d(\kappa)$ and $X_r(\kappa)$ represent the demand and response signal magnitudes obtained from **FFT** analysis at the κ^{th} frequency index of signal frequency f under investigation. The **REM** values for all the controllers at their respective demand frequencies are shown in Table 2.4, which seems to indicate that the lowest **REM** values (shown in bold) for all three frequencies are obtained for the proposed scheme. This suggests that the proposed scheme is an improvement over the established control schemes [9], [48] up to 1.0Hz in terms of control performance of the system.

Table 2.4: Comparison of REM values for four control configurations and three sinusoidal demand signals as shown in Fig. 2.10(a)

Demand Type	Freq (Hz)	Controllers (REM values)			
		PI	GA-OFFB	AFSMC-AB	ARHNN-SMC-AB (Proposed)
Sinusoid (0.04m ampl.)	0.1	9.338	7.509	5.080	0.372
	0.5	8.123	7.900	5.042	0.358
	1.0	8.113	7.904	5.285	0.740

2.4.5.3 Qualitative Analysis

The proposed **ARHNN-SMC-AB** controller complements the state-of-the-art techniques used for motion control of **EHAS** by bridging the gaps, some of which are mentioned below.

- *Study 1*: Bakhshande *et al.* reported a new approach for improving the closed loop tracking performance of a hydraulic cylinder utilizing a combination of well-known **SMC** method with a linear high-gain Proportional-Integral-Observer, which experimentally outperformed the proportional and **SMC** schemes for 0.05Hz reference sinusoidal signal [155].
- *Study 2*: Cheng *et al.* proposed a second order **SMC** scheme based on the fractional-order proportional integral derivative sliding surface and a sliding mode state observer for tracking control of electrohydraulic servo-systems, which shows more effectiveness and efficiency compared with **PID** and 2-**SMC** methods in simulations conducted with 0.2Hz sinusoidal signal [156].
- *Study 3*: Yang developed a novel nonlinear robust controller via backstepping technique utilizing two linear extended state observers in order to ensure

time-varying output constraints for a double-rod hydraulic actuator system, which exhibits better transient and steady state responses compared to industrial **PID** and high gain disturbance observer-based controllers in simulations pertaining to a 0.5Hz sinusoidal reference signal [157].

- *Study 4*: An adaptive nonlinear integral controller with anti-windup feature coupled with an input-linearized first-order **SMC** was proposed by Dasmahapatra *et al.* for motion control of an angular **EHAS**, which seems particularly suitable for high frequency applications by virtue of the experiments performed on a laboratory scale test rig with primarily sinusoidal demand signals within the frequency range of 1Hz to 10Hz [77].

The structure of our proposed scheme bears some similarity with the works reported in *Studies 1* and *2* in the sense that a robust **SMC** scheme is combined with an estimator or observer to achieve the desired performance for an electro-hydraulic system. However, this work proposes hybridization of **RHNN** with **ISMC** for a similar purpose and hence the adaptive laws developed in this work are completely different. The control structure employed in *Study 3* is much different from the proposed scheme and utilized backstepping technique. However in all *Studies 1*, *2*, and *3*, the usefulness was demonstrated using a single low frequency sinusoidal reference signal, of frequency 0.05Hz, 0.2Hz, and 0.5Hz respectively. On the other hand, our system has been successfully implemented for a varied range of demand waveforms (sinusoidal, square and trapezoidal), with the frequency range extending up to 1.0Hz. In case of *Study 4*, the control configuration also employed a variety of **SMC** design with adaptive features. However that design was employed for an angular **EHAS**, as opposed to the present solution proposed for linear **EHAS**, and hence the application areas and desired bandwidth are quite different for that application. While those experiments conducted in *Study 4* focused on robust performance for the high velocity regime beginning from 1.0Hz frequency, the present study focuses on achieving the same for the low velocity regime up to 1.0Hz.

These promising results are indicative of the fact that the universal approximation capability of **RHNN** can be successfully used in conjunction with an integral **SMC** scheme in motion control strategies of nonlinear dynamic systems for practical applications.

2.5 Summary

In this chapter, the development of an **RHNN** based control architecture synergized with an integral **SMC** has been described, with the intent of achieving

significant improvement in tracking performance for low cost industry grade rugged electrohydraulic systems which tend to suffer from a myriad of nonlinearities and uncertainties. Real-time experiments conducted on a representative single actuator test bench reveals that the proposed control scheme is capable of tracking demands within the frequency zone of 1.0Hz, which is notable in comparison to previous investigations [99], [152]. The contribution of the adaptive scheme designed for the **RHNN-ISM**C has also been established experimentally. A comparison of the performance measures and spectral analysis of the system responses for the proposed scheme with previously established methodologies (**GA-OFFB** & **AFSMC**) involving optimized feedforward components and integral sliding mode control with fuzzy estimators respectively [9], [48], suggest a significant improvement in performance. It is noteworthy that the proposed controller is able to achieve acceptable position tracking performance for various periodic demands in the low velocity regime of a dynamic nonlinear **EHAS** with much ease despite beginning the operation from the same initial values for the controller parameters, which evinces the robust nature of the proposed controller.

It should suffice to say that this chapter deals with the development of a neuro-sliding mode controller based on the feedback signals from a conventional **LVDT**. However, situations may arise which demand a non-attached position transducer for controlling the actuator motion. Such a scenario has been investigated in the upcoming chapter.

Visual Sensing Based Adaptive Sliding Mode Position Control

The widespread use of **EHAS** in industries has been thoroughly discussed in Chapter 1 and their closed-loop control aspects, involving a novel synergistic combination of **RNN** and **SMC** have been elaborated in Chapter 2. However, it is necessary to consider applications of **EHAS** with limited scope of employing sensors like **LVDT**, which needs to be attached with the actuator piston, in order to form a closed-loop system. Such a concern may arise in applications with space constraints for sensor fixtures or attachment issues of **LVDT** with the piston rod. In case of such applications, the use of non-attachment type sensors may play a major role in the implementation of **EHAS**. In this chapter, a vision sensor has been considered as a potential solution for addressing this problem and its feasibility for a linear **EHAS** has been investigated.

3.1 Background

The existing literature offers some insights into the use of vision guided motion tracking systems. For instance, Liu *et al.* suggested visual tracking of a moving target with a hydraulic robotic manipulator by means of a velocity and force control scheme. The real time experimental results of the study indicated that a high degree of stability and accuracy can be achieved for the manipulators by using vision integrated control approaches [158]. Not long after that, Firouzi *et al.* presented a vision based real-time object tracking system consisting of a **2-DoF** manipulator with a **CCD** camera mounted on its end effector. An image based vi-

sual servo control **IBVS** control method was used to estimate the distance and the motion of an object utilizing just the 2D target object position in the image plane [159]. Around the same time, González-Galván *et al.* developed a vision based calibration-free camera-space manipulation method for positioning and precise path tracking of industrial robots. They performed a comparative study to address the problem associated with resolution of the camera per unit of workspace, over large surfaces. This was done by employing two approaches, one involving multiple camera units and the other by applying cameras mounted over sensor less pan/tilt units [160]. Later, Hulke *et al.* proposed a single stationary camera based visual motion tracking scheme for minimally invasive surgery (**MIS**) using multiple markers attached to the laparoscopic tools. The authors used a camshift algorithm along with Kalman filtering techniques to achieve occlusion-free tracking. The markers were tracked in 3D by acquiring position and depth information based on variations in the position and size of the markers respectively, in image coordinates which were then converted to world coordinates using a calibration matrix. The authors reported a highly accurate tracking performance based on the experimental results of the scheme [161].

Motivated by the studies presented above and the notion of sliding mode control presented earlier in *Section 1.4.1*, the prime objective of this study is to develop a linear **EHAS** setup based on non-contact (or visual) sensing principles with the aid of an integral sliding mode control scheme owing to its robustness to parameter variations and external disturbances on the system. The control scheme is further strengthened by the addition of a deadband controller and the use of Lyapunov function based adaptive laws to update the dynamic (or free) parameters of the system, thus tackling the continuous variation of the parameters.

The rest of the chapter has been organised as follows. The formulation of the adaptive sliding mode based control scheme has been elaborated in *Section 3.2*. The developed control strategy is then employed on a vision enabled linear **EHAS**, whose workflow and specifications have been explained in *Section 3.3*. The results obtained from the real-time experiments conducted on the laboratory setup and the significant observations of this investigation are described in *Section 3.4*.

3.2 Adaptive First-order Sliding Mode Controller with Bias

Considering the n^{th} order, single-input-single-output (**SISO**), nonlinear system as represented by (2.1a) in *Chapter 2*, with $\mathbf{x}(t)$ as the state vector in (2.1b), $y(t)$ as the scalar output in (2.1c) and the scalar input $v(t)$. In order to formulate

the adaptive control scheme, the input-linearized-form of the system dynamics equation represented in (2.2) is again taken into account, such that,

$$\mathbf{x}^{(n)} = f(\mathbf{x}(t), t) + bu(\mathbf{x}(t), t) + d(\mathbf{x}(t), t) \quad (3.1)$$

where, $u(\mathbf{x}(t), t)$ represents the control variable of the system and $d(\mathbf{x}(t), t)$ stands for the disturbances affecting the system as described in *Section 1.2*. The control objective for designing the control variable u remains the same as in (2.3), whereby the error vector $\mathbf{e}(t)$ between the $\mathbf{x}(t)$ and the demanded vector, $\mathbf{x}^*(t)$ should asymptotically reach zero, expressed as,

$$\mathbf{e}(t) = \mathbf{x}^*(t) - \mathbf{x}(t) \rightarrow 0 \quad \text{as } t \rightarrow \infty \quad (3.2)$$

However, keeping in mind the arguments provided in *Section 2.2.2.1* regarding the impracticality of designing the ideal control law for the variable $u^*(t)$ as represented in (2.7b) to fulfill the above objective, it becomes obvious that a feasible estimate of the same may be found by designing a suitable control law, as mentioned earlier.

3.2.1 Proposed Control Scheme

In order to design a practical estimate of $u^*(t)$, a first-order sliding mode controller is proposed here owing to its robustness to deviations in system parameters and external disturbances, as explained in *Chapter 1*. In order to begin the construction of the proposed law, a modified form of the input-linearized system dynamic equation represented in (2.20) is considered, such that,

$$\ddot{x} = -\gamma\dot{x} + u + d \quad (3.3)$$

where, u is the control variable, d is the disturbance associated with the system and γ is assumed to be a positive quantity so as to stabilize the system dynamics. Since, the control objective as shown in (3.2) is to converge the system output x with the system demand x^* , the system dynamic equation in (3.3) can also be recast as,

$$\ddot{x}^* = -\gamma\dot{x}^* + u^* + d + \delta d \quad (3.4)$$

where δd is the additional disturbance emerging from the tracking difference, which would be resolved by the sliding mode controller. Then, by combining (3.3) and (3.4), and neglecting the effect of δd a second order error dynamics equation can be written in the form,

$$\ddot{e} = -\gamma\dot{e} + (u^* - u) \quad (3.5)$$

The error dynamics gets further reduced to a terminal form as u converges to u^* , yielding the equation,

$$\ddot{e} + \gamma\dot{e} = 0 \quad (3.6)$$

which clearly indicates an asymptotically stable error dynamics.

Based on these formulations, the control input in (3.3) can be configured as a summation of a *first-order sliding mode control* (or **1-SMC**) $u_{sm}(t)$ and a tan-hyperbolic deadband controller or bias $u_{db}(t)$. So,

$$u(t) = u_{sm}(t) + u_{db}(t) \quad (3.7)$$

The **1-SMC**, responsible for taking care of the disturbances and modelling approximations is expressed as,

$$u_{sm}(t) = -\eta_{sm} \text{sgn}(e(t)) \quad (3.8)$$

where, $\eta_{sm} > 0$ defines the confines of the SMC within which it may toggle ($\pm\eta_{sm}$). The tan-hyperbolic deadband controller or bias is introduced to compensate for the hard nonlinearities encountered in an **EHAS**, and is in the form,

$$u_{db} = u_{sat} \tanh(\alpha_b e(t)) + \chi \quad (3.9)$$

where, u_{sat} represents the saturation level of the controller and α_b is a large fixed parameter determining the rate at which the saturation level is approached. The second term consisting of the small positive number χ is used to overcome singularity during voltage extraction.

Since, the exact values of u cannot be determined for practical purposes, their estimate denoted by the hat symbol ($\hat{\cdot}$) over the variables, may be used to write the estimate of the proposed law in (3.7) in the form,

$$\hat{u}(t) = \hat{u}_{sm}(t) + \hat{u}_{db}(t) \quad (3.10)$$

where, $\hat{u}_{sm}$ and $\hat{u}_{db}$ are the estimates of the **1-SMC** and the deadband controller.

3.2.2 Formulation of Adaptive Laws

In order to form the adaptive structure, the estimated control variables in (3.10) may be used to express the desired control in such a form, where, Δ_{sm} and Δ_{db} are the first-order terms in Taylor series expansions of the **SMC** and deadband controls, respectively. Δ_h contains all the higher-order terms in the expansion and may be neglected for practical purposes. Also, $\hat{u}_{sm}$ and $\hat{u}_{db}$ are the estimates of

u_{sm} and u_{db} , respectively, as mentioned earlier, which are further formed by the estimates $\hat{\eta}_{sm}$ and $\hat{u}_{sat}$ for η_{sm} and u_{sat} in (3.8) and (3.9), respectively. This desired form of control can be expressed mathematically as,

$$u^* = \hat{u}_{sm} + \Delta_{sm} + \hat{u}_{db} + \Delta_{db} + \Delta_h \quad (3.11)$$

In order to evaluate the estimates $\hat{\eta}_{sm}$ and $\hat{u}_{sat}$ at various sampling instants, a method is assumed such that difference between the estimates and their respective convergence limits given by,

$$\tilde{\eta}_{sm} = \eta_{sm}^* - \hat{\eta}_{sm} \quad (3.12)$$

$$\tilde{u}_{sat} = u_{sat}^* - \hat{u}_{sat} \quad (3.13)$$

continually tend to zero. By combining (3.8) and (3.12), Δ_{sm} can be written as,

$$\Delta_{sm} = (\partial u_{sm} / \partial \eta_{sm})(\eta_{sm}^* - \hat{\eta}_{sm}) = \tilde{\eta}_{sm} \text{sgn}(e) \quad (3.14)$$

Similarly, combining (3.9) and (3.13) yields Δ_{db} in the form,

$$\Delta_{db} = (\partial u_{db} / \partial u_{sat})(u_{sat}^* - \hat{u}_{sat}) = \tilde{u}_{sat} \tanh(\alpha_b e) \quad (3.15)$$

Based on these formulations, the following theorem may be stated.

Theorem 3 *Based on the proposed control law in (3.10) for a class of nonlinear systems, the following adaptive laws will lead to bounded closed loop signals, such that the tracking error converges to zero asymptotically.*

$$\dot{\hat{\eta}}_{sm} = \sigma_{sm} \dot{e}(t) \text{sgn}(e(t)) \quad (3.16a)$$

$$\dot{\hat{u}}_{sat} = \sigma_{sat} \dot{e}(t) \tanh(\alpha_b e(t)) \quad (3.16b)$$

where, σ_{sm} and σ_{sat} are the positive learning rates.

Proof of Theorem 1: Let us consider a positive definite Lyapunov function for the system [9], [145], in order to design a convergent adaptive strategy in the form,

$$W = (\dot{e}^2 + \tilde{u}_{sat}^2 / \sigma_{sat} + \tilde{\eta}_{sm}^2 / \sigma_{sm}) / 2 \quad (3.17a)$$

Differentiating (3.17a) with respect to t and substituting (3.5), (3.9), (3.10), (3.11), (3.14) and (3.15), the following relation may be obtained,

$$\dot{W} = \dot{e}[-\gamma \dot{e} + \tilde{u}_{sat} \tanh(\alpha_b e) + \tilde{\eta}_{sm} \text{sgn}(e)] + \tilde{u}_{sat} \dot{\tilde{u}}_{sat} / \sigma_{sat} + \tilde{\eta}_{sm} \dot{\tilde{\eta}}_{sm} / \sigma_{sm} \quad (3.17b)$$

Further, by substituting (3.16a) and (3.16b) in (3.17b), leads to,

$$\dot{W} = -\gamma \dot{e}^2 \leq 0 \quad (3.17c)$$

Equation (3.17c) suggests that $\dot{W}$ is negative definite, since γ is a positive quantity. Hence, based on Lyapunov theorem it is quite clear that (3.17a) and (3.17c) together provide a convergent adaptation scheme with the estimates converging to their respective limits, which in turn guarantees a stable error dynamics based on the equation (3.6). Also, this ensures that the closed loop system signals are bounded and non-increasing.

3.3 Control of Real-time Electrohydraulic Actuation System

The control scheme in (3.10) along with its adaptive laws in (3.16a) and (3.16b), or in other words, the proposed Adaptive Sliding Mode Control with Deadband (**ASMCDB**) scheme is implemented on a vision enabled linear **EHAS** comprising a monocular CCD camera and a proportional valve (**PV**) driven single-rod actuator (**SRA**), for assessing its position tracking performance.

3.3.1 Vision enabled Linear EHAS Test Setup

The **EHAS** test setup used for conducting this particular experimental study in position control is similar to that presented in Fig. 2.2, with the addition of a ball shaped marker (**BSM**) and a monocular CCD camera (**CAM**) as a replacement for the position transducer (**LVDT**). The photographic view of this laboratory setup is shown in Fig. 3.1 with its component specifications listed in Table 3.1. The position of the piston rod is captured by the **CAM** which is kept orthogonal to the actuator plane in an *eye-on-hand* configuration and tracked with the assistance of the marker **BSM** placed on top of the Spring Holder cup (**HC**) as shown in the side view in Fig. 3.1. The *eye-on-hand* configuration of the **CAM** has been chosen since the rotational motion of the piston rod of the **SRA** prevents the **CAM** from being mounted on the end-effector in an *eye-in-hand* configuration. If the rotational motion were to be arrested mechanically, it may have resulted in breaking of the piston seals. Hence, an *eye-on-hand* configuration has been employed for this study.

The continuous video feed from the **CAM** of the side view of the Marker (**BSM**) as shown in Fig. 3.1, is used to track **BSM** in the image plane by employing machine vision algorithms. This tracking data can in turn be used to

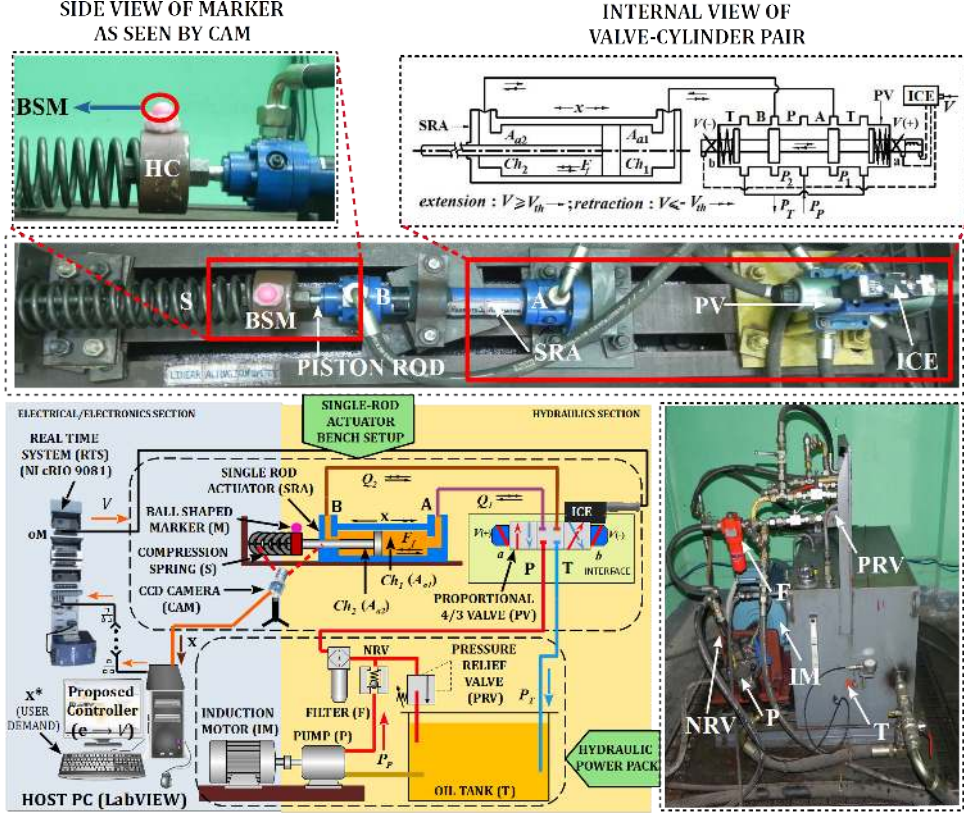


Figure 3.1: Laboratory scale electrohydraulic position tracking setup employing a visual sensor along with an internal view of valve-cylinder pair.

identify the actual position of the piston rod ($\mathbf{x}(t)$) in the real world units (meters) with suitable scaling. The proposed controller (**ASMCDB**) designed within the LabVIEW environment on the host PC for real time operation, computes a voltage signal $V(t)$ corresponding to the error in the piston rod position data ($\mathbf{e}(t)$) as represented in (3.2). This voltage is then continually fed to the **PV** via the output module of the real time system (**RTS**) ensuring closed-loop operation. The flow through the **PV** and the **SRA** along with the piston rod motion is depicted schematically within the inset attached to the figure where $\pm V_{th}$ represents the range of threshold value of the valve deadband voltage beyond which the piston motion begins. The flow paths for piston extension and retraction have been represented by single arrowhead lines and double arrow headed lines respectively within the inset. The operation of rest of the hydraulic circuit follows the principles described earlier in *Section 2.3.1*.

3.3.2 Implementation of Control Methodology on test setup

The control methodology for this study can be divided into two distinct stages following each other. The first stage deals with the machine vision algorithms used to track the target marker **BSM** and the second stage utilizes the error in

Table 3.1: Component specifications of the Vision enabled EHAS Test Setup given in Fig. 3.1

Component	Brand	Specifications
Double-acting single-rod actuators (SRA)	Rexroth	CD250D40/20-200A1X/01-CGDHT; Full Stroke: 0.200m; Bore diameter: 0.040m; Rod diameter: 0.020m
Proportional valve (PV)	Rexroth	4WREE-10-E50-23/G24K31/A1V with Integrated Control Electronics (ICE), Command Signal: $\pm 10V$
Charge-Coupled Device Camera (CAM)	Basler	ACE, ac A640-120uc, 120 fps
Ball Shaped Marker (BSM)	Local	Diameter: 2.96 cm; Color: Red
Compression spring (S)	Local	20kN/m Stiffness
Spring Holder Cup (HC)	Local	Inner diameter: 9 cm; Outer diameter: 10 cm; Length: 5.8 cm
Induction Motor (IM)	ABB	3-phase; 4-pole; 30 kW; 1470 rpm
Axial-Piston Pump (P)	Rexroth	A10VSO 45DR/31;60 lpm
Inline Filter (F)	Hydac	DFBN/HC30TB3A1.X/12-VB6; Working Pressure: 300 bar (max)
Pressure-relief Valve (PRV)	Yuken	EBG-03-H-11; 24.5MPa
Hydraulic Oil Tank (T)	Special Technique	Dimensions: 1.0m x 0.6m x 0.5m; Capacity: 200L
Real-time system (RTS)	National Instruments	NI-cRIO 9081 with 1.06GHz Intel Celeron U3405 & Spartan-6 LX75
Output-module (oM)	National Instruments	NI 9263 (4 Channel; $\pm 10V$ 16-bit simultaneous analog output)

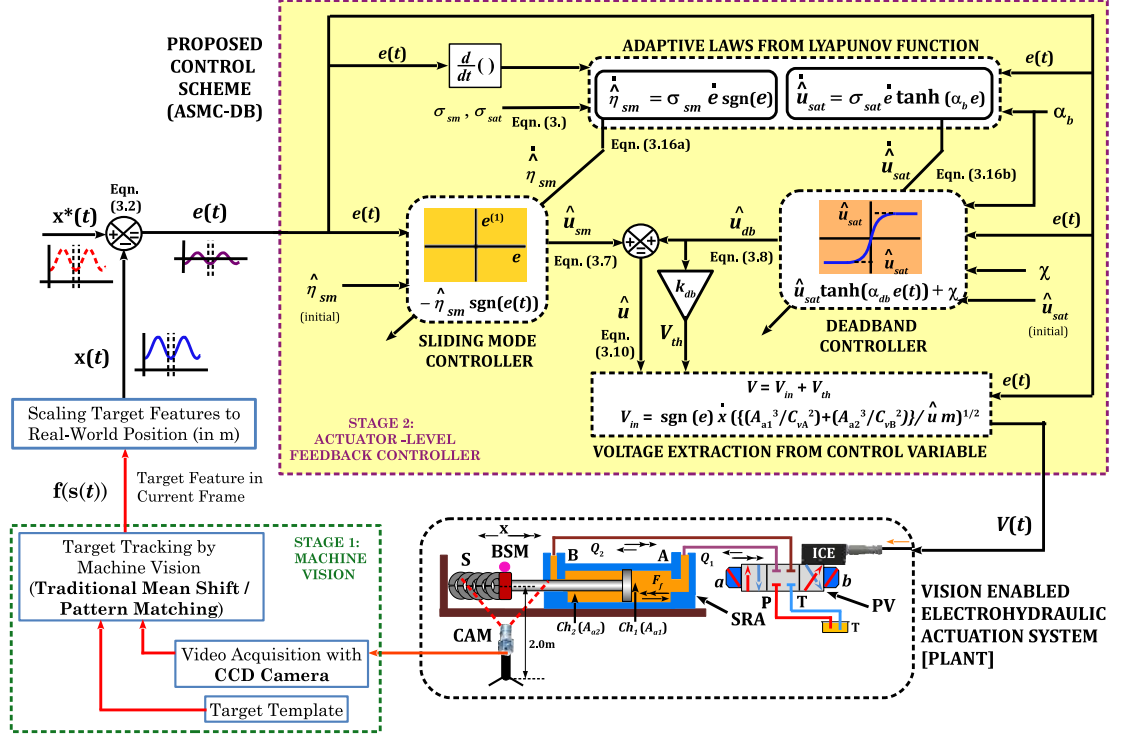


Figure 3.2: Schematic illustration of the proposed actuator level control architecture with a vision enabled **EHAS** test setup.

piston position in (3.2) with the proposed control scheme in (3.10) as an actuator-level feedback controller. The interplay amongst the stages has been shown in detail in Fig. 3.2.

3.3.2.1 Machine Vision Algorithm

The prime objective of this stage of the control methodology is to track the image feature coordinates (r, c) of the centroid of the target marker **BSM**, within the field of view of the camera **CAM**, where (r, c) represents the pixel row and pixel column in the image plane. A vector $\mathbf{s}(t)$ can be defined which represents the time-varying set of image coordinates (r, c) of the centroid of the target marker **BSM**. The current image feature vector, defined as $\mathbf{f}(\mathbf{s}(t))$, is detected with the aid of the **CAM**. In order to perform this tracking, two standard template-based machine vision algorithms are employed viz. the traditional mean-shift [162] and the geometric pattern matching as indicated in Fig. 3.2, which are deployed within the LabVIEW environment. It is observed during experiments that the mean-shift algorithm, a conventionally used tracking algorithm, is able to track the target efficiently and continuously for all image frames in a well-lit environment. However, in a dim-lit environment, the mean-shift based target tracking becomes skewed and discontinuous. For such a dim-lit setting, it is found that the geometric

pattern matching algorithm provides efficient tracking results and is comparatively easy to implement, compared to other available algorithms. Next, a suitable scaling operation is performed to convert the current image feature vector, $\mathbf{f}(\mathbf{s}(t))$, from pixel coordinates to real-world coordinates (in meters). This in turn provides the current position of the actuator piston, $\mathbf{x}(t)$.

3.3.2.2 Actuator-Level Feedback Controller

In this second stage of the control methodology, the current actuator position, $\mathbf{x}(t)$ obtained with the aid of the machine vision algorithms in the first stage is used to determine the position error, $\mathbf{e}(t)$ as described in (3.2). This position error is then utilized to form the control variable $\hat{u}(t)$ as proposed in (3.10), which can then be used in turn to extract the control voltage, $V(t)$ fed to the **PV** as shown in Fig. 3.2. To begin with, the vision enabled **EHAS** setup shown in Fig. 3.1 is considered as a second order nonlinear system ($n=2$), where the motion of the cylinder piston is expressed by the modified form of the motion dynamic equation of **SRA** of piston mass m as represented in (2.16) in the previous chapter, beginning from an initial position, x_0 , against the compression spring (**S**) of stiffness k with pre-compression δ . Based on equation (2.16), the proposed control scheme (**ASMCDDB**) may be implemented in real-time by following the couple of steps provided below.

- (a) *Formation of the control variable ($\hat{u}$):* The proposed control variable in (3.10) is formed by the summation of the **1-SMC** controller ($\hat{u}_{sm}$) and the dead-band controller ($\hat{u}_{db}$), which are in turn functions of the error signal ($\mathbf{e}(t)$) and the fixed parameters α_{db} and τ , which are assumed as positive quantities. The controller components also depend on the free parameters $\hat{\eta}_{sm}$ and $\hat{u}_{sat}$, for which adaptive laws have been formulated in (3.16a) and (3.16b) as functions of the error signal ($\mathbf{e}(t)$) and the rate of change of error ($\dot{\mathbf{e}}(t)$). The learning rates, σ_{sm} and σ_{sat} , for these laws are determined experimentally with an educated guess serving as the initial value of these free parameters.
- (b) *Extraction of Control Voltage (V) from ($\hat{u}$):* The control voltage (V) to be sent to the **PV** can be extracted from the variable $\hat{u}$ using the system equations as mentioned sequentially in **Step 2** of *Section 2.3.2*.
 1. *Actuator chamber pressures:* The chamber pressures P_1 and P_2 in (2.16) can be extracted from the pressure-discharge relation of the proportional valve [8], [9]. Assuming incompressible flow and negligible leakage, for the extension stroke of the actuator piston, the chamber pres-

tures are represented as,

$$P_1 = P_P - [(A_{a1}\dot{x})/\{C_{vA}(V - V_{th})\}]^2 \quad \text{for } V \geq V_{th} \quad (3.18a)$$

$$P_2 = P_T + [(A_{a2}\dot{x})/\{C_{vB}(V - V_{th})\}]^2 \quad \text{for } V \geq V_{th} \quad (3.18b)$$

For the retraction stroke of the actuator piston, they are formulated as,

$$P_1 = P_T + [(A_{a1}\dot{x})/\{C_{vA}(V + V_{th})\}]^2 \quad \text{for } V < V_{th} \quad (3.19a)$$

$$P_2 = P_P - [(A_{a2}\dot{x})/\{C_{vB}(V + V_{th})\}]^2 \quad \text{for } V < V_{th} \quad (3.19b)$$

where, C_{vA} and C_{vB} are the orifice coefficients of Ports A and B, respectively with $C_{vA} = 2C_{vB} = 2.19 \times 10^{-7} \text{ m}^3/(\text{s} \cdot V \sqrt{\text{Pa}})$, extracted from the flow-pressure conditions as mentioned in the **PV** specifications given in Table 3.1.

2. *Control Voltage*: Based on the modified form of the input-linearized system dynamic equation in (3.3), (2.16), (3.18a), (3.18b), (3.19a), (3.19b), the generalised form of the input linearized control (u) and the disturbance (d), can be represented mathematically as,

$$u = -\text{sgn}(V)(\dot{x}^2/m)\{(A_{a1}^3/C_{vA}^2) + (A_{a2}^3/C_{vB}^2)\}/\{V - \text{sgn}(V)V_{th}\}^2 \quad (3.20a)$$

$$\begin{aligned} d &= \frac{[k(x_0 - \delta) + P_P A_{a1} - P_T A_{a2} - F_f]}{m} + \gamma \dot{x} - \left(\frac{k}{m}\right)x \quad \text{for } V \geq V_{th} \\ &= \frac{[k(x_0 - \delta) + P_T A_{a1} - P_P A_{a2} - F_f]}{m} + \gamma \dot{x} - \left(\frac{k}{m}\right)x \quad \text{for } V < V_{th} \end{aligned} \quad (3.20b)$$

The control variable (V) in (3.20a) can be extracted from it in the form,

$$V = V_{in} + V_{th} \quad (3.20c)$$

where, the input linearized voltage term (V_{in}) has the form,

$$V_{in} = \text{sgn}(e)[\dot{x}^2\{(A_{a1}^3/C_{vA}^2) + (A_{a2}^3/C_{vB}^2)\}/(um)]^{1/2} \quad (3.20d)$$

and the threshold voltage (V_{th}) for deadband has the form,

$$V_{th} = k_{db}u_{db} \quad (3.20e)$$

with $k_{db} \approx 1V s^2/m$.

The execution of the both these steps involving the formation of $\hat{u}$ and the extraction of the control voltage (V) has been illustrated in detail in the schematic diagram of the entire control architecture in Fig. 3.2, along with relevant equation numbers. It is quite evident from Fig. 3.2 that the positioning of the **CAM** with respect to the piston rod and the marker (**BSM**) also plays a crucial role in the efficient working of the system. Hence, this factor should be kept in mind, before investigating the performance of the system.

3.4 Results and Discussion

3.4.1 Free and Fixed Parameters of the System

The parameter vector of the system can be defined as,

$$\mathbf{p} = [\hat{u}_{sat} \quad \hat{\eta}_{sm} \quad \alpha_b \quad \tau \quad \sigma_{sm} \quad \sigma_{sat}]^T \quad (3.21a)$$

with $\hat{u}_{sat}$ and $\hat{\eta}_{sm}$ denoting the initial values of these free or adaptive parameters, whereas, α_b , τ , σ_{sm} and σ_{sat} represent the values set for these fixed parameters. The nominal settings for the parameter vector ($\mathbf{p}$) along with their units are chosen as,

$$\mathbf{p} = [2\text{m/s}^2 \quad 150\text{m/s}^2 \quad 140\text{m}^{-1} \quad 0.1\text{m/s}^2 \quad 0.5\text{s}^{-1} \quad 2.5\text{s}^{-1}]^T \quad (3.21b)$$

The value of $\hat{u}_{sat}$ is chosen as 2m/s^2 keeping in the mind the 10% deadband and the uncertainties present in the **EHAS**. The initial value of $\hat{\eta}_{sm}$ and the values of α_{db} , σ_{sm} and σ_{sat} are chosen based on experimental observations. The value of τ is chosen as 0.1m/s^2 considering a small positive value for overcoming the singularity in (3.20d).

3.4.2 Variation of System Response based on Sensor Type

In order to assess the effectiveness of a visual sensor, a comparison study is performed between the closed loop **EHAS** responses for a standard demand signal utilizing an in-line position transducer (**LVDT**) and a monocular visual sensor (**CAM**), separately. The sinusoidal demand signal of 0.5Hz is used as the test signal with a conventional **PI** controller.

The study is conducted for two different cases. In the first case, the orthogonal **CAM-to-BSM** distance is fixed at 1.0m and demand amplitude at 0.03m. In the next case, the **CAM-to-BSM** distance is 2.0m and the demand amplitude is chosen to be 0.04m. However, in both the cases, the responses for the **LVDT** and

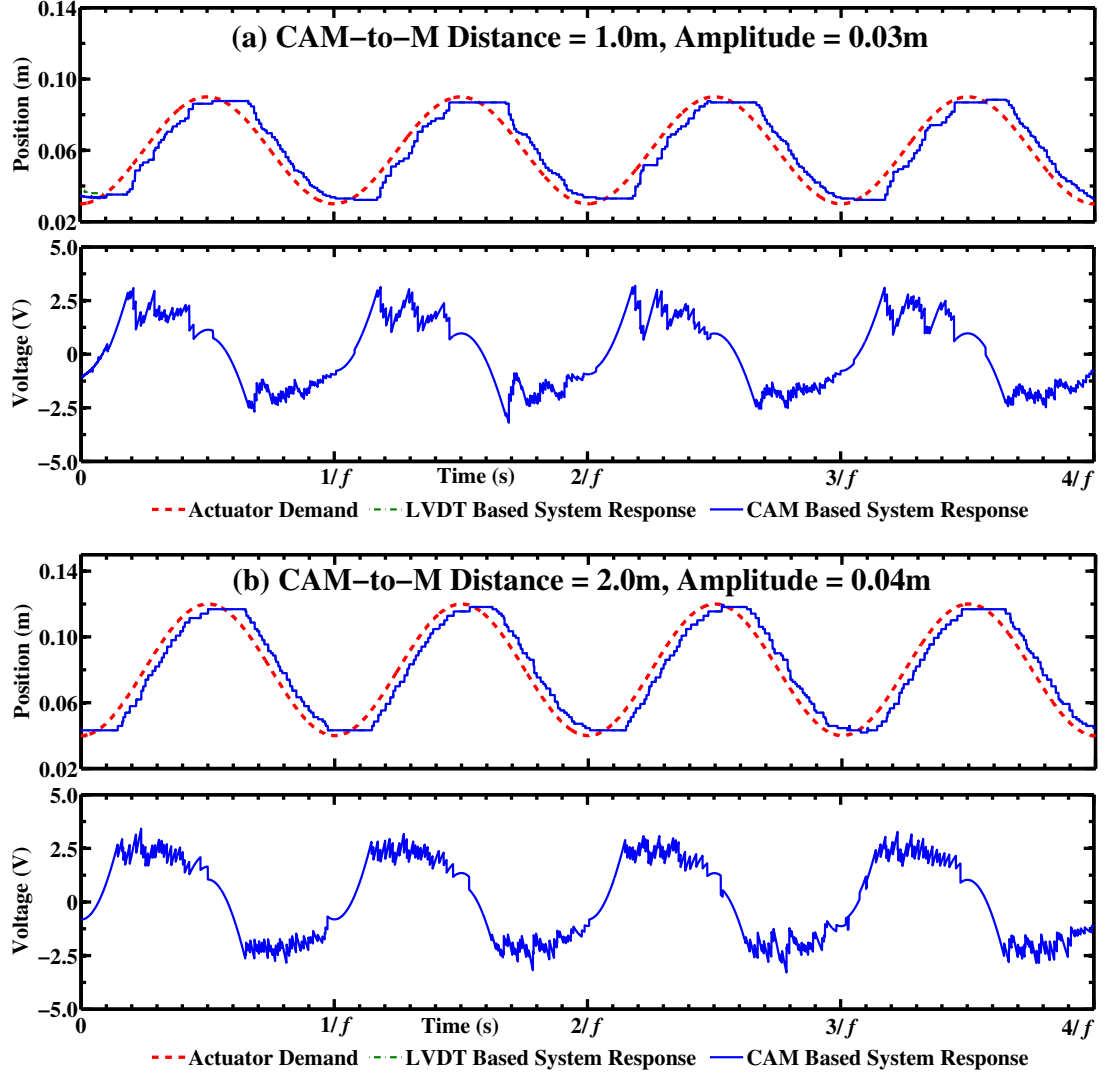


Figure 3.3: Comparison of System Responses for in-line position transducer (LVDT) and visual sensor (CAM) for varying orthogonal distances and demand amplitude.

the **CAM** nearly overlap as shown in Fig. 3.3. It may be convincingly concluded from this study that for an **EHAS**, visual sensors may be readily used instead of contact sensors.

3.4.3 Performance of the Proposed Control Scheme

The performance of the proposed controller (**ASMCDB**) has been studied for three conventional demand types and their results have been compared with those of a conventionally tuned **PI** controller under similar excitations. The fixed proportional and integral gain parameters of the **PI** controller are tuned at $k_P = 175\text{V/m}$ and $k_I = 90\text{V}/(\text{m.s})$, respectively [139]. The axial piston pump (**P**) used for driving the hydraulic fluid through the circuit is kept in the fixed displacement mode and the pressure relief valve (**PRV**) is set at 11MPa during the entirety of

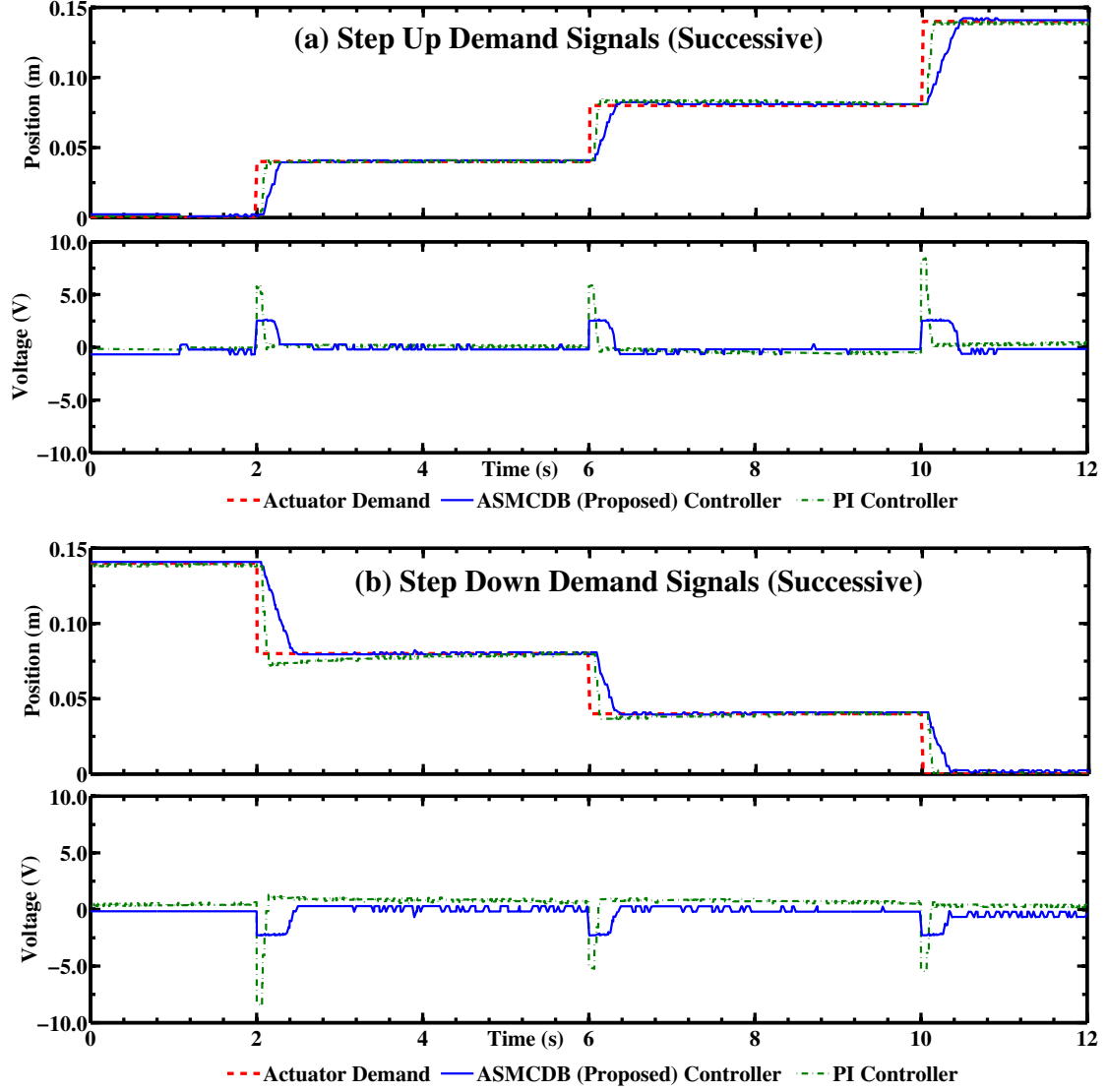


Figure 3.4: Comparison of System responses of the **PI** controller with the proposed controller (**ASMCDB**) for successive (a) Step Up Demands (b) Step Down Demands.

these experiments. The results obtained from these experiments are presented in Fig. 3.4 to Fig. 3.6.

1. **Step Demands:** The response of the vision enabled **EHAS** test setup for three consecutive step up and step down demands each with a duration of 4 seconds and amplitude of 40mm, 40mm and 60mm respectively in Fig. 3.4. The position response indicates that the proposed controller has lower overshoot and slightly higher rise time as compared with the **PI** controller. The presence of voltage oscillations can be attributed to the limited processing speed of the **CAM** being used. Despite this constraint, the proposed controller shows good performance as compared to the **PI** controller, which displayed comparatively more oscillations than **ASMCDB**.

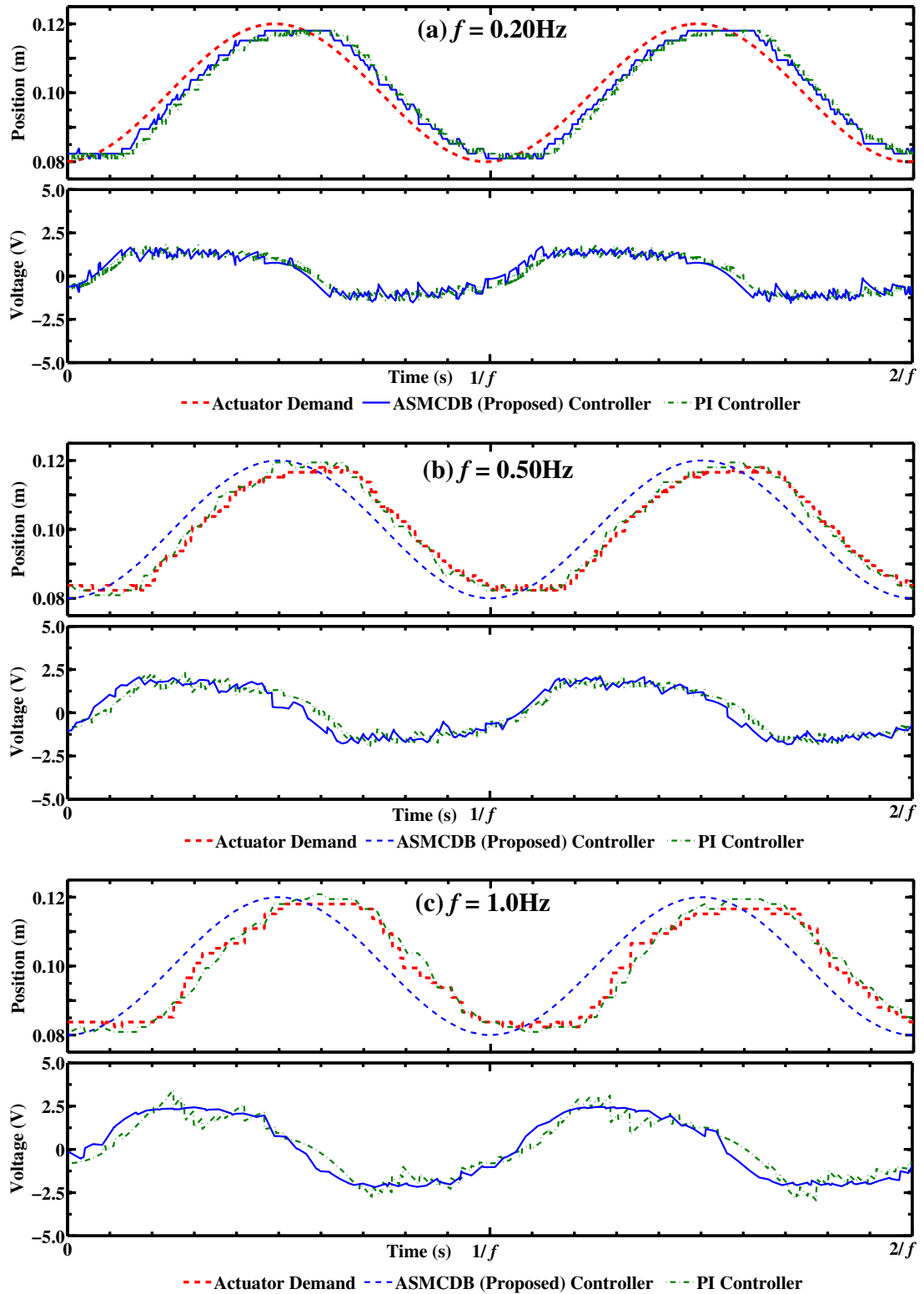


Figure 3.5: Comparison of System responses of the **PI** controller with the proposed controller (**ASMCDB**) for Sinusoidal demand signals of amplitude 0.04m and fundamental frequencies of (a) 0.2Hz (b) 0.5Hz (c) 1.0Hz

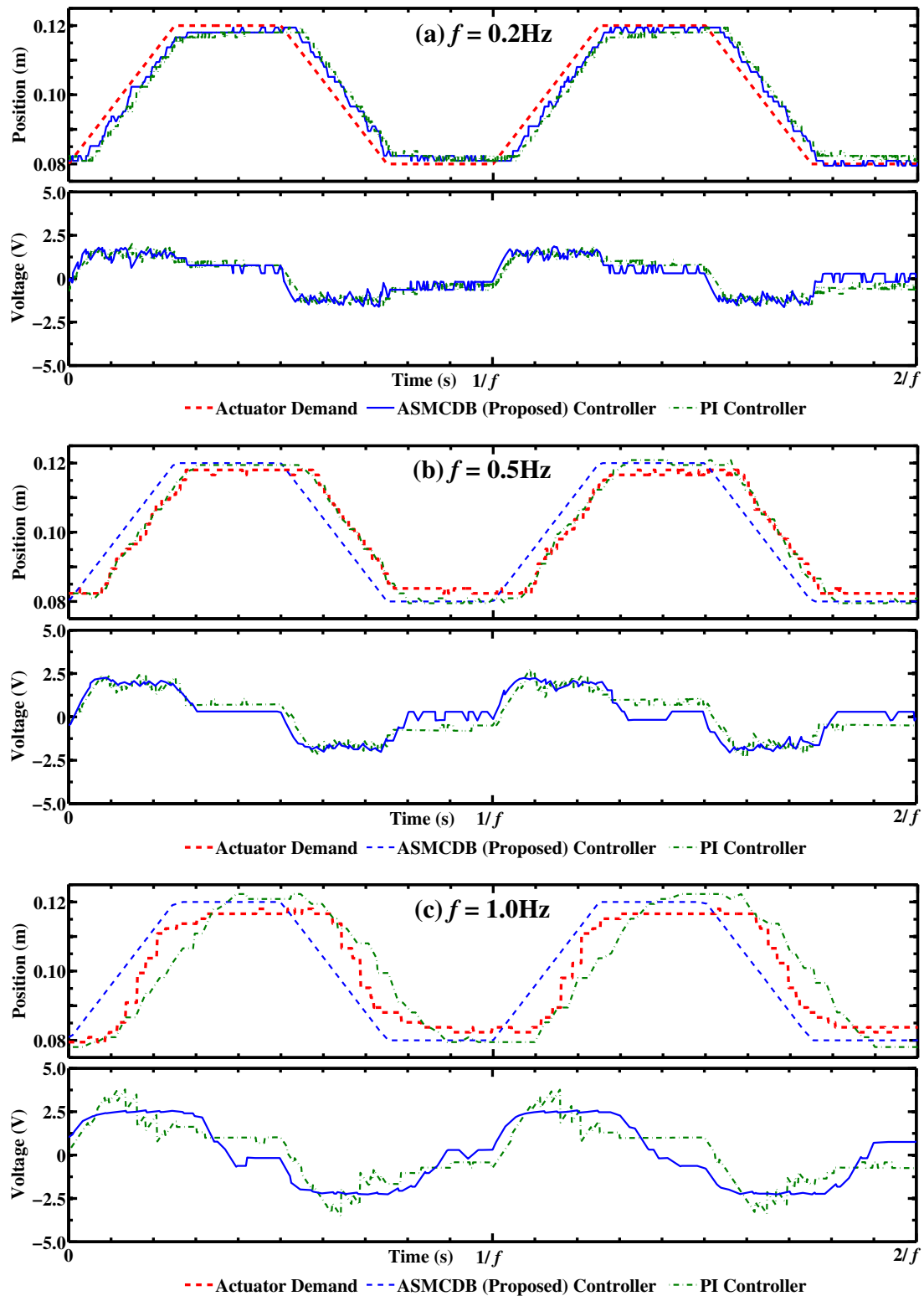


Figure 3.6: Comparison of System responses of the **PI** controller with the proposed controller (**ASMCDB**) for Trapezoidal demand signals of amplitude 0.04m and fundamental frequencies of (a) 0.2Hz (b) 0.5Hz (c) 1.0Hz

2. **Sinusoidal Demands:** The test setup is subjected to tracking demands of sinusoidal nature of peak-to-peak amplitude 0.04m and frequencies of 2Hz, 0.5Hz and 1.0Hz, respectively. The system response and control voltages for these demand signals for both the proposed controller (**ASMCD**) and the **PI** controller are shown in Fig. 3.5. It is quite obvious from the figure that increase in the demand frequency results in a phase shift in the response. However, it can also be noted that the amplitude attenuation observed for the **PI** response has been satisfactorily reduced by **ASMCD** controller.
3. **Trapezoidal Demands:** Several tracking demands of trapezoidal nature with peak-to-peak amplitude of 0.04m and frequencies of 0.2Hz, 0.5Hz and 1.0Hz, are also used to test the performance of the system. Fig. 3.6 shows a comparison of system responses and control voltages for all these demands with both the proposed controller (**ASMCD**) and the **PI** controller. It can be observed from the figure that amplitude attenuation begins to occur for higher frequencies with the **PI** Controller, whereas, the proposed controller (**ASMCD**) seems to have taken care of this problem despite the presence of certain phase shift.

3.4.4 Performance Analysis

- (a) *Regulation Demands:* In order to quantitatively evaluate the performance of the proposed controller compared to the **PI** controller, for regulation or step demands shown in Fig. 3.4, performance indices may be calculated for them as defined in (2.22) of Section 2.4.4. The metrics considered for this evaluation are the **IAE**, **ITAE** and **CE**. Normalization of these metrics are not necessary since aperiodic motions are considered. In this case, these metrics are evaluated as a whole for the successive step-up and step down scenarios shown in Fig. 3.4 and tabulated in Table 3.2. The performance indices for the successive step demands show that the values of the proposed **ASMCD** controller are lower by one order of magnitude compared to those of the **PI** controller, thus establishing the robustness of the **ASMCD** controller.
- (b) *Tracking Demands:* In order to evaluate the real-time tracking performance of the **ASMCD** Controller for the various periodic demands *viz.* Sinusoidal and Trapezoidal, as shown in Figs. 3.5 and 3.6, the normalised performance metrics in (2.23), namely, Normalised Discrete Absolute Error (**NDAE**), Normalised Discrete Time-weighted Absolute Error (**NDTAE**) and Normalised Control Effort (**NCE**) are calculated. As mentioned in Sec-

Table 3.2: Comparison of performance metrics for ASMCDB and PI controller based system responses for successive step-up and step-down demands in Fig. 3.4

Demand Type	Controllers	Performance Indices		
		IAE (ms)	ITAE (ms ²)	CE (V ² s)
Step-Up (Successive)	PI	0.0182	0.1250	7.3746
	ASMCDB	0.0047	0.0272	0.8318
Step-Down (Successive)	PI	0.0199	0.3416	9.9968
	ASMCDB	0.0044	0.0773	0.7209

Table 3.3: Comparison of performance metrics for ASMCDB and PI controller based system responses for sinusoidal and trapezoidal demands in Fig. 3.5 & Fig. 3.6

Demand Type	Freq (Hz)	Controllers (Performance Indices)	
		PI	ASMCDB (Proposed)
NDAE values			
Sinusoid (0.04m Amplitude)	0.2	0.0294	0.0035
	0.5	0.0368	0.0048
	1.0	0.0444	0.0080
Trapezoid (0.04m Amplitude)	0.2	0.0277	0.0030
	0.5	0.0345	0.0044
	1.0	0.0442	0.0097
NDTAE values			
Sinusoid (0.04m Amplitude)	0.2	0.1103	0.0130
	0.5	0.6182	0.0807
	1.0	1.7675	0.3195
Trapezoid (0.04m Amplitude)	0.2	0.1025	0.0109
	0.5	0.5760	0.0729
	1.0	1.7562	0.3856
NCE values			
Sinusoid (0.04m Amplitude)	0.2	0.0152	0.0024
	0.5	0.0237	0.0039
	1.0	0.0371	0.0068
Trapezoid (0.04m Amplitude)	0.2	0.0146	0.0022
	0.5	0.0232	0.0038
	1.0	0.0395	0.0069

tion 2.4.4, these normalised form of the metrics are necessary to obtain frequency and amplitude independent performance values. Based on the system specifications in Table 3.1, the maximum available stroke of the **SRA**(s_{max}) and the maximum voltage to the **PV**(V_{max}) are 0.20m and $\pm 10V$, respectively for performing the normalisation operation of these metrics. These normalised performance metrics are evaluated and tabulated in Table 3.3, for the respective time windows of the sinusoidal and trapezoidal demand types as shown in Figs. 3.5 and 3.6. It is quite obvious from the comparison of the performance metrics presented in Table 3.3 that the proposed **ASMCDB** scheme exhibits lower values by almost an order of 10, for all three metrics compared to those of the **PI** controller, as shown in bold. Hence, it can be concluded from these metrics that the performance of the proposed controller is considerably better than a conventional **PI** controller for the tracking demands considered in this study. The performance of the proposed controller up to 1.0Hz is also significant considering the fact that the frame rate of the vision sensor limits the data acquisition to around 100 samples/sec which in turn restricts the system bandwidth.

3.5 Summary

In this chapter, a position control scheme for a vision enabled linear **EHAS** has been described utilizing a combination of first-order sliding mode (**1-SMC**) controller with a tan-hyperbolic deadband controller. A monocular CCD camera placed orthogonal to the plane of the piston motion has been used as the motion sensor in place of a conventional contact type sensor, in order to demonstrate its effectiveness in scenarios with design constraints. The piston motion is tracked by marker-based machine vision algorithms with a ball-shaped marker placed (**BSM**) on the spring holding cup. A convergent adaptive strategy based on a Lyapunov function has also been established in order to make the controller robust to changes in demand. The performance of the controlled is assessed in comparison to a tuned **PI** controller based on several performance metrics. The performance metrics for regulation and conventional tracking demands used for the investigation, show a lower value compared to the **PI** controller, which in turn indicates an improvement in performance.

It is quite evident from this chapter that a vision sensing apparatus or camera can be effectively used with a **1-DoF** electrohydraulic system for closed-loop operation. However, it is also possible to extend this symbiotic relationship to a structure with higher degrees of freedom and in turn, develop a motion sensing system. This possibility has been explored in the succeeding chapter.

Development of a Vision based Motion Sensing System

This chapter, the background of the motion control aspects of the **SP** has been presented. This is followed by a detailed description of the real-time motion sensing system and its components. The control objective and the employed methodology involving five separate stages, have been briefly described thereafter. Discussions regarding the results of the real-time experiments conducted by employing the control methodology to the motion sensing system, involving several applicable object motion trajectories has been presented at the end.

4.1 Background

The paradigm of object tracking in computer vision has been widely researched over the past few decades, emphasizing its role in aerospace, medical and military applications with numerous tracking algorithms proposed by researchers in recent times [163]–[167]. A paradigm shift occurred, when the field of visual servoing emerged in robotics as a consequence of utilizing visual information from computer vision algorithms in dynamic feedback loops to control robotic systems [105], [112], [123], [124], [168]–[172]. Most visual servoing applications available in literature utilize the capability of serial robots or manipulators, whereas, parallel manipulators are seldom utilized for visual servoing tasks owing to their restricted workspace [42], [131]. Although, parallel manipulators have actually been in use across industrial & research fields for over five decades, since their inception by Gough [173] as a Universal tire-testing machine and subsequent academic popularization by Stewart [174] as a 6-**DoF** flight simulator capable of generating any generalized motion in 3D space. The most commonly used parallel manipulator, the Stewart Platform, has been utilized as an effective vibration isolator [175], in

machine tools applications [176], [177], in force and torque sensing applications [178], [179] and even as a workbench for 3D printing[180], primarily owing to their high load carrying capacity, decent dynamic performance and precision positioning capability due to non-cumulative nature of the joint errors [131], [181].

However, in a chapter of the Handbook of Robotics [182], Chaumette *et al.* points out that accurate positioning of a parallel manipulator by visual servoing may have potential applications. It is well known from literature that amongst the two commonly used servoing principles, namely, position-based and image-based servoing; the former is highly sensitive to camera calibration errors, has poor robustness of pose estimation and requires accurate 3D target model [132], thus, diminishing the positioning accuracy of the system. Depending on this fact, prior research on the topic seems to have diverged into two different approaches. In one approach, the researchers [134], [135] used an eye-on-hand configuration for vision-based control of parallel manipulators by observing the position of its legs, which revealed that this approach may be suitable for high-speed tasks. In another approach, Qi & McInroy proposed [132], [136], [137] the use of parallel robot based pointing systems for position regulation, utilizing eye-in-hand image-based visual servoing (**IBVS**). Simulation studies of the proposed system demonstrated that such systems are robust and can handle large initial feature errors.

Based on such facts and following the works presented above, the objective of the present work is to implement a real-time experimental setup involving a 6-Spherical-Prismatic-Spherical (**SPS**) configured electrohydraulic Stewart platform with an eye-in-hand camera configuration as shown in Fig.4.1. It is intended to work as a motion sensing and trajectory replicating system for moving objects, utilizing a dynamic image based look-and-move visual servoing scheme. At this point, it is curious to note that such a study, involving a hydraulic setup, seems to be absent from literature, to the best of my knowledge. The system is designed with hydraulic components in order to exploit the advantages offered by them, as previously mentioned in *Chapter 1*, especially their accurate positioning capability, fast response time and speed reversal capabilities. Therefore, the novelty of this study lies in the fact that it serves, not only as a proof of concept extended by prior research[136], [137], but also evaluates the capability of such electrohydraulic systems for purposes of real-time application.

The rest of the chapter has been organised in the stated manner. The real-time motion sensing system and its components has been described in detail in Section 4.2. The control objective and the employed methodology involving five separate stages, have been explained in Section 4.3. Section 4.4 discusses the results of the real-time experiments conducted by employing the control methodology to the motion sensing system, involving several applicable object motion trajectories.

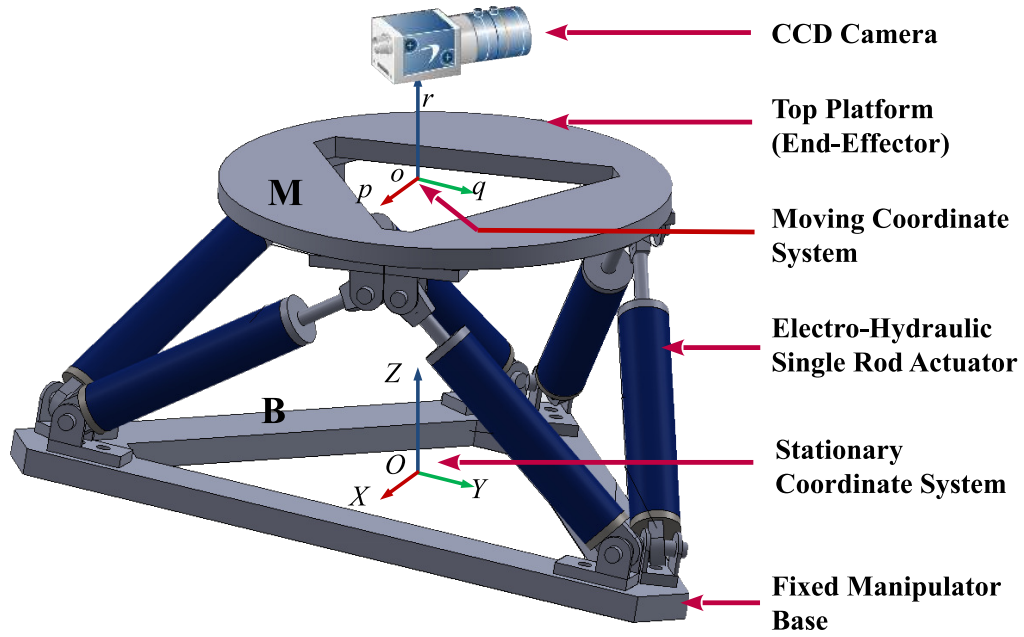


Figure 4.1: Illustration of 6-DoF Electrohydraulic Parallel Manipulator with eye-in-hand monocular camera configuration.

4.2 The Real-Time Motion Sensing System

The experimental motion sensing system devised for this study can be segregated into two sections: (i) Parallel Manipulator Based Servoing System, and (ii) The Virtual Moving Target System; each of which are discussed below in brief.

4.2.1 Parallel Manipulator based Servoing System (PMBSS)

The visual servoing aspect of the study is accomplished with the aid of a laboratory scale six-legged electrohydraulic parallel manipulator or Stewart platform as represented in Fig.4.1. Hydraulic single-rod actuators **SRA** driven by electronically enabled proportional valves (**PV**) and linear variable differential transformers (**LVDT**) as in-line position transducers, form the six legs of the Stewart Platform **SP**. These legs are affixed to a stationary semi-triangular base platform **B** at the base and to a circular moving platform **M** at the top. The top platform **M** acts as the end-effector of the manipulator. A monocular camera (**C**) is mounted on the transverse plane atop the end-effector following the traditional eye-in-hand approach, at a height of 450 mm from the origin o of the moving platform **M**, keeping in mind, the approximate height of the moving test target. The six hydraulic actuators are supplied with pressurised oil from the supply manifold (**SM**) placed centrally around origin O of the stationary base **B**. The **SM** is in turn supplied via an external hydraulic power pack consisting of a 3-phase induction motor **IM** driven axial piston pump **P** with pressure compensator [153], connected to a

200L hydraulic oil tank **T** and accompanied by accessories like the pressure-relief valve **PRV**, Inline filter **F** and and non-return valves (**NRV**) as shown in Fig.4.2.

The pose of the moving platform **M** is dictated by the combined motion of the six legs, achieved by the extension or retraction of the six actuator piston rods. The position of the piston rod of each actuator obtained via the in-line position sensor **LVDT**, is continuously fed to a controller designed within the LabVIEW environment on a host PC with the aid of the input modules **iM** of a real-time data acquisition system **RTS**. The controller, in turn, generates distinct voltages V_i ($i = 1$ to 6) for each actuator-valve pair based on the error in the current position reading of the piston rods and the present demand of actuator positions, determined by the desired pose of the end-effector (**q**). These voltages are then fed to their respective proportional valves **PV** through the output modules **oM** of the **RTS**. This leads to spool movement within the proportional valves, which in turn causes high pressure oil to flow to the rod end (Ch_1) or cap end (Ch_2) of the actuators, resulting in extension or retraction of the piston rods, denoted by single arrowhead and double-arrowhead lines, respectively around the actuator rod illustration in Fig. 4.2. A temporal pattern of such piston rod movements for all the six legs, generates the desired pose (**q**) of the moving platform.

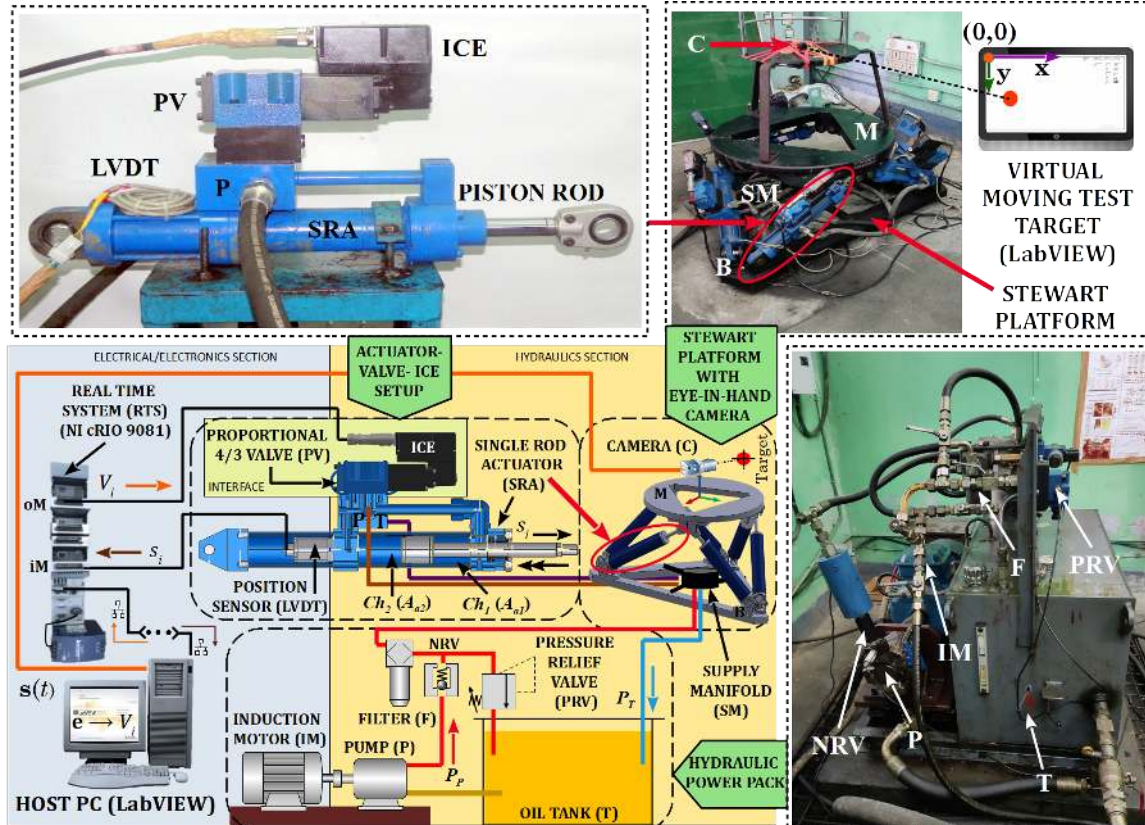


Figure 4.2: Laboratory scale electrohydraulic parallel manipulator based servoing system (PMBSS) with a virtual moving test target setup.

Table 4.1: Component specifications of the laboratory scale Parallel Manipulator based Servoing System

Component	Brand	Specifications
Stewart Platform structure (SP)	Special Technique	System weight: 700kg; Top platform weight: 250kg; Supply Manifold (SM): 1P, 1T, 12 C ports
Monocular CCD Camera (C)	Basler	ACE; acA640-120uc; 120 fps
Double-acting single-rod actuators (SRA)	Eaton-Vickers	ST52/E355/080M40/EN8; Full Stroke: 0.150m; Bore diameter: 0.050m; Rod diameter: 0.028m; Closed Length: 0.570m
Proportional valve (PV)	Eaton-Vickers	KBSDG4V-5-92L-50-M1-PE7-H7-11 with Integrated Control Electronics (ICE), Zero underlap, Command Signal: $\pm 10V$
Linear variable differential transformer (LVD T)	MTS Temposonics	0-0.15m range with 0.5-10V analog output; Resolution: 16bit-0.0015% (Min. $10\mu m$)
Induction Motor (IM)	ABB	3GBA201044-ADEIN; 3-phase; 2-pole; 37 kW; 2945 rpm
Axial-Piston Pump (P)	Rexroth	3A10VSO/32 ; 280 bar; 60 lpm
Inline Filter (F)	Stauff	SF045B/TB/B; Working Pressure: 420 bar (max)
Pressure-relief Valve (PRV)	Rexroth	DB10-1-41/315-IN001; Max. Pressure - 315 bar
Hydraulic Oil Tank (T)	Special Technique	Dimensions: 1.0m x 0.6m x 0.5m; Capacity: 200L
Real-time system (RTS)	National Instruments	NI-cRIO 9081 with 1.06GHz Intel Celeron U3405 & Spartan-6 LX75
Input-module (iM)	National Instruments	NI 9215 (4 Channel; $\pm 10V$ 16-bit simultaneous analog input)
Output-module (oM)	National Instruments	NI 9263 (4 Channel; $\pm 10V$ 16-bit simultaneous analog output)

The desired pose of the moving platform, however, is not defined by a user for the parallel manipulator based servoing system **PMBSS**. Instead, the desired pose at any time instant is automatically determined by the machine vision and visual servoing algorithms, from the motion of a virtual moving test target (**TT**) captured by the monocular camera **C** mounted on the moving platform. Hence, the **PMBSS** is classified as a dynamic look-and-move end-point open loop system, according to visual servoing terminology [124], [168]–[170]. The specifications of the major components of this system are enlisted in Table 4.1.

4.2.2 Virtual Moving Test Target System

The test target (**TT**) to be tracked by the **PMBSS**, is rendered on another PC monitor with the aid of the LabVIEW environment, in the form of a red circle of diameter 100mm on an otherwise white background. The upper left corner of the monitor is considered as the origin as shown in Fig. 4.2. The monitor sits at a height of 800mm from the ground and at a depth Z from the eye-in-hand camera which may be varied for carrying out parametric studies. The virtual target can be programmed in LabVIEW to follow various 1D and 2D trajectories on the PC monitor with much accuracy.

4.3 Control Methodology

The control objective of this real-time motion sensing system is to minimize the image pixel error $\mathbf{e}(t)$ between the current image feature of the test target $\mathbf{f}(\mathbf{s}(t))$ as detected with the aid of the monocular camera **C** and the desired image feature $\mathbf{f}^*(\mathbf{s}_c)$, by moving the end-effector **M** of the parallel manipulator in the direction of target motion. This essentially leads the end-effector to continuously follow the trajectory traced by the test target on a plane parallel to the frontal plane of the camera in real-time. The vector $\mathbf{s}(t)$ represents the time-varying set of image coordinates (r, c) of the centroid of the test target. The vector $\mathbf{s}_c$ represents the coordinates of the image center (r_c, c_c) . In mathematical terms, the control objective can be stated as,

$$\mathbf{e}(t) = (\mathbf{f}(\mathbf{s}(t)) - \mathbf{f}^*(\mathbf{s}_c)) \rightarrow 0 \quad \text{as } t \rightarrow \infty \quad (4.1)$$

In order to achieve this, a multi-faceted control methodology is employed, involving five distinct stages which may be hierarchically organized as: Geometric Camera Calibration, Machine Vision Algorithm, Image-based Visual Servoing Scheme, Inverse Kinematics of the parallel manipulator and Joint-level feedback control. A detailed view of the interplay amongst these stages has been shown in Fig. 4.3.

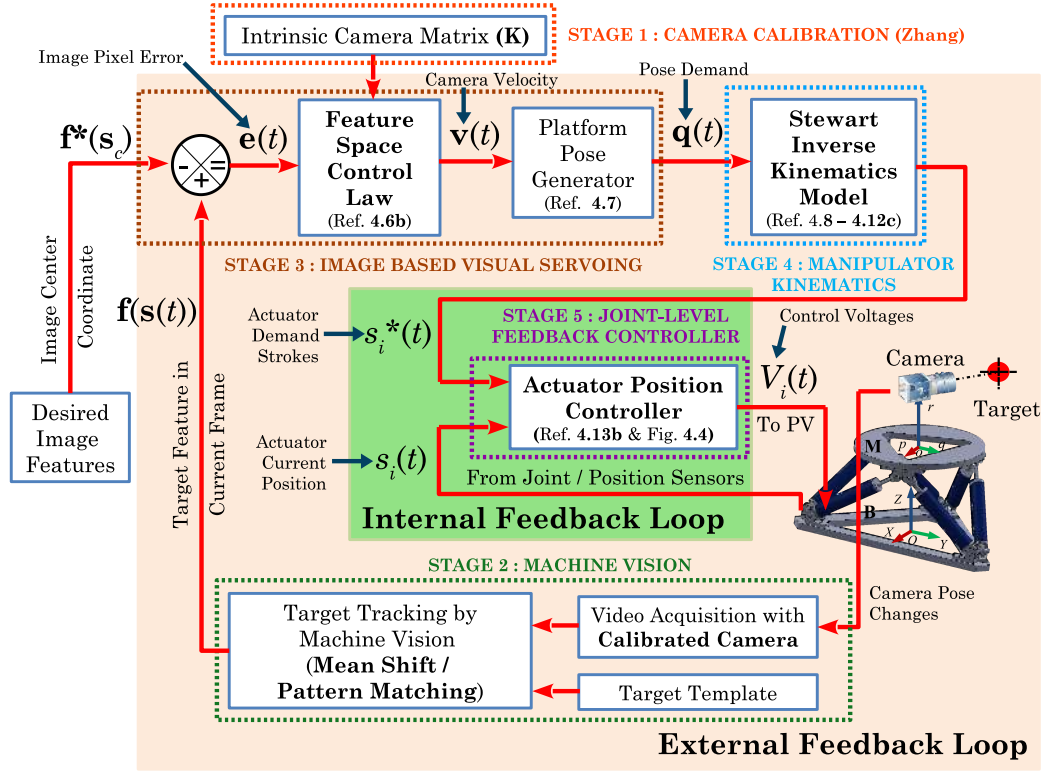


Figure 4.3: Image-based Visual Servo Control architecture of an eye-in-hand Stewart Platform for tracking moving objects

4.3.1 Geometric Camera Calibration

The geometric camera calibration is performed to estimate the parameters of the lens and image sensor, which are necessary to predict the image coordinates (r, c) of the projection of a point within the field of view of the camera $\mathbf{C}$, given that the coordinates of that point with respect to the world coordinate frame are known. Hence, a calibration algorithm is used to calculate: (i) intrinsic camera matrix $\mathbf{K}$ and, (ii) radial and tangential lens distortion. For the purpose of our study, we used Zhang's calibration algorithm [183], which is the most commonly used camera calibration method available in **MATLAB**, as indicated in Fig. 4.3. Images from one such calibration session is shown in Fig. 4.4. Based on this calibration, the intrinsic camera matrix for the Basler ACE CCD camera (indicated in Table 4.1), comprising of five *intrinsic* parameters, namely, the optical center (Principal Point) in pixels (O_r, O_c) , the focal length in pixels (f_x, f_y) and the skew coefficient (s_c) . is found to be,

$$\mathbf{K} = \begin{pmatrix} f_x & 0 & 0 \\ s_c & f_y & 0 \\ O_r & O_c & 1 \end{pmatrix} = \begin{pmatrix} 1382 & 0 & 0 \\ -2.67 & 1384 & 0 \\ 325.43 & 329.95 & 1 \end{pmatrix} \quad (4.2)$$

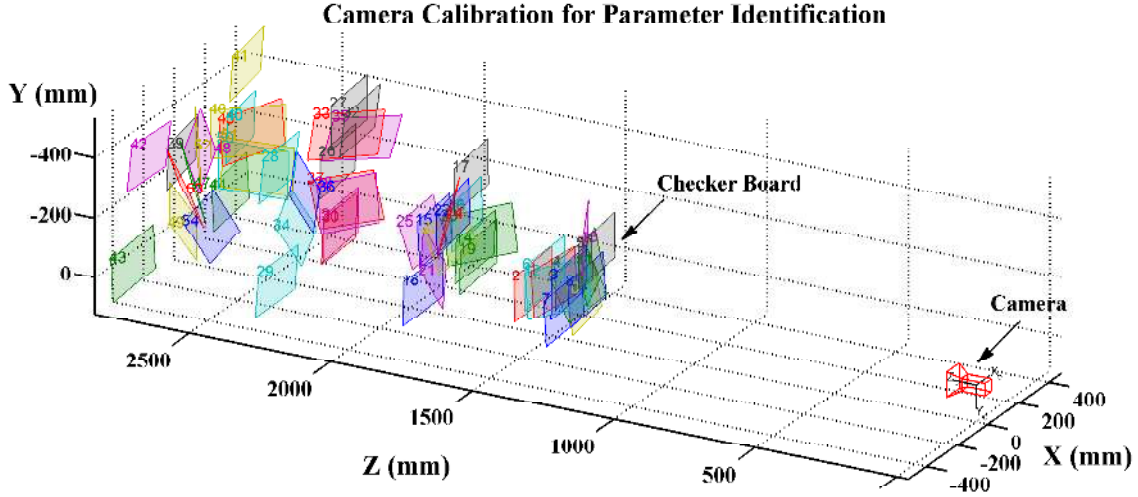


Figure 4.4: Camera Calibration Session for determining camera parameters

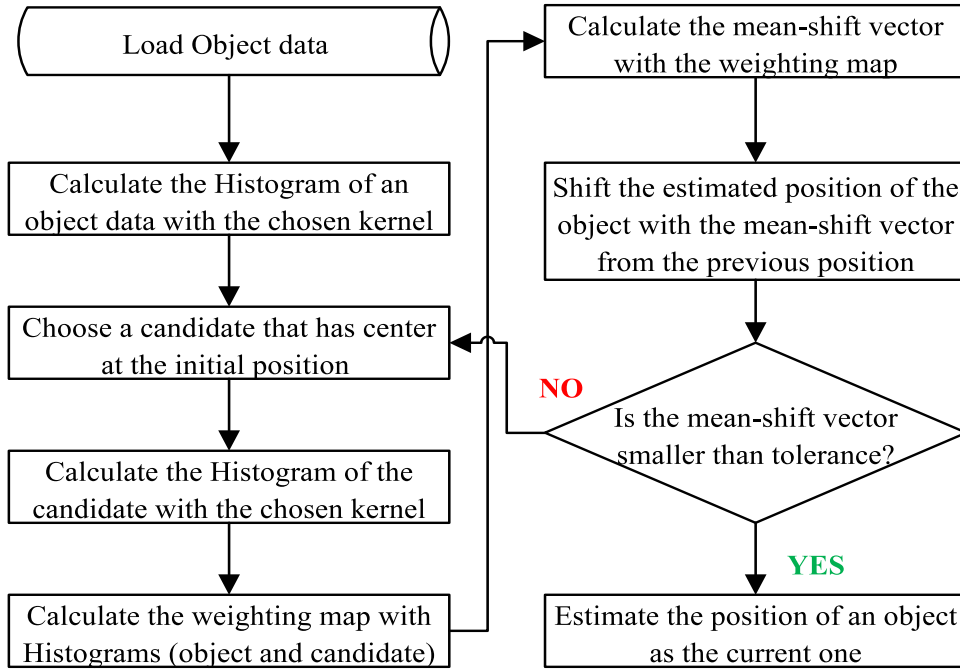


Figure 4.5: Flowchart of Meanshift algorithm [162]

4.3.2 Machine Vision Algorithm

The primary objective of this step is to track the image feature coordinates (r, c) of the test target (**TT**), within the field of view of the camera **C**, which in turn forms the current feature vector $\mathbf{f}(\mathbf{s}(t))$ as discussed prior to (4.1). In order to perform this tracking, machine vision algorithms are employed. Although, numerous state-of-the-art machine vision algorithms are available in literature [163]–[167], for this study, we chose two standard template-based machine vision algorithms viz. mean-shift [162] and geometric pattern matching as indicated in

Fig 4.3, which are deployed within the **LabVIEW** environment, after the camera calibration. Based on the preference of machine vision algorithms as mentioned in Section 3.3.2.1 of the previous chapter, the mean-shift algorithm is utilised for well-lit environments based on the algorithm shown in Fig. 4.5, whereas the geometric pattern matching is used for dim-lit environments, due to skewness and discontinuity observed during target tracking in dim-lit settings by the mean-shift algorithm.

4.3.3 Image-based Visual Servoing Scheme

The objective of the visual servoing scheme is to determine an error correcting pose ($\mathbf{q}$) for the manipulator end-effector based on the image pixel error ($\mathbf{e}$) calculated from the information available from the visual sensors regarding the target. This can be accomplished by following couple of steps. In the first step, a mapping of the position and orientation of the end-effector to the corresponding image feature parameters is required, since the image error is defined in the image parameter space, whereas the the manipulator pose vector ($\mathbf{q}$) is defined either in terms of world coordinates or task space coordinates. This is accomplished by using the principles of perspective projection, in the form of an Interaction matrix or Feature Jacobian matrix $\mathbf{J}$. In the second step, formulation of a feature space control law is necessary since the visual sensor is mounted atop the end-effector in an *eye-in-hand* configuration and any change in the end-effector pose, in turn, effects the image error. This control law takes the form of a velocity controller for the visual sensor $\mathbf{C}$ based on the Jacobian matrix $\mathbf{J}$ as shown in Fig 4.3.

4.3.3.1 Mapping between image parameters and end-effector position

Let us consider that the end-effector pose vector ($\mathbf{q}$) of the manipulator is a combination of the translational and rotational vectors and can be represented as [184],

$$\mathbf{q} = (\mathbf{d}^T \quad \boldsymbol{\theta}^T)^T = (x \quad y \quad z \quad \alpha \quad \beta \quad \gamma)^T \quad (4.3a)$$

where, the translation movements x_0 , y_0 and z_0 , together describe the instantaneous displacement of the platform in the form of $\mathbf{d}$ as,

$$\mathbf{d} = \mathbf{x}^T \hat{\mathbf{e}}_0 \quad (4.3b)$$

with the translation vector and the unit vectors along X, Y and Z directions respectively, given by,

$$\mathbf{x} = (x \quad y \quad z)^T \quad \text{and} \quad \hat{\mathbf{e}}_0 = (\hat{i} \quad \hat{j} \quad \hat{k})^T. \quad (4.3c)$$

The angular displacements, α , β and γ , describe the instantaneous orientation of the platform in the form of the vector,

$$\boldsymbol{\theta} = (\alpha \quad \beta \quad \gamma)^T. \quad (4.3d)$$

Since the camera is mounted atop the end-effector, the camera (or end-effector) velocity $\mathbf{v}$ [104] can be represented based on (4.3a) as,

$$\mathbf{v}(t) = \dot{\mathbf{q}} = (\dot{\mathbf{d}}^T \quad \dot{\boldsymbol{\theta}}^T) \quad (4.4)$$

where $\dot{\mathbf{d}}$ represents the instantaneous translational velocity and $\dot{\boldsymbol{\theta}}$ denotes the instantaneous angular velocity of the camera frame. Based on the principles of perspective projection [112], [123], the relationship between the current image feature $\mathbf{f}$ of the target and the camera velocity $\mathbf{v}$ may be expressed as,

$$\dot{\mathbf{f}}(\mathbf{s}(t)) = \mathbf{J}^T \mathbf{v}(t) \quad (4.5a)$$

where, the image feature rate vector $\dot{\mathbf{f}}$ can be represented as,

$$\dot{\mathbf{f}} = (\dot{r} \quad \dot{c})^T \quad (4.5b)$$

with (r, c) being the row and column pixel coordinates of the target centroid. Also, the feature Jacobian matrix $\mathbf{J}$ can be represented as [112],

$$\mathbf{J} = \begin{pmatrix} f_x/Z & 0 \\ 0 & f_y/Z \\ (r - O_r)/Z & (c - O_c)/Z \\ (r - O_r)(c - O_c)/f_y & -(f_y^2 + (c - O_c)^2)/f_y \\ (f_x^2 + (r - O_r)^2)/f_x & -(r - O_r)(c - O_c)/f_x \\ f_x(c - O_c)/f_y & f_y(r - O_r)/f_x \end{pmatrix} \quad (4.5c)$$

with the values of the parameters f_x , f_y , O_r and O_c obtained from the intrinsic camera matrix $\mathbf{K}$ in (4.2). Z represents the camera depth from the target parallel to the ground. However, similar to most visual servoing applications, the present study also demands the computation of 'camera velocity $\mathbf{v}$ for a given image feature rate vector $\dot{\mathbf{f}}$. Hence, (4.5a) can be inverted to obtain,

$$\mathbf{v}(t) = \mathbf{J}^+ \dot{\mathbf{f}}(\mathbf{s}(t)) \quad (4.5d)$$

where, $\mathbf{J}^+$ is the Moore-Penrose pseudoinverse for $\mathbf{J}$, in case $\mathbf{J}$ is not a square matrix and is non-invertible, when the number of visual features is greater than the degrees of freedom available in the task space. It can be represented in the form,

$$\mathbf{J}^+ = (\mathbf{J}^T \mathbf{J}^T)^{-1} \mathbf{J}^T \quad (4.5e)$$

4.3.3.2 Formulation of the Feature Space control law

The relationship between the camera velocity and the time variation of image pixel error can be obtained by combining (4.1) and (4.5d) in the form,

$$\mathbf{v}(t) = \mathbf{J}^+ \dot{\mathbf{e}}(t) \quad (4.6a)$$

Considering the camera velocity $\mathbf{v}$ as the control variable and ensuring a decoupled decrease of image pixel error $\mathbf{e}$, i.e., $\dot{\mathbf{e}} = -\lambda \mathbf{e}$, the feature space control law can be designed based on (4.6a) in the form [112],

$$\mathbf{v}(t) = -\lambda \mathbf{J}^+ \mathbf{e}(t) \quad (4.6b)$$

where, λ represents a constant gain of appropriate dimension. Based on the feature space control law in (4.6b), the pose vector $\mathbf{q}$ can be computed as,

$$\mathbf{q}(t) = \int_{t_0}^{t_i} \mathbf{v}(t) dt \quad (4.7)$$

where, t_0 and t_i represent values of the initial and i^{th} time instant.

4.3.4 Manipulator Kinematics

The inverse kinematics of the manipulator under investigation is used to calculate the actuator-level demand variables, i.e., actuator strokes with the aid of the camera velocity obtained from the visual servoing scheme as shown in Fig. 4.3. The inverse kinematics model begins operation by attaining the end-effector pose demand ($\mathbf{q}$) from the camera velocity $\mathbf{v}$ as mentioned in the previous subsection. In order to achieve this error-correcting pose $\mathbf{q}$ of the end-effector $\mathbf{M}$, a leg vector is defined as the position of the center of the top joint relative to center of the bottom joint representing the six electrohydraulic single-rod actuators forming the six parallel legs of the manipulator. This leg vector can be expressed mathematically in the form,

$$\mathbf{l}_i(t) = \hat{\mathbf{e}}_0^T \mathbf{x}_i = \hat{\mathbf{e}}_0^T (\mathbf{R}_m(t)^T \mathbf{p}_{T_i} + \mathbf{d}(t) - \mathbf{x}_{B_i}), \quad (4.8)$$

where, the leg-length vector, the position vector of B_i in the global coordinate system and the position vector of T_i in the moving coordinate system are written respectively as,

$$\mathbf{x}_i = \begin{pmatrix} x_i = x_{T_i} - x_{B_i} \\ y_i = y_{T_i} - y_{B_i} \\ z_i = z_{T_i} - z_{B_i} \end{pmatrix} \hat{\mathbf{e}}_0 \quad (4.9a)$$

$$\mathbf{x}_{B_i} = (x_{B_i} \quad y_{B_i} \quad z_{B_i})^T \hat{\mathbf{e}}_0 \quad (4.9b)$$

$$\mathbf{p}_{T_i} = (p_{T_i} \quad q_{T_i} \quad r_{T_i})^T \hat{\mathbf{e}}_0 \quad (4.9c)$$

$\mathbf{R}_m$ is the rotation matrix for transforming a vector from the moving to the stationary coordinate system by successive applications of roll transformation $\mathbf{R}_\alpha$ due to rotation by α about p axis, pitch transformation $\mathbf{R}_\beta$ due to rotation by β about the most recent q axis and yaw transformation $\mathbf{R}_\gamma$ due to rotation by γ about the most recent r axis yielding,

$$\mathbf{R}_m = \mathbf{R}_\gamma \mathbf{R}_\beta \mathbf{R}_\alpha \quad (4.10a)$$

where, the individual transformations are related as,

$$\hat{\mathbf{e}}_o = \mathbf{R}_\alpha \hat{\mathbf{e}}_O \quad ; \quad \hat{\mathbf{e}}_o = \mathbf{R}_\beta \hat{\mathbf{e}}_O \quad \text{and} \quad \hat{\mathbf{e}}_o = \mathbf{R}_\gamma \hat{\mathbf{e}}_O. \quad (4.10b)$$

The overall rotation of the moving platform M in (4.10a) can be obtained in the form,

$$\mathbf{R}_m^T = \begin{pmatrix} c\beta c\gamma & c\beta s\gamma & -s\beta \\ -c\alpha s\gamma + s\alpha s\beta c\gamma & c\alpha c\gamma + s\alpha s\beta s\gamma & s\alpha c\beta \\ s\alpha s\gamma + c\alpha s\beta c\gamma & -s\alpha c\gamma + c\alpha s\beta s\gamma & c\alpha c\beta \end{pmatrix} \quad (4.11)$$

where, $s\alpha = \sin\alpha$, $c\alpha = \cos\alpha$ and so on. The inverse kinematics model uses the lengths of the six legs of the parallel manipulator in terms of the platform pose ($\mathbf{q}$). Hence, (4.8) is used to express the square of the leg length as,

$$l_i^2 = (\mathbf{p}_{T_i}^T \mathbf{R}_m + \mathbf{d}^T - \mathbf{x}_{B_i}^T)(\mathbf{R}_m^T \mathbf{p}_{T_i} + \mathbf{d} - \mathbf{x}_{B_i}), \quad (4.12a)$$

The initial leg length vectors for the six legs can be calculated by the following relation,

$$\mathbf{l}_0 = (\mathbf{p}_{T_i} + \mathbf{H} - \mathbf{x}_{B_i}), \quad (4.12b)$$

where, $\mathbf{H} = [0 \ 0 \ h]$ with h as the height at which the origins o (of the platform $\mathbf{M}$) and O (of the bottom frame $\mathbf{B}$) lie along a vertical line as mentioned earlier. Then, the actuator demand strokes can be calculated as the difference of the 2-norm of

the current leg-length vector ($\mathbf{l}_i$) in (4.8) and the 2-norm of the initial leg length vector ($\mathbf{l}_0$) in (4.12b) and can be represented mathematically as,

$$s_i^*(t) = \|\mathbf{l}_i(t)\| - \|\mathbf{l}_0\|; \quad i = 1 \quad \text{to} \quad 6. \quad (4.12c)$$

The initial leg length in our case is assumed to be at the mid-stroke of actuators i.e. 0.075m, considering the fact that the maximum actuator strokes are 0.15m as mentioned in Table 4.1. This pose is chosen as the neutral position for the moving platform during our studies.

4.3.5 Joint-level Feedback Control

The architecture of the actuator-position controller, indicated in Fig. 4.3, is shown in detail in Fig. 4.6.

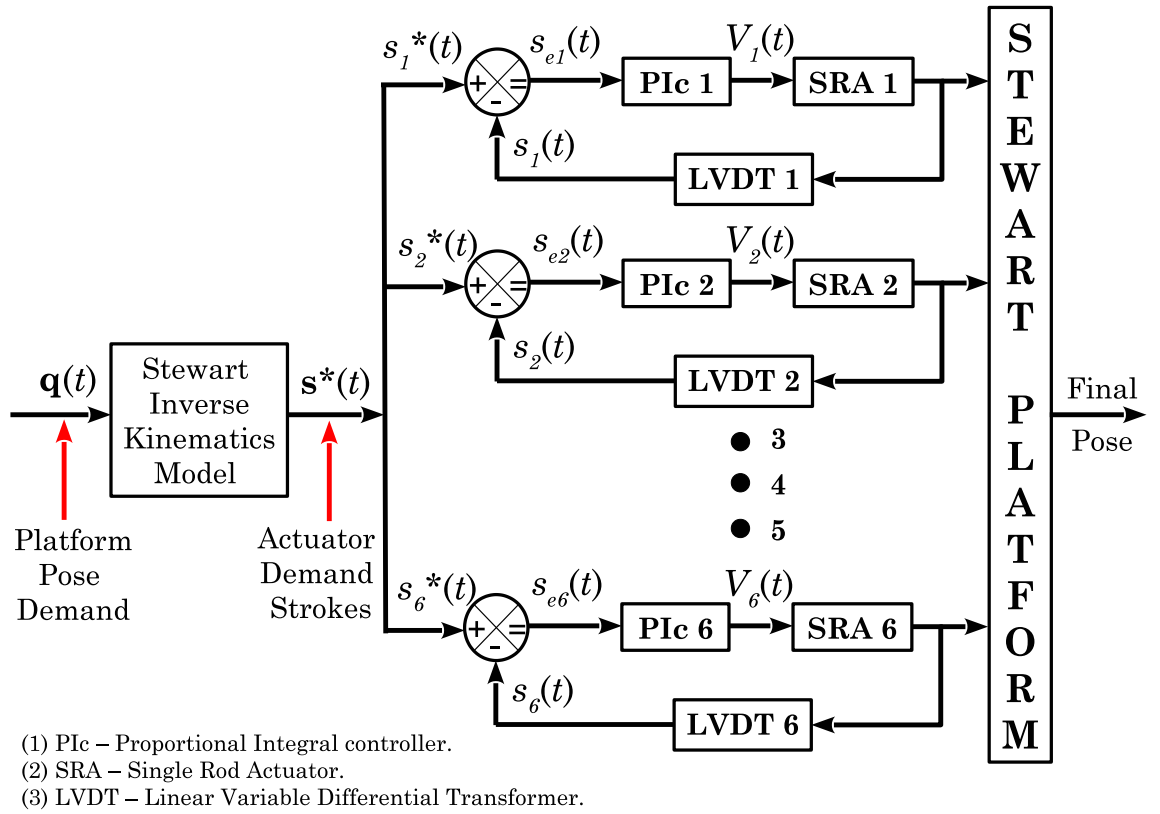


Figure 4.6: Joint-level position control schematic of the Stewart Platform.

It can be observed that each manipulator leg **SRA** followed by the respective leg number, has a closed loop operating system, whereby the error in stroke of each leg is computed based on the difference between the corresponding demanded stroke s_i^* in (4.12c) and the instantaneous position response of the actuator piston rod, say s_i measured directly by the **LVDT** mounted at the base of each actuator. The

leg-length error for each leg may be denoted as,

$$s_{e_i}(t) = s_i^*(t) - s_i(t); \quad i = 1 \quad \text{to} \quad 6. \quad (4.13a)$$

The leg length error corresponding to each leg is then fed to a distinct PI controller (**PIc**), which computes the voltage signals to be sent to each of the six proportional valves with the assistance of OM of the RTS. The mathematical form for computing each voltage signal can be represented as,

$$V_i(t) = k_P * s_{e_i}(t) + k_I * \int s_{e_i}(t) dt, \quad (4.13b)$$

where, V_i represents the voltage provided to the i^{th} leg **PV**. Here, k_P and k_I denote the proportional and integral gains, respectively for each of the **PIc**. The voltage signals are constrained within the limits of $\pm 10V$ for the zero-underlap proportional valve which is used in the present study.

4.4 Results and Discussion

In the present investigation, the center of the unmasked section of the acquired images from the **PMBSS** is considered as the image center, denoted by the red cross in Fig. 4.7 and is chosen as the desired image feature $\mathbf{f}^*(\mathbf{s}_c)$ for the virtual target **TT**. The centroid of the virtual test target **TT** is considered as the current image feature $\mathbf{f}(\mathbf{s}(t))$ to be tracked by the machine vision algorithm. Now, if we suppose that the test target **TT** is fixed at any location within any of the four quadrants, as denoted by the blue crosses in Fig. 4.7, it becomes quite obvious that in order to minimize the pixel error $\mathbf{e}(t)$, the camera **C** mounted atop the **PMBSS** needs to move forward/backward along the x-axis and up/down along the y-axis of the image plane. These motions of the camera, in turn, are achieved by the respective motion of the moving platform **M** along the Y and Z axes of the stationary coordinate system shown in Fig. 4.1, which are technically termed as *Sway* and *Heave* motions. It can be also be hypothesized that each quadrant requires a distinct set of sway and heave directions in order to achieve the error-correcting pose $\mathbf{q}(t)$ as shown in Fig. 4.7.

The real-time experiments are conducted for two reference motions of the test target **TT**, one along the horizontal x-axis and the other along the vertical y-axis of the image plane in Fig. 4.7. These experiments are initiated with certain values of the fixed parameters of the algorithm, which were arrived at, after several trials in real-time with the servoing system, except for the camera depth (Z) which was fixed based on the location of the virtual target PC from the **PMBSS**.

These parameters can be denoted in the form of vector $\mathbf{k}$ as,

$$\mathbf{k} = (k_P \quad k_I \quad \lambda_x \quad \lambda_y \quad Z) = (200\text{V/m} \quad 100\text{V}/(\text{m.s}) \quad 25\text{s}^{-1} \quad 3\text{s}^{-1} \quad 3\text{m}) \quad (4.14)$$

where, λ_x and λ_y represents the constant gains of the feature space control law in (4.6b) for the x and y axes of the image plane, respectively. Z represents the depth of the camera frame from the target in millimeters. Additionally, a pixel boundary of 3x3 pixels is also considered around the image center, within which the error $\mathbf{e}(t)$ is assumed to be zero, so as avoid any chatter. It is also be noted that the feature Jacobian $\mathbf{J}$, represented in (4.5c), is reduced to an order of 2, since only sway and heave motions are considered, thus eliminating the Moore-Penrose psuedoinverse in (4.5e).

In the hydraulic section of the **PMBSS**, the system pressure P_p of the power pack is maintained at 100 bar and the system flow rate at 54 lpm. Also, before commencing the experiments, the parallel manipulator is brought to its neutral position characterised by mid-stroke position of each actuator at 0.075m and **B**-to-**M** height of 0.473m. This arrangement allows the end-effector to move both ways along the moving coordinate system centered on **M**. These experimental studies on the target reference motions are discussed in the following sections.

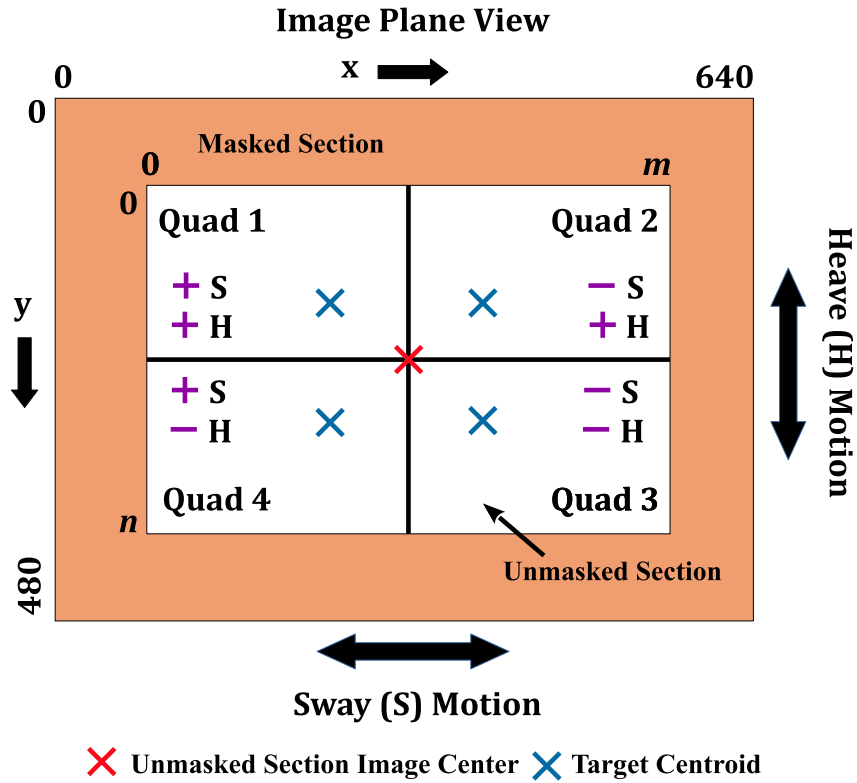


Figure 4.7: Image Plane Quadrants and respective sway-heave motions of the end-effector of the parallel manipulator.

4.4.1 Horizontal motion of moving target

In this particular study, the target **TT** is moved along the horizontal axis of the Virtual target monitor as shown in Fig. 4.2. A temporal sinusoidal motion of peak-to-peak amplitude 0.1m is imposed on the target. In order to assess the real-time performance of the **PMBSS**, sinusoidal motions of three different frequencies *viz.* 0.05Hz, 0.10 Hz and 0.15 Hz, are imposed on the target. The results for each of these frequencies are shown in Figs. 4.8, 4.9 and 4.10, respectively.

Based on the platform pose demands ($\mathbf{q}$) as shown in Figs. 4.8(b), 4.9(b) and 4.10(b), for all the three frequencies and from the image lane quadrant view in Fig. 4.7, it can be deduced that the target motion in the image plane occurs from Quadrant 1 to Quadrant 2. This leads the **PMBSS** into a periodic sway and an almost steady positive heave as evident from Figs. 4.8(c), 4.9(c) and 4.10(c). The actuator demand strokes (s_i^*) and the actuator response strokes (s_i) of Figs. 4.8, 4.9 and 4.10 indicate that the upper stroke limit of 150mm for Actuator 4 (**SRA 4**) is reached for the 0.15Hz frequency, thus, leading to a mechanical constraint for higher frequencies of target motion. The control voltage provided to the proportional valves (V_i) shown in Figs. 4.8(e), 4.9(e) and 4.10(e), evinces an increase in magnitude from 2 Volts to around 5 Volts as the frequency is increased. It should also be noted that the platform response trajectory shown in Figs. 4.8(a), 4.9(a) and 4.10(a), is calculated based on the inverse of the manipulator kinematics model expressed in (4.8), from the recorded responses of the **LVDT** attached to each of the six **SRA**.

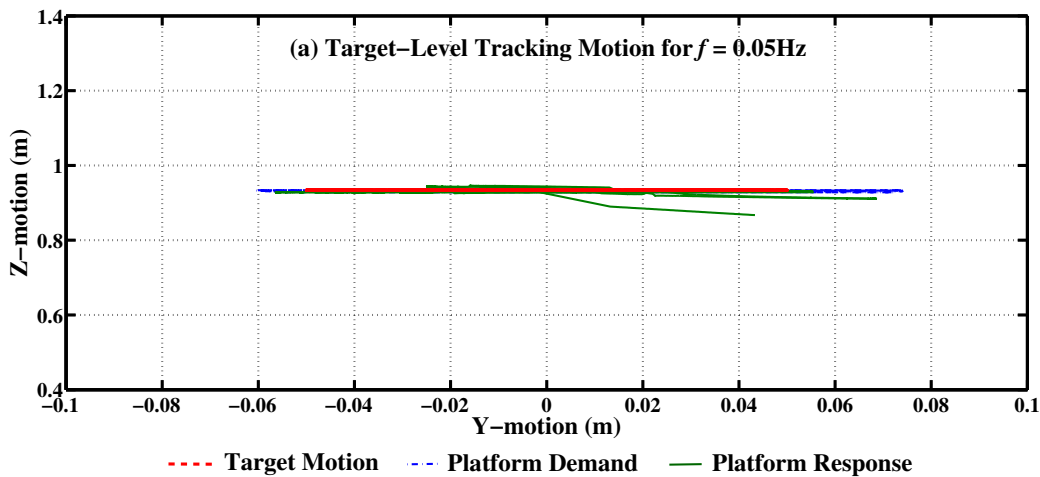


Figure 4.8: Horizontal movement of **PMBSS** for tracking virtual target moving at 0.05Hz frequency

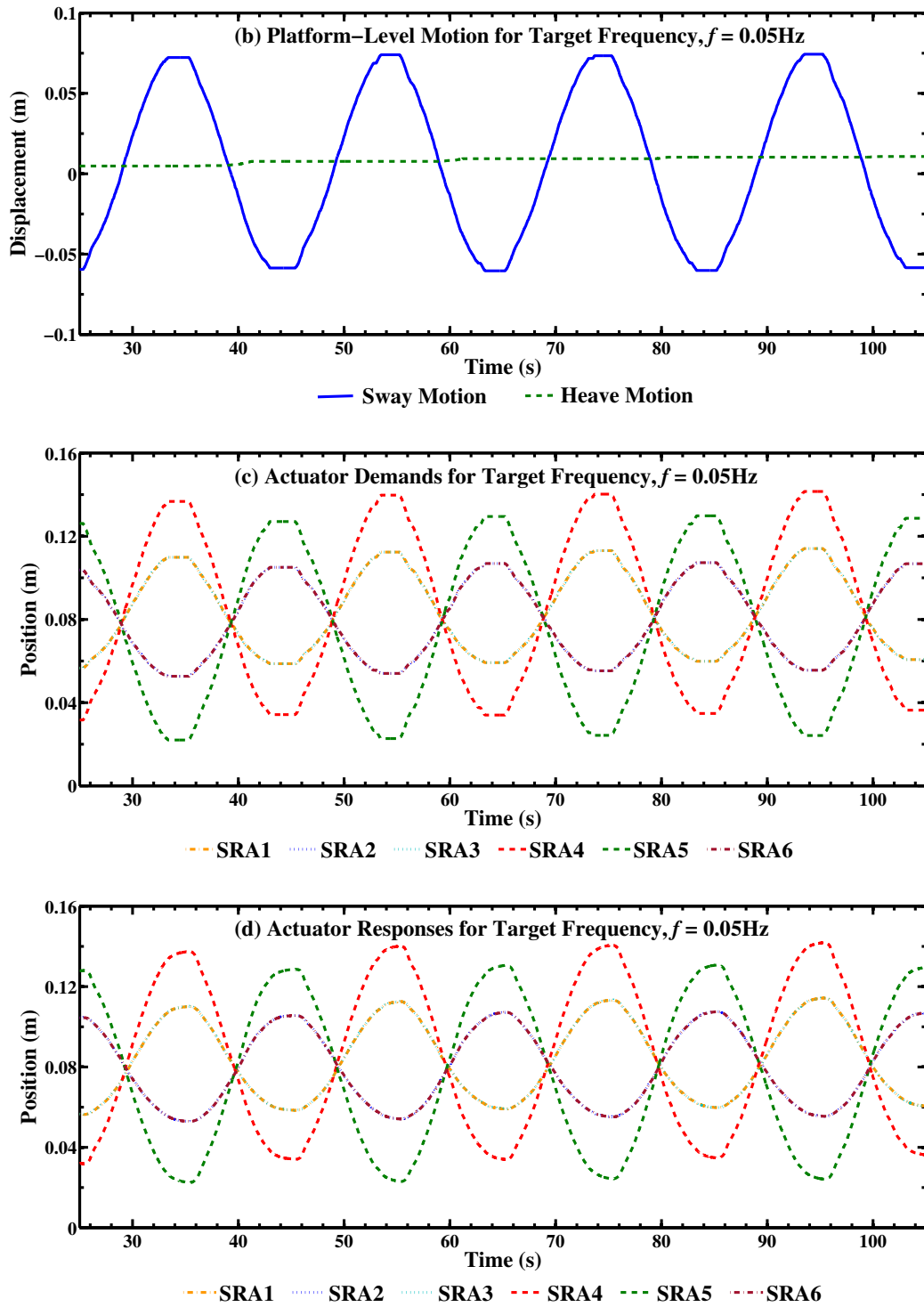


Figure 4.8: Horizontal movement of **PMBSS** for tracking virtual target moving at 0.05Hz frequency (contd.)

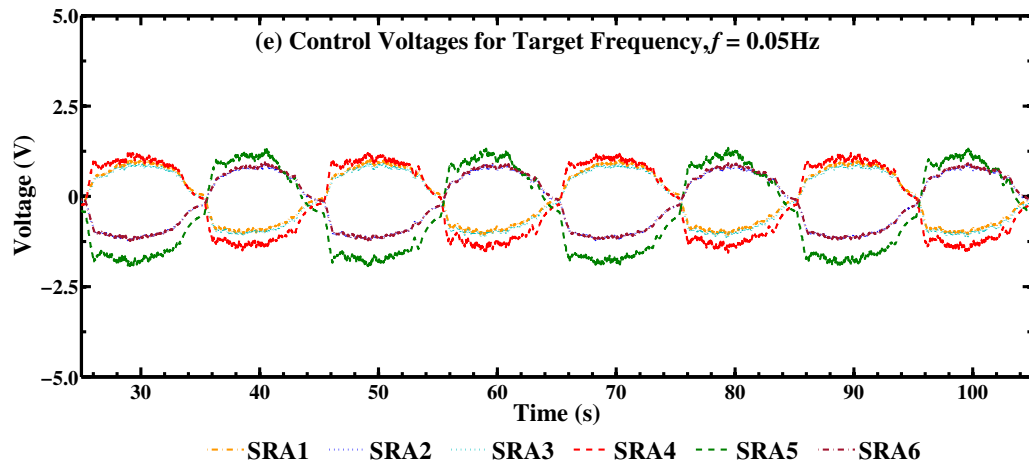


Figure 4.8: Horizontal movement of **PMBSS** for tracking virtual target moving at 0.05Hz frequency (contd.)

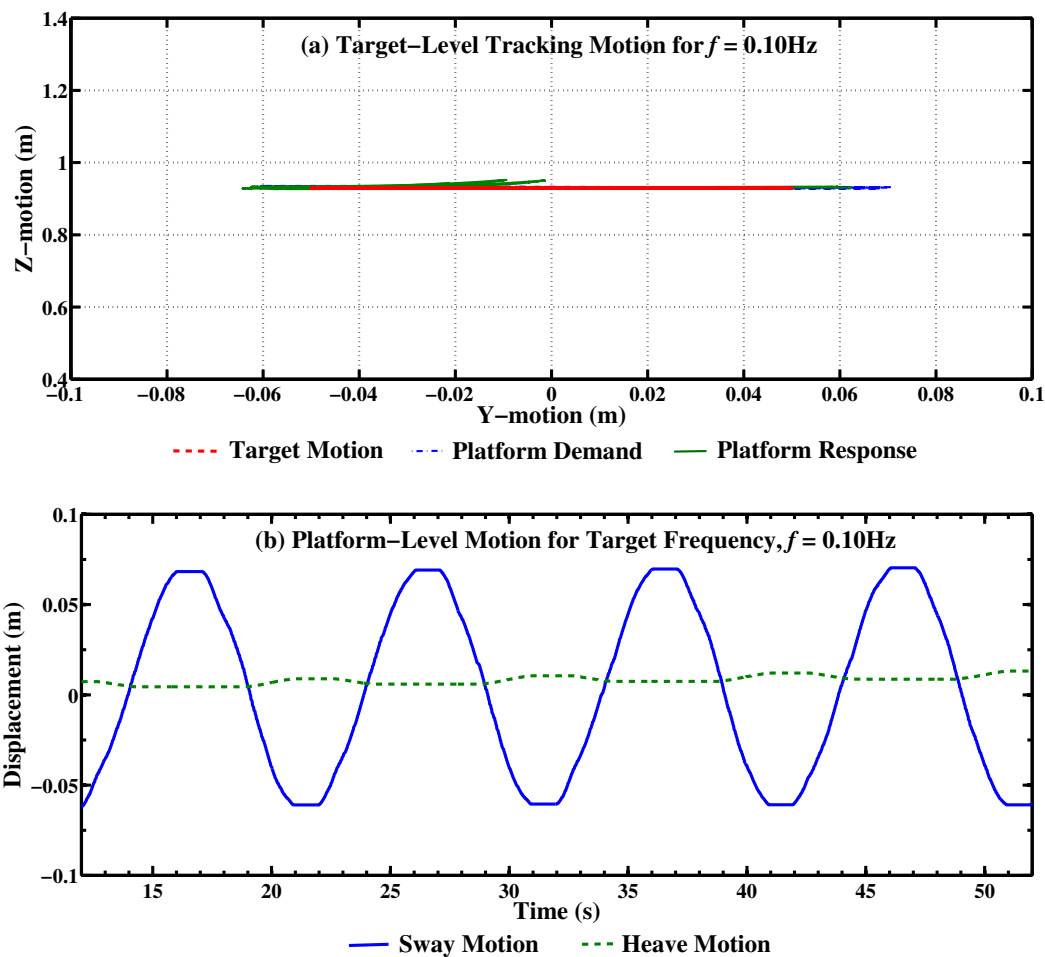


Figure 4.9: Horizontal movement of **PMBSS** for tracking virtual target moving at 0.10Hz frequency

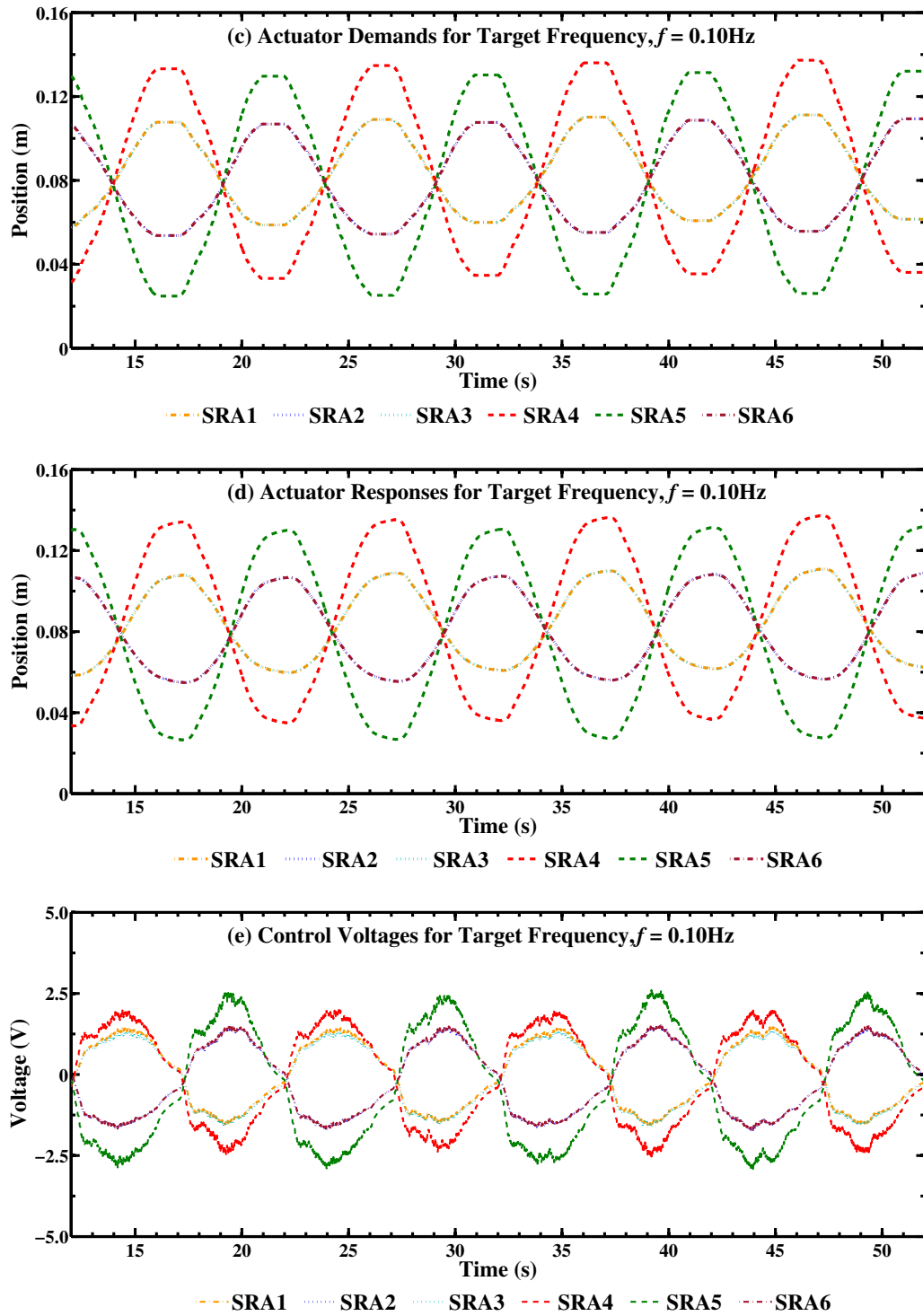


Figure 4.9: Horizontal movement of **PMBSS** for tracking virtual target moving at 0.10Hz frequency (contd.)

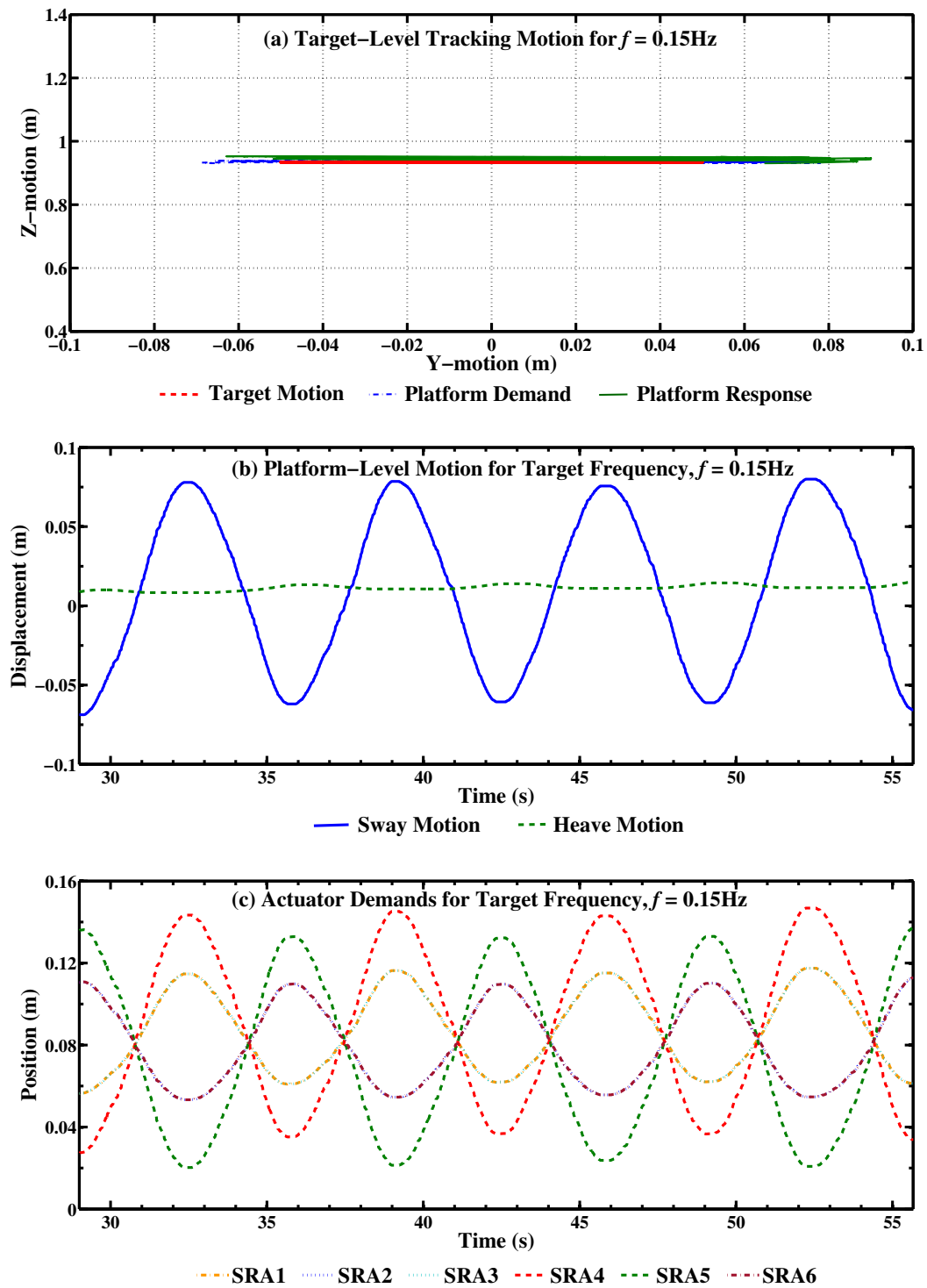


Figure 4.10: Horizontal movement of **PMBSS** for tracking virtual target moving at 0.15Hz frequency

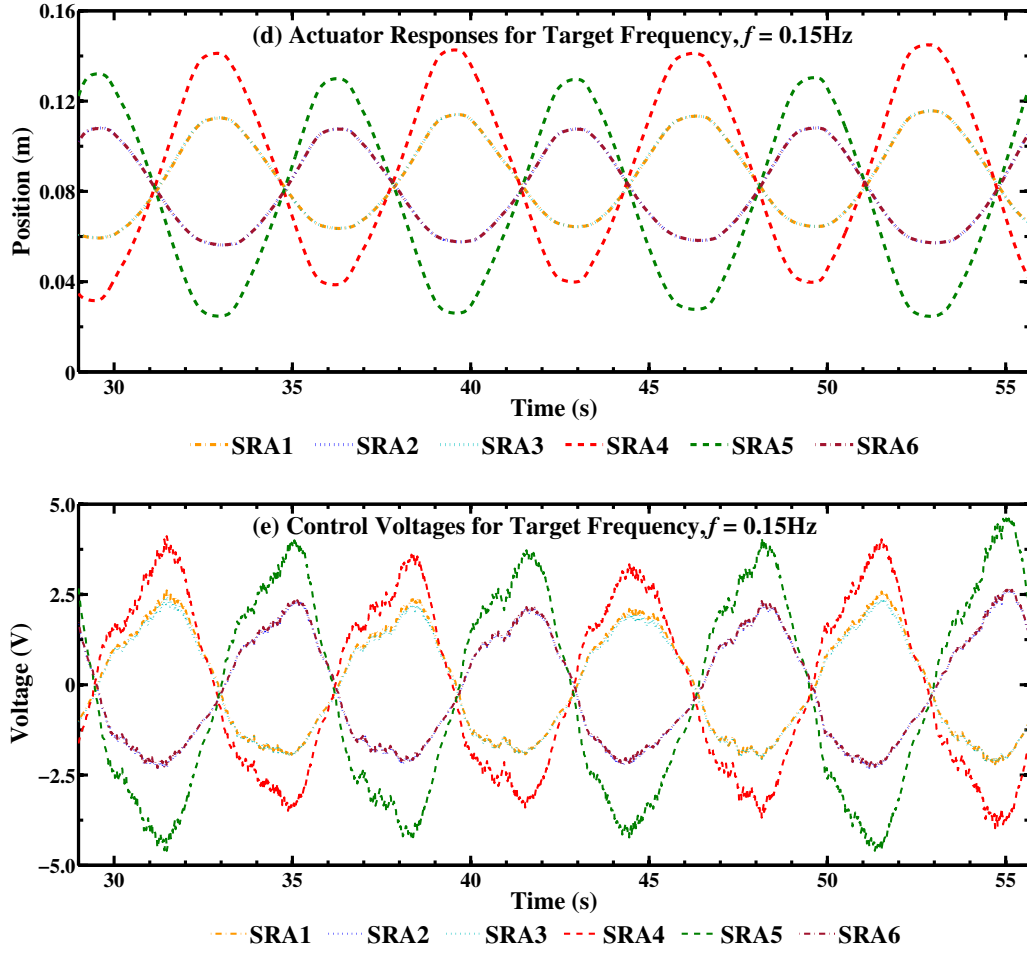


Figure 4.10: Horizontal movement of **PMBSS** for tracking virtual target moving at 0.15Hz frequency (contd.)

4.4.2 Vertical motion of moving target

In this case, the target **TT** is moved along the vertical axis of the Virtual target monitor, following a temporal sinusoidal motion of peak-to-peak amplitude 0.1m. Again, three different frequencies of this sinusoidal motion *viz.* 0.05Hz, 0.10 Hz and 0.15 Hz, are imposed on the target to assess the vertical motion performance of the **PMBSS**. The results for each of these frequencies are shown in Figs. 4.11, 4.12 and 4.13, respectively.

Again, based on the platform pose demand (**q**) as shown in Figs. 4.11(b), 4.12(b) and 4.13(b), for all the three frequencies and from the image plane quadrant view shown in Fig. 4.7, we may infer that the target motion in the image plane takes place entirely within Quadrant 2. This causes the **PMBSS** to hold onto a steady negative sway and leads to a positive periodic heave, as evident from Figs. 4.11(c), 4.12(c) and 4.13(c).

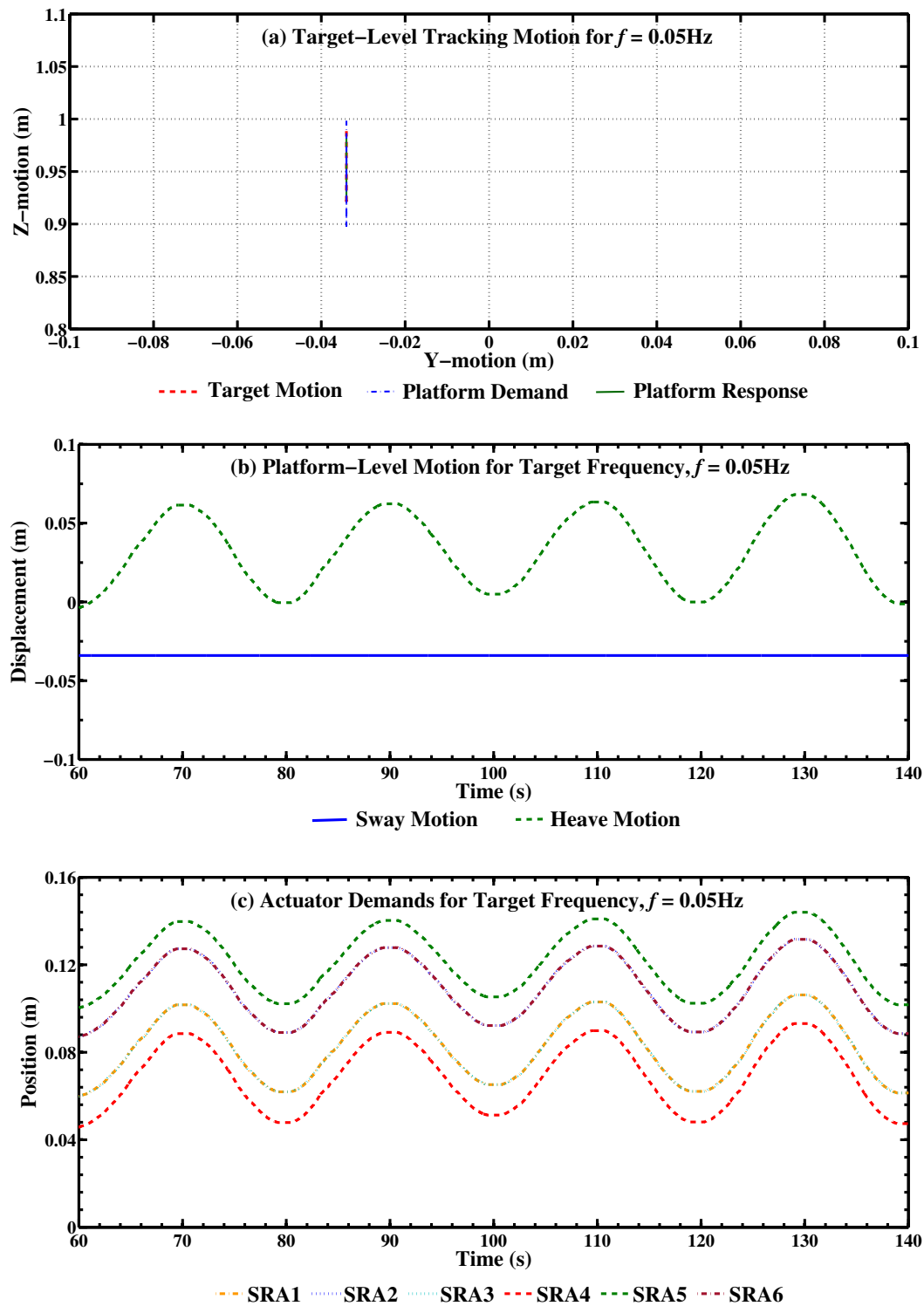


Figure 4.11: Vertical movement of PMBSS for tracking virtual target moving at 0.05Hz frequency

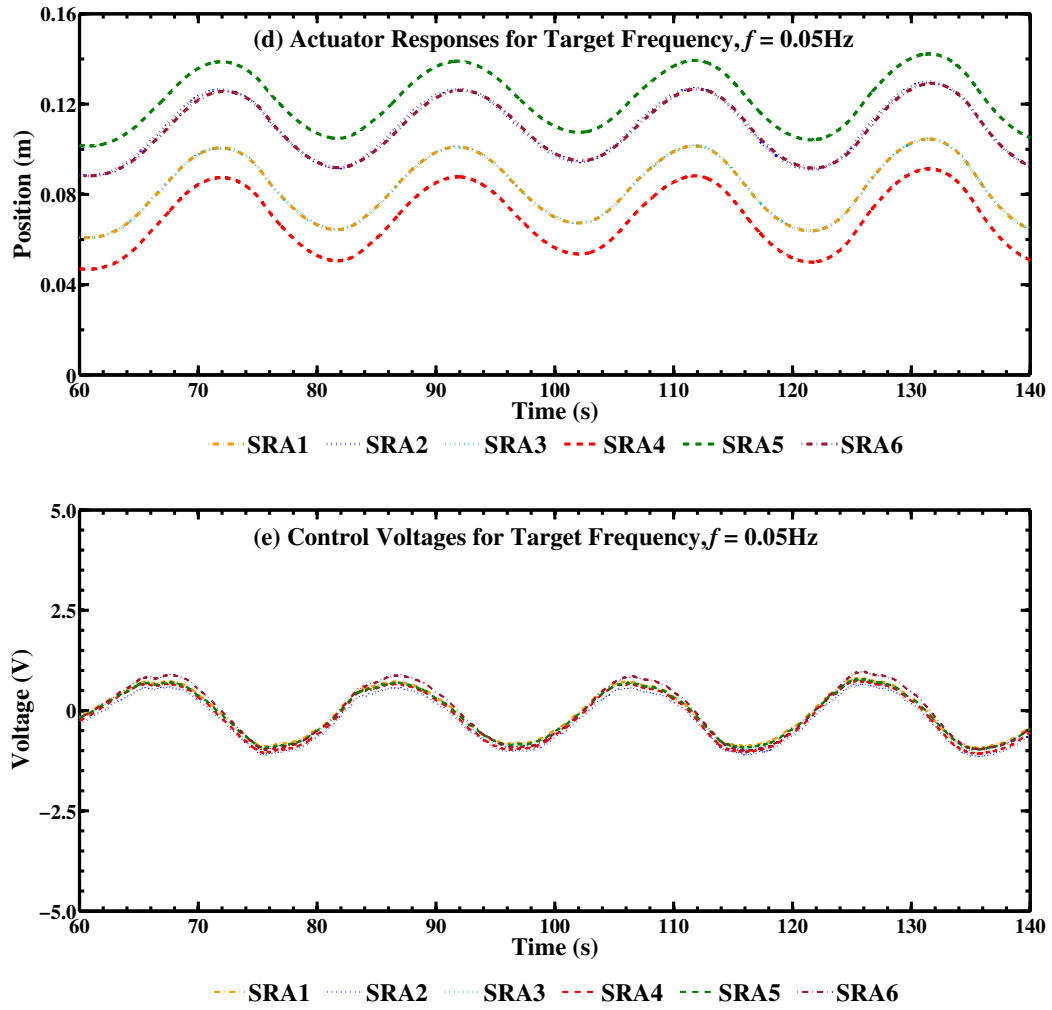


Figure 4.11: Vertical movement of PMBSS for tracking virtual target moving at 0.05Hz frequency (contd.)

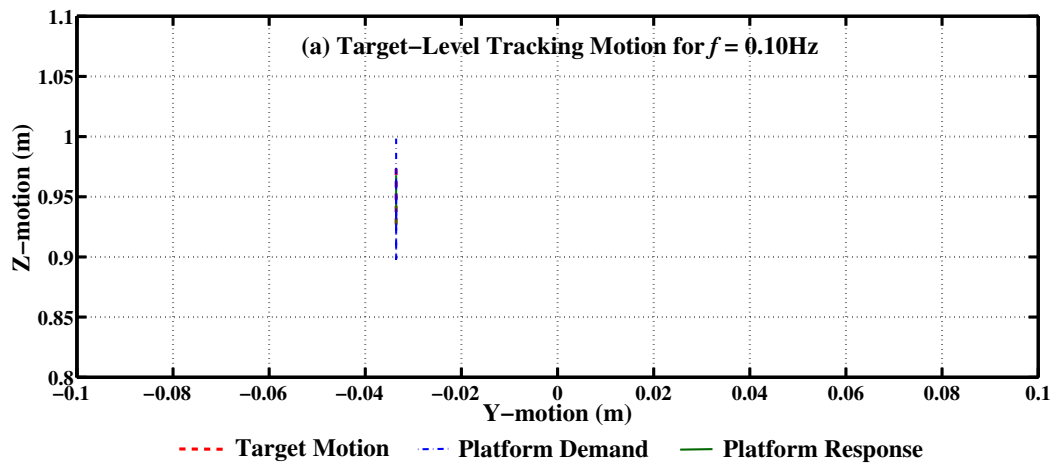


Figure 4.12: Vertical movement of PMBSS for tracking virtual target moving at 0.10Hz frequency

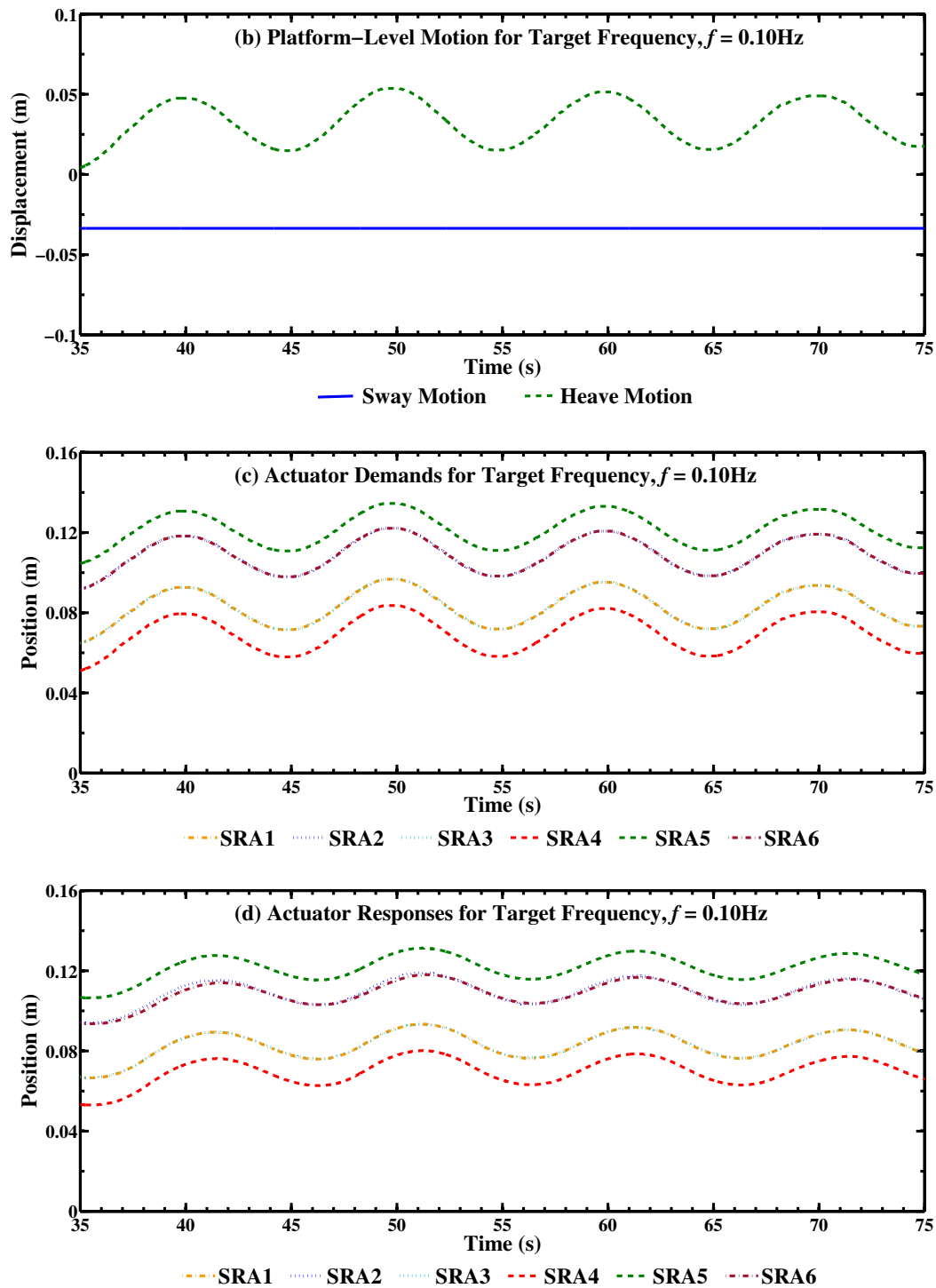


Figure 4.12: Vertical movement of **PMBSS** for tracking virtual target moving at 0.10Hz frequency (contd.)

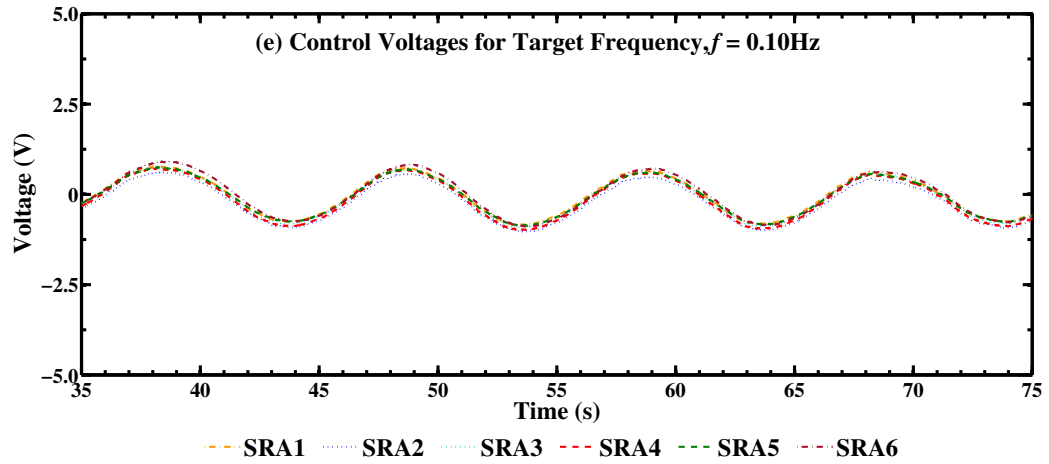


Figure 4.12: Vertical movement of **PMBSS** for tracking virtual target moving at 0.10Hz frequency (contd.)

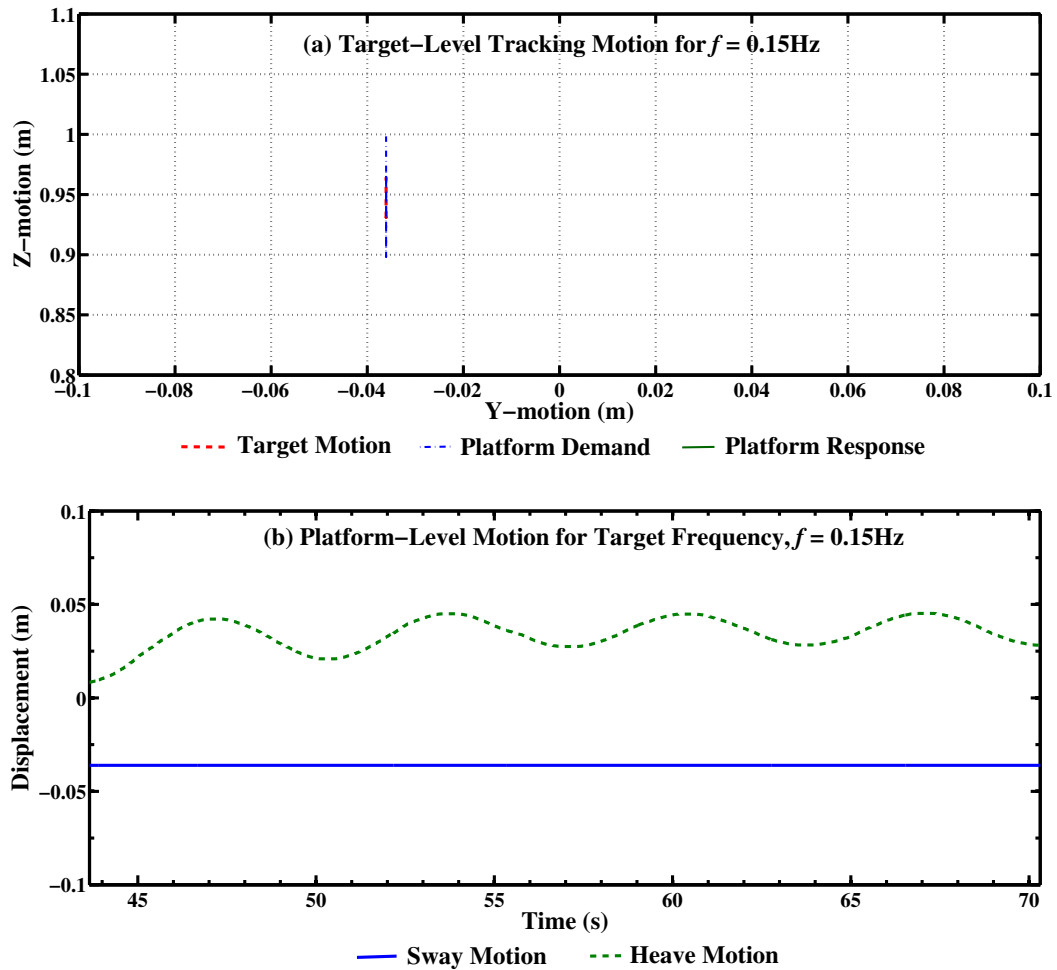


Figure 4.13: Vertical movement of **PMBSS** for tracking virtual target moving at 0.15Hz frequency

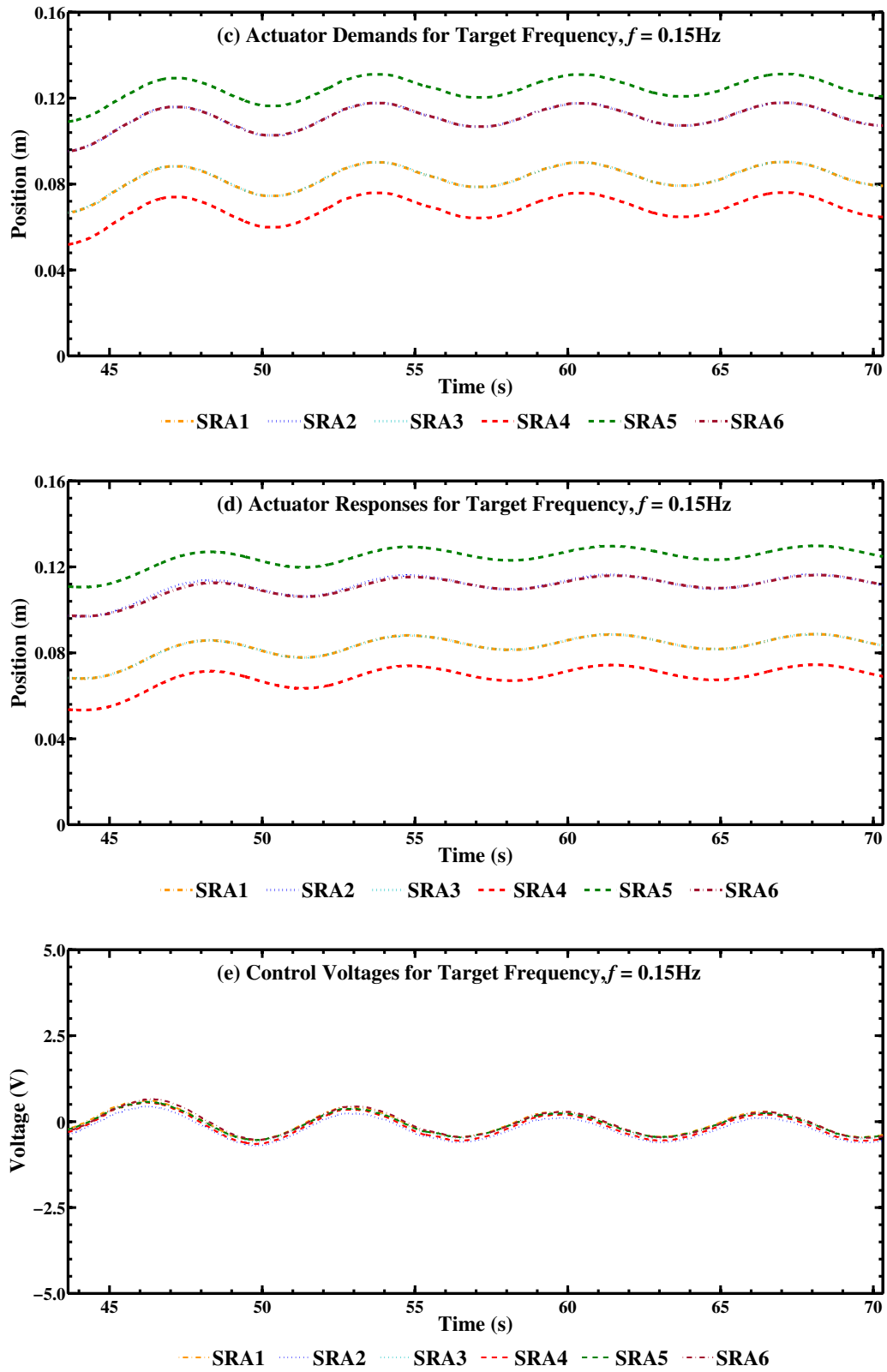


Figure 4.13: Vertical movement of **PMBSS** for tracking virtual target moving at 0.15Hz frequency (contd.)

The actuator demand strokes (s_i^*) and the actuator response strokes (s_i) in Figs. 4.11, 4.12 and 4.13 indicate that the Actuator 5 (**SRA 5**) reaches the upper stroke limit of 150mm for 0.05Hz frequency. However, the same phenomenon is not observed for higher frequencies owing to a reduction in the platform pose demand ($\mathbf{q}$) magnitude compared to the target magnitude. This is further evinced by the fact that the control voltages (V_i) in Figs. 4.11(e), 4.12(e) and 4.13(e) show magnitude reduction with increase in the target frequency within the ± 1 Volt range.

4.4.3 Diagonal motion of moving target

In this simple 2D motion case, the target **TT** is moved diagonally across the virtual target monitor, following a temporal sinusoidal motion along both the x and y-axis of the monitor, with peak-to-peak amplitude of 0.1m. The performance for the diagonal motion of the **PMBSS** is assessed with the aid of three different frequencies of this sinusoidal motion *viz.* 0.05Hz, 0.10 Hz and 0.15 Hz, of the target. The results for each of these frequencies are shown in Figs. 4.14, 4.15 and 4.16, respectively.

Based on the platform pose demand ($\mathbf{q}$) as shown in Figs. 4.14(b), 4.15(b) and 4.16(b), for all the three frequencies under investigation and from the image plane quadrant view shown in Fig. 4.7, we may infer that the target motion in the image plane takes place along the edges of Quadrant 2. Therefore, the **PMBSS** exhibits a periodic negative sway with near zero peak, and a periodic positive heave.

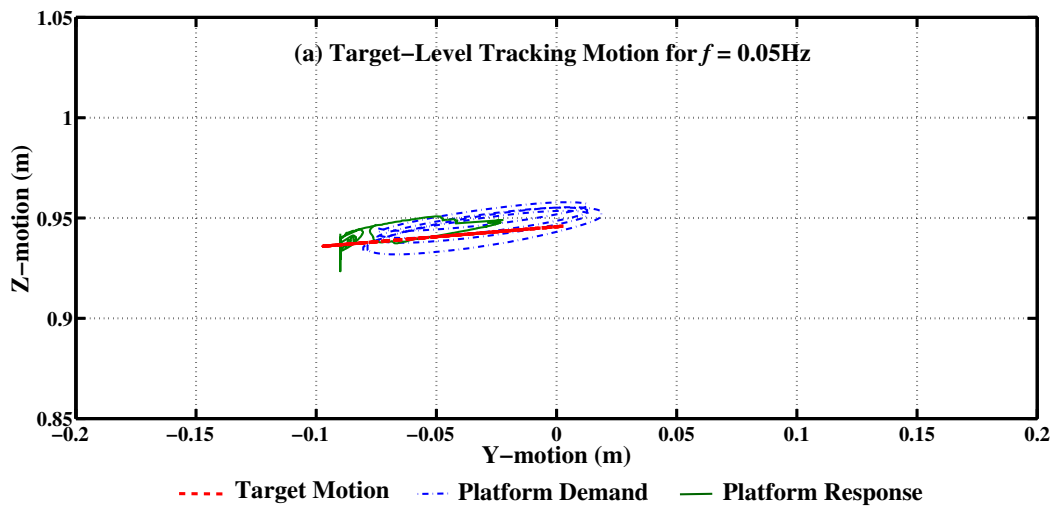


Figure 4.14: Diagonal movement of **PMBSS** for tracking virtual target moving at 0.05Hz frequency

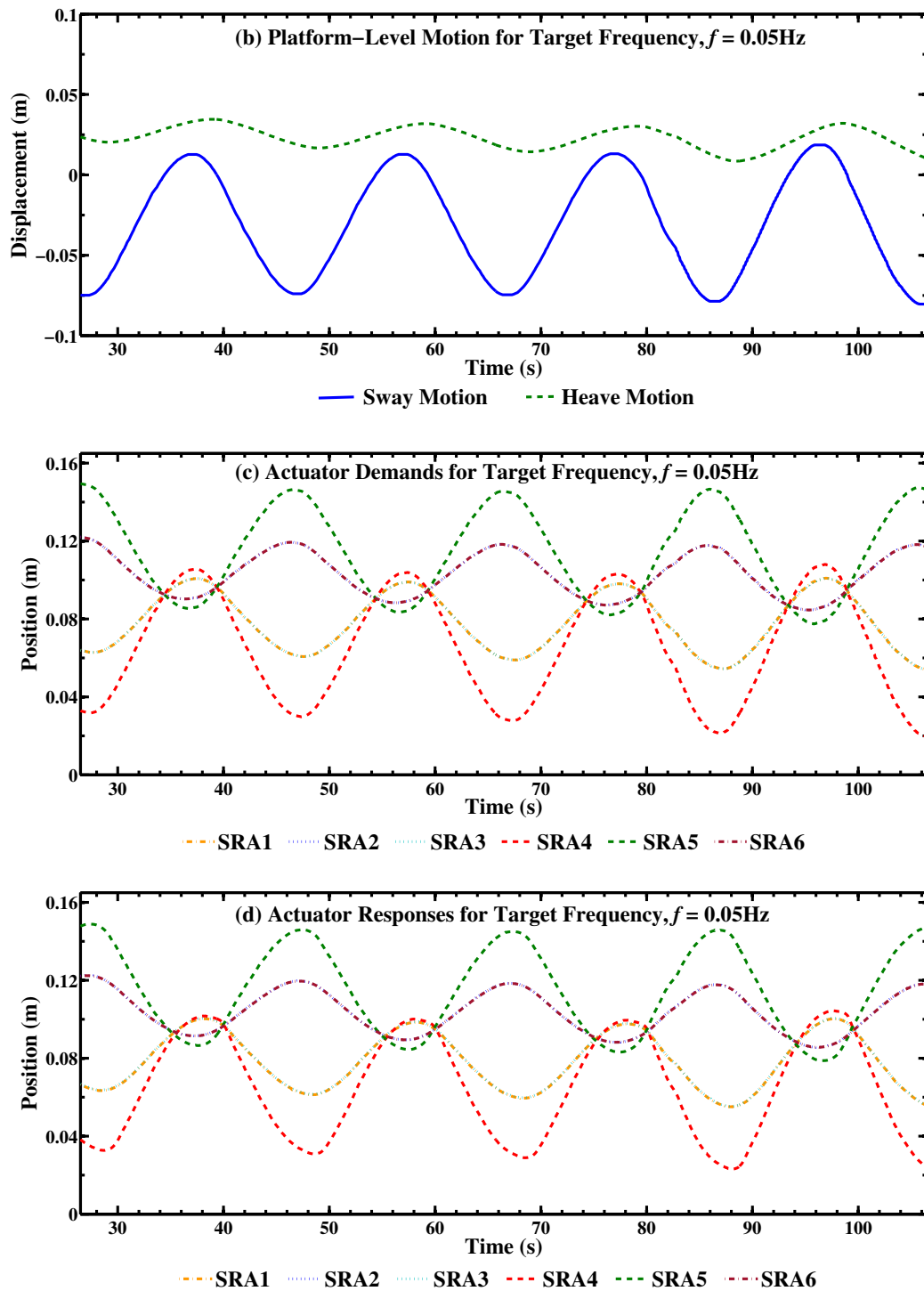


Figure 4.14: Diagonal movement of **PMBSS** for tracking virtual target moving at 0.05Hz frequency (contd.)

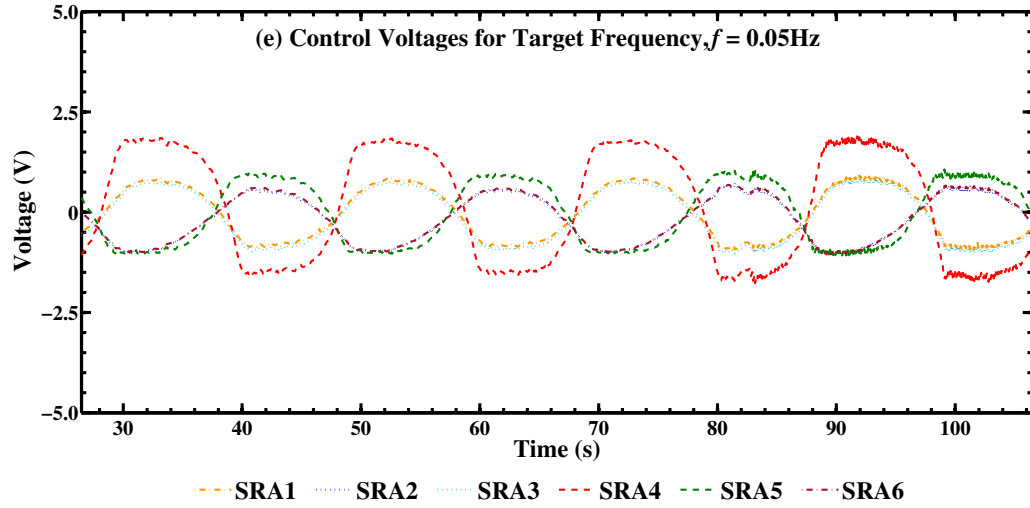


Figure 4.14: Diagonal movement of PMBSS for tracking virtual target moving at 0.05Hz frequency (contd.)

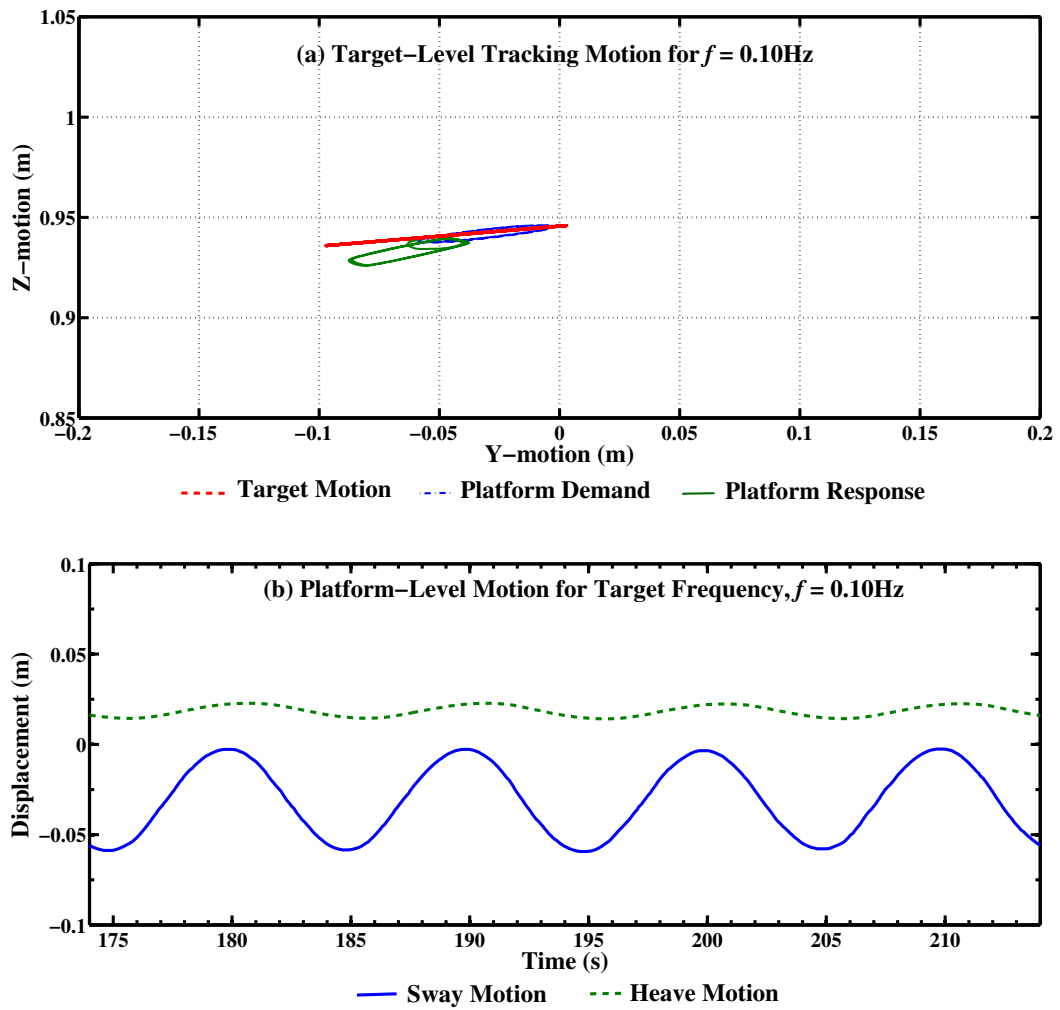


Figure 4.15: Diagonal movement of PMBSS for tracking virtual target moving at 0.10Hz frequency

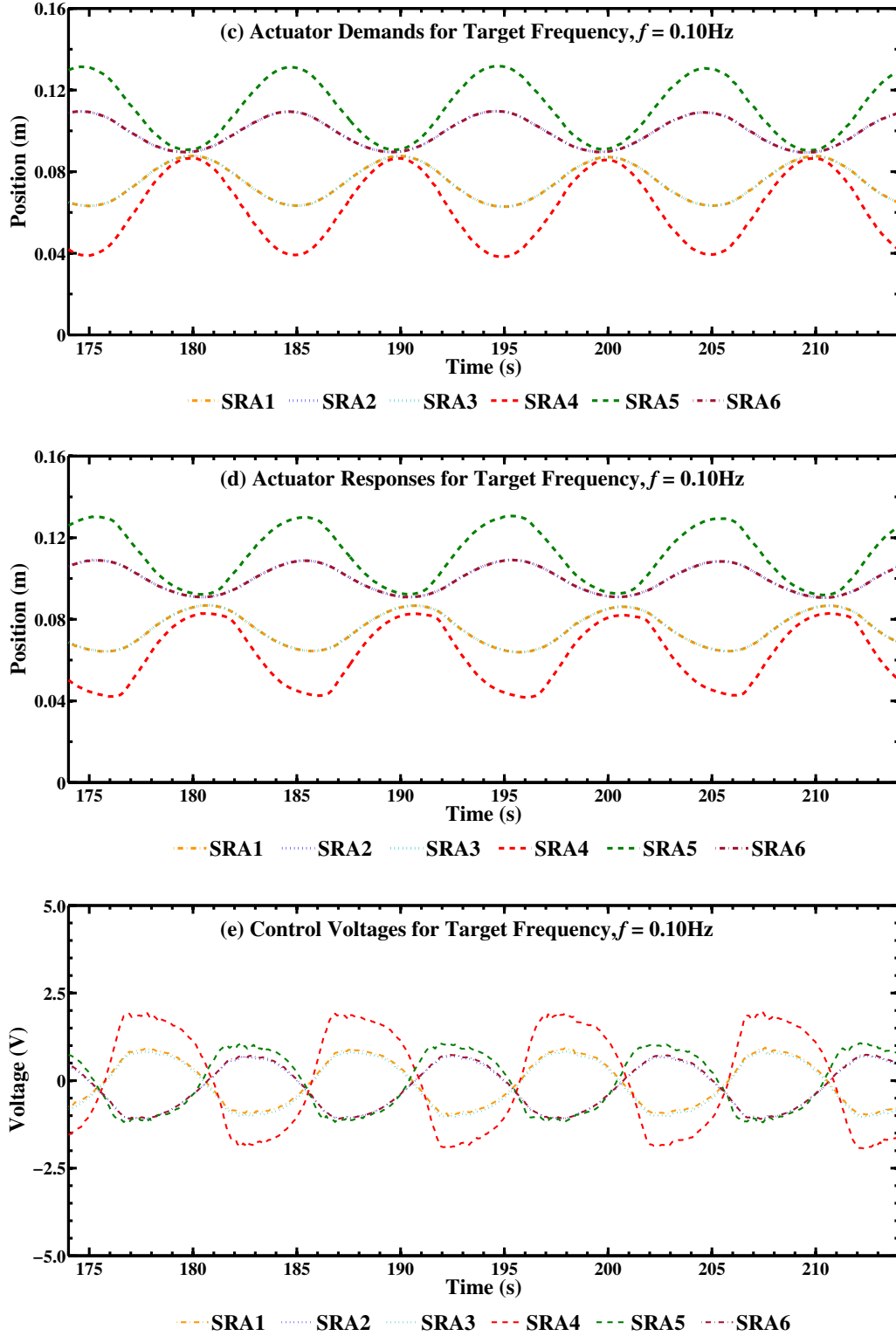


Figure 4.15: Diagonal movement of **PMBSS** for tracking virtual target moving at 0.10Hz frequency (contd.)

The actuator demand strokes (s_i^*) and the actuator response strokes (s_i) in Figs. 4.14, 4.15 and 4.16 indicate that both Actuator 5 (**SRA 5**) and Actuator 6 (**SRA 6**), reach the upper stroke limit of 150mm for 0.05Hz frequency. This

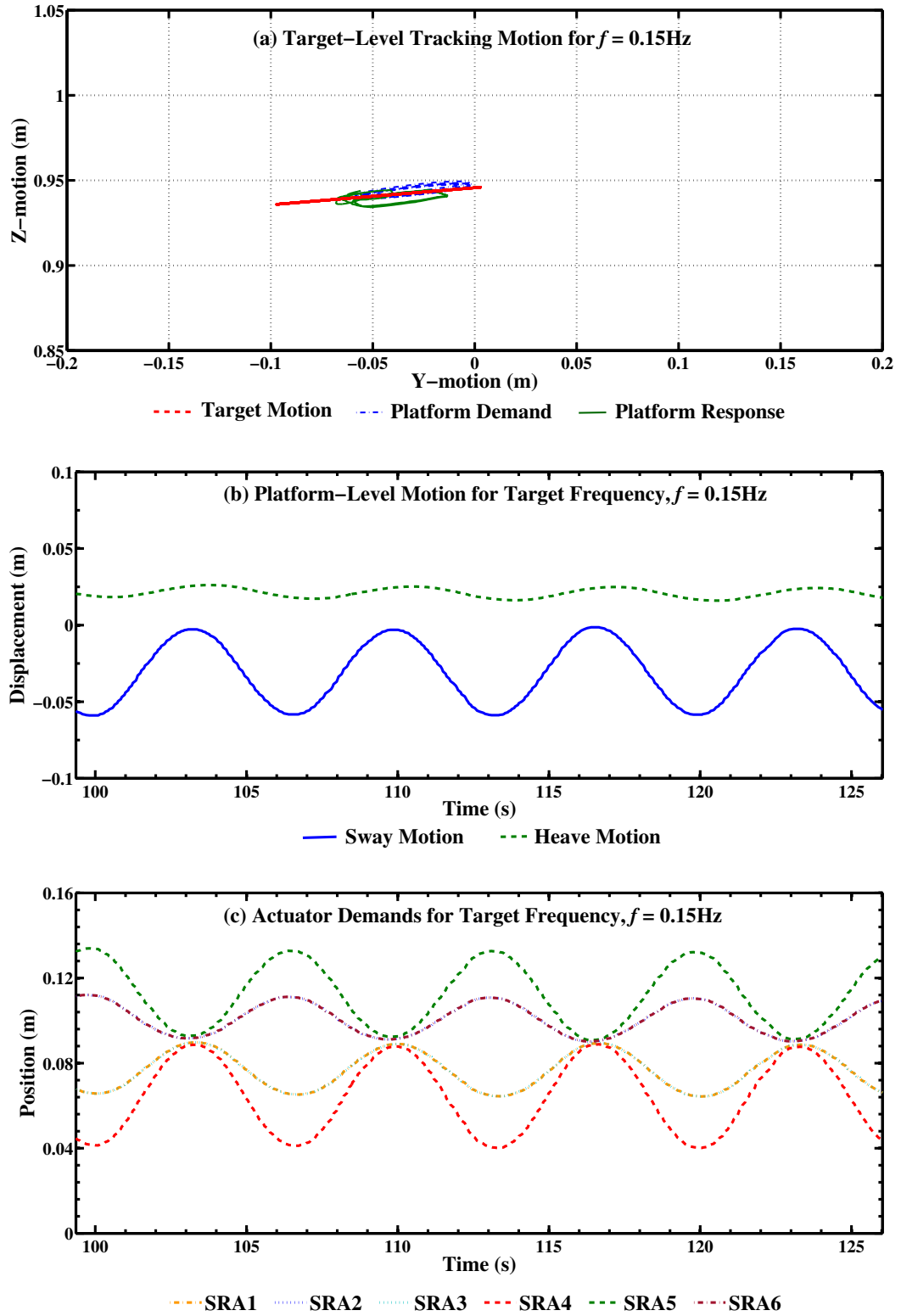


Figure 4.16: Diagonal movement of PMBSS for tracking virtual target moving at 0.15Hz frequency

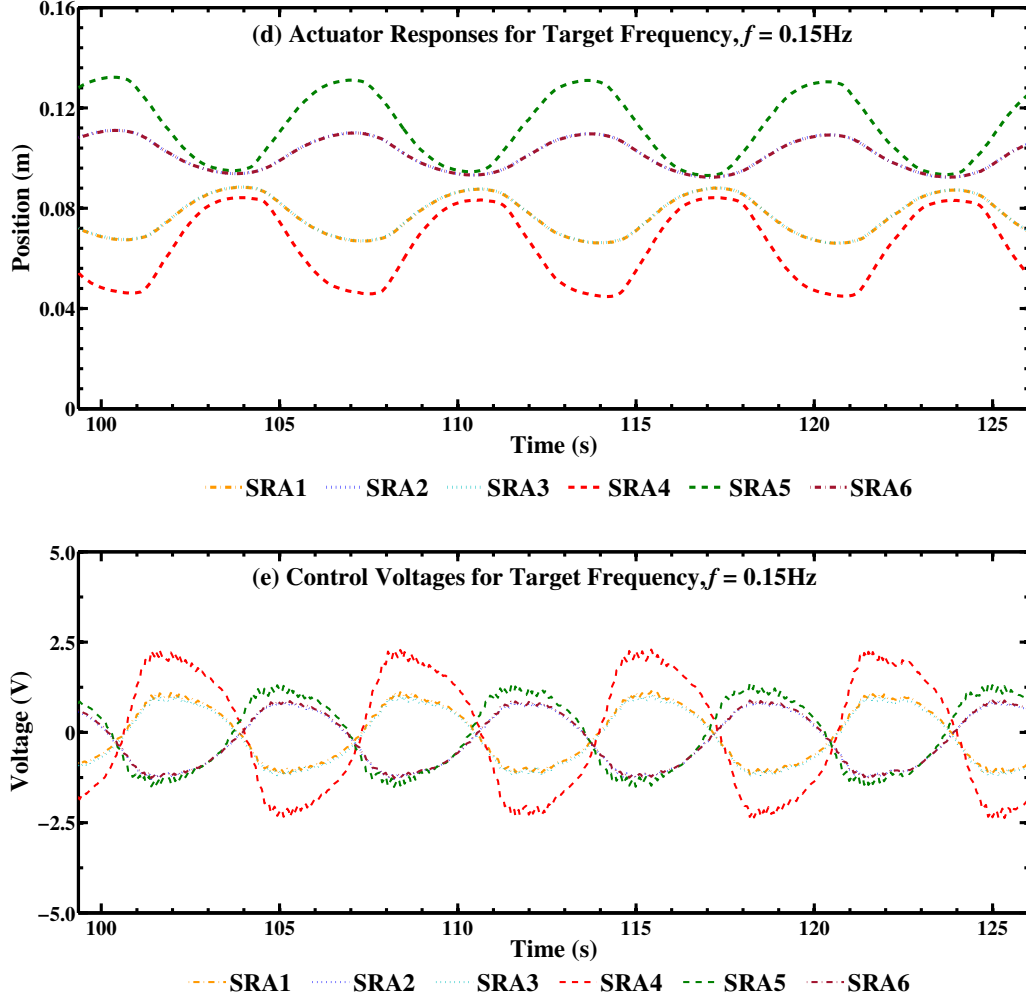


Figure 4.16: Diagonal movement of **PMBSS** for tracking virtual target moving at 0.15Hz frequency (contd.)

behaviour disappears as the frequency is increased, which may be attributed to the reduction in the platform pose demand ($\mathbf{q}$) magnitude. However, the control voltages (V_i) in in Figs. 4.14(e), 4.15(e) and 4.16(e) show similar magnitudes for all the target frequencies within the ± 2 Volt range.

4.4.4 Performance Analysis

The performance of the **PMBSS** for the experimental cases conducted during this investigation are evaluated based on three performance indices. The first index, namely, the Discrete Absolute Trajectory Error (**DATE**) takes into account the deviation of the platform motion from the intended target trajectory based on the Target Tracking Motion plots in Figs. 4.8(a), 4.9(a), 4.10(a), 4.11(a), 4.12(a), 4.13(a), 4.14(a), 4.15(a) and 4.16(a). The other two indices, namely the Discrete Average Absolute Error (**DAAE**) and Discrete Average Control Effort (**DACE**)

are concerned with the actuator-position control errors and voltages averaged over the six actuators. These three indices can be defined in the discrete form with non-dimensionalised actuator stroke and frequency as,

$$\mathbf{DATE} = \frac{1}{\max(t^*)T} \sum_{j=0}^{N_s-1} |t_e(j)| \tau \quad (4.15a)$$

$$\mathbf{DAAE} = \frac{1}{num} \sum_{i=1}^{num=6} \left\{ \frac{1}{\max(s_i^*)T} \sum_{j=0}^{N_s} |s_{e_i}(j)| \tau \right\} \quad (4.15b)$$

$$\mathbf{DACE} = \frac{1}{num} \sum_{i=1}^{num=6} \left\{ \frac{1}{\max(|V_i|)T} \sum_{j=0}^{N_s} V_i(j)^2 \tau \right\} \quad (4.15c)$$

where, t_e represents the trajectory error between the platform demand trajectory (t^*) obtained from $\mathbf{q}$ and the platform response trajectory (t) obtained from the inversion of the manipulator kinematics model for the recorded samples (j), s_{e_i} represents the leg length errors of the i^{th} leg of the platform as denoted in equation (4.13a), V_i represents the **PI** control voltage sent to the i^{th} leg valve as denoted in equation (4.13b), N_s denotes the total number of samples, num represents the total number of actuators i.e. six, τ is the sample time and T represents the time-period of the target frequency under investigation.

Table 4.2: Comparison of performance indices of the PMBSS for the reference target trajectories depicted in Figs. 4.8 and 4.9

Target Frequency (Hz)	DATE	DAAE	DACE
Horizontal Motion of Target			
0.05	0.1699	0.1513	0.5642
0.10	0.1179	0.5370	2.6024
0.15	0.3187	0.5855	4.0573
Vertical Motion of Target			
0.05	0.3037	0.2377	0.2149
0.10	0.2645	0.2408	0.2130
0.15	0.3178	0.5746	0.6342
Diagonal Motion of Target			
0.05	0.0318	0.0334	0.0797
0.10	0.0298	0.0890	0.2147
0.15	0.0254	0.0746	0.2266

The maximum values of the actuator demands (s_i^*) and control voltages (V_i) for non-dimensionalisation are chosen to be the maximum available stroke of the

SRA, i.e. 0.15m and maximum voltage range of the **PV**, i.e. $\pm 10V$, respectively as provided in Table 4.1. The maximum value of the desired target trajectory (t^*) is 0.1m for all the three cases as mentioned in their respective sections. In order to quantitatively analyse the effectiveness of the **PMBSS** for the considered motions, the performance indices for all these motions are tabulated in Table 4.2 for the respective time windows shown in Figs. 4.8 to 4.16.

- (a) *Horizontal Motion*: The values for the horizontal motion with increase in target frequency provided in Table 4.2 seem to validate the observations made based on the plots shown in Figs. 4.8 to 4.10, whereby, a rise in control voltage is observed as suggested by the **DACE** values, while the actuator displacement error seems to have reached a saturation zone as indicated by the **DAAE** values owing to the mechanical constraint of the actuator stroke. However, the **DATE** values suggest that target trajectory tracking operation is least deviated for 0.1Hz frequency.
- (b) *Vertical Motion*: Similar observations can be made from the values of the vertical reference motion in Table 4.2 for the plots shown in Figs. 4.11 to 4.13. The **DAAE** and **DACE** values seem to suggest drastic increase in actuator displacement error and control effort from 0.1Hz to 0.15Hz. It is significant to note that in this case as well, the **DATE** values show that the target trajectory tracking operation works best for 0.1Hz frequency, out of the frequencies investigated here.
- (b) *Diagonal Motion*: In the case of the diagonal motion as shown in Figs. 4.14 to 4.16, the values of the performance indices, **DAAE** and **DACE** in Table 4.2 seem to suggest that the actuator displacement error and control effort are lowest for the 0.05Hz frequency. However, the **DATE** values show that the tracking error is least for the 0.15Hz scenario.

4.5 Summary

In this chapter, the development of a motion sensing system based on the visual servoing capabilities of an industry-grade electrohydraulic parallel manipulator based on eye-in-hand visual servoing is discussed. The parallel manipulator based servoing system (**PMBSS**) is developed and experimented upon with the aid of a virtual moving test target system, in a laboratory environment. Real-time experiments are performed on the system with the aid of state-of-the art NI systems for three different cases of the target trajectory motion with varying temporal velocities. The experimental data is quantified in the form of performance indices,

which suggest that this particular system under investigation seems to operate optimally, from the trajectory tracking point of view, for the target frequency of 0.1Hz for most of the investigated reference trajectories.

In order to further increase the applicability of this real-time technology, investigations regarding various other aspects of this motion sensing system need to be carried out in the future. Some of these notions have been highlighted as future scope of work in the final chapter.

Conclusions

The present thesis is focused on the development of real-time control solutions for electrohydraulic actuation systems utilized in electrohydraulic systems and manipulators, with the aid of neural networks, sliding mode control and vision based feedback control techniques. This chapter presents a consolidated conclusion to the thesis work by drawing inferences from the studies discussed in the preceding chapters. In addition to that, several research directions for future studies emerging from this present thesis have also been delineated.

5.1 Conclusions of the Present Work

In the context of the present thesis work, detailed studies regarding the design of an adaptive recurrent neural network with Hermite polynomial based activation functions augmented with an integral sliding mode scheme (**RHNN-SMC**) and a vision enabled adaptive first-order sliding mode scheme with a deadband controller (**ASMCDDB**), have been carried out through real-time experiments on a laboratory-scale linear **EHAS** with a large deadband proportional valve and industry-grade cylinder. Adaptive formulations for several control variables based on the minimization of Lyapunov functions have been developed for both the proposed control schemes. These formulations have been shown to be theoretically convergent, thus constraining these parameters within a bound.

The real-time results for the **RHNN-SMC** scheme have demonstrated its ability to precisely control the actuator position of the **EHAS** with contact-type position transducer (**LVD**), in the low-frequency regime which is generally plagued by discontinuous nonlinearities and actuator-friction. The **RHNN-SMC** scheme has yielded the lowest values for all the normalized performance metrics (**NDAE**, **NDTAE** and **NCE**) compared to previously established methodologies like BCGA-Optimized Fuzzy-Feedforward-Bias Controller (**GA-OFFB**), Adap-

tive Fuzzy-Sliding Mode Controller with Adaptive Bias (**AFSMC-AB**) and industrially common **PI** controller for the single **DoF** linear **EHAS** in the low-frequency region. This feature of the controller may be attributed to the superior dynamic mapping capability of the **RHNN** and the disturbance rejection capability of **SMC**. Additionally, the use of an integral **SMC** scheme provides global robustness to the scheme, which is evident from the system responses to a range of conventional tracking excitations. Similar observations were made for the **ASM-CDB** controller used for position tracking of a vision enabled linear **EHAS**. It is noted during this investigation that the use of visual sensors to track the piston motion alongside control schemes such as the **SMC** result in similar performances as those obtained with a contact-type position transducer. This **ASMCDB** scheme also outperforms a conventional **PI** controller for regulation and tracking demands upto 1.0Hz, which is evident from the lowest values of the normalized performance metrics obtained for the proposed controller. The tracking results of the control scheme upto 1.0Hz for this visually enabled **EHAS** is significant considering the fact that the frame rate of the vision sensor limits the data acquisition to around 100 samples/sec which in turn restricts the system bandwidth. Therefore, based on experimental results it is quite evident that both the proposed control schemes show stability and robustness for low-cost single-actuator **EHAS** designed with **PVs** and both types of sensing apparatus, viz. contact and visual, despite many nonlinear and uncertain features of these systems as mentioned in Chapter 1.

Another crucial aspect of this thesis has been the development of a motion sensing system using a six-actuator parallel configured robotic manipulator or Stewart Platform with an eye-in-hand visual sensor. Although, such a system has been theorized in limited studies by Qi & McInroy, a real-time laboratory scale working system has not been reported in literature until now. Hence, this investigation also serves as a proof of concept. The developed parallel manipulator based visual servoing system with a vision sensor attached to the end-effector of the manipulator (**PMBSS**), tracks and replicates target trajectories displayed on a virtual moving test target system. The performance of the motion sensing system is analysed by subjecting it to simple 1D and 2D target trajectories with different temporal velocities. The corresponding performance indices (**DATE**, **DAAE** and **DACE**) calculated for each of these trajectories indicate that this particular laboratory scale **PMBSS** is particularly suited to operating at 0.1Hz for 1D and 2D trajectories, which shows the lowest values for these performance metrics. It is also obvious from the experimental results that the operational bandwidth of this **PMBSS** is limited to 0.15Hz. This may be attributed to the mass of the moving platform of the system (250kgs), which restricts it from tracking higher frequency target motions.

5.2 Scope for Future Works

In the present scenario, where social distancing and non-contact measurements are fast becoming global trends, spatial target tracking and positioning systems using visual cues in industrial environments are sure to have a positive impact on the workforce. Based on this premise, it can be concluded that research on such topics will gain popularity not just amongst academicians but also industrialists. Hence, it becomes necessary to keep improving the technologies associated with **EHAS**. Several such notions regarding future directions for research as an extension of the present work, are enumerated below.

1. **Upgradation of the RNN-SMC scheme to track higher frequency demands and achieve energy savings:** In the present study, the advantages of the **RNN-SMC** scheme are limited to the lower frequency regime. Hence, there is room for upgrading the **RNN-SMC** architecture so as to achieve high-frequency demand tracking by augmenting the feedback controller with a feedforward algorithm. Additionally, the studies presented in this thesis work consider a fixed-displacement operation of the pump. Therefore, there is a possibility of including energy savings in the above control scheme with the aid of variable displacement pump-control mode and/or proportional relief valves with active control.
2. **Implementing advanced Machine Vision Algorithms:** State-of-the-art computer vision algorithms may be used to capture targets of different shapes and sizes with pre-defined or random trajectories, which may be utilized to access the performance of the system to such stimulus. Also, the effect of illumination conditions may be considered for these vision algorithms.
3. **Developing various operational modes of the Motion Sensing System:** Currently, the system works only in the translation y-z plane of the parallel manipulator. However, the Stewart Platform is a 6-**DoF** system and this capability of the manipulator should be explored with respect to visual servoing, so that it may be utilized even in scenarios where translational movements are insufficient.
4. **Upgradation of the Visual Servoing Scheme:** The visual servoing scheme used in the motion sensing system may be improved by developing adaptive laws for the servoing parameters and utilizing nonlinear image based visual control schemes. Other visual servoing techniques as discussed in Chapter 1, may also be used to assess the variation in system performance.

Also, in the present study, the test target for the servoing scheme has been simulated on a computer monitor. It is possible to include real-time targets within the scope of the experiments and therefore, evaluate the performance of the servoing schemes for tracking real-time target motions.

5. **Incorporating nonlinear controls schemes at the actuator level controller in the Motion Sensing System:** The nonlinear control schemes described in the first two chapters of this thesis for a linear **EHAS**, viz. **RHNN-SMC** and **ASMCDB** may be incorporated as the Joint-level feedback controller in the 6-DoF **PMBSS**, thus providing a much rigorous validation of these proposed control schemes.
6. **Developing a Simulation environment for the EHAS with single and multiple cylinder arrangements:** Before implementing the algorithms in real-time experiments, it may be necessary to develop a simulation environment for the single-cylinder and multiple-cylinder **EHAS** (serial and parallel manipulators) encompassing all the areas involved in their operation. This may be done with the aid of MATLAB/Simulink/ Automation Studio or any other suitable software package or their combinations.

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